LiveBoard



LiveBoard:
The revolutionary software for the advanced management of audio-video contributions and animated graphics.

User Manual

rev. 06/30/2025

Style and Innovation with ClassX

ClassX srl
Via Del Tiglio 91
56012 Calcinaia (PI) ITALY
Tel. +39 0587 48 81 28
E-Mail: support@classx.it
Home-Page: www.classxcg.com

ORIGINAL EDITION: 2008/2025

© Copyright 2025 ClassX srl

No part of this manual may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying and recording, for any purpose without the express written consent of a duly authorized representative of ClassX srl.
The information contained in this manual is believed to be accurate and correct at the time of publication. All rights reserved. Unpublished rights reserved under international copyright laws, under international copyright laws.

ClassX, CastaliaCG, MoreCG, CoralCG, LiveBoard, SGL, TemplateActions, LiveStadium, FrameMarker, plugin, SocialServer, SportsGraphics are trademarks of ClassX srl.

All other brand names are trademarks of their respective holders.

1 Introduction and provisions

Congratulations for having purchased LiveBoard, the revolutionary software for the advanced management of audio-video clips and animated graphics.
The name “LiveBoard” was born starting from the following concept: the representation and execution of a content is entrusted to a button. All buttons are arranged in a grid that forms a “board”, similar to a keyboard.

1.1 What is LiveBoard

The answer is easy: LiveBoard is the ideal environment for the advanced management of audio, video contents and live graphics. It is not only a simple playout of graphical templates, but an advanced system that allows the synchronous execution together with audio and video clips, tickers, count-down, logo ect..
LiveBoard integrates the most advanced technologies developed by ClassX in a complete frame-based, advanced graphic engine able to manage the multiple broadcast of more than one graphic template realized with CastaliaCG and MoreCG, and of the contributions created through the MediaPlugs, the advanced suite of plugins, integrated with all ClassX programs.
The special focus in developing the WYSIWYG graphic interface allows the user to exploit the whole potential of the product in a short time and with ease.

1.2 Who is LiveBoard intended for

LiveBoard is a proficient software to be used in those directions that need to show precise and reliable instant dynamic graphic contents for Live events, Sports events, News editions, Talk Shows, Presentations and any such event that needs to be managed in real-time.
LiveBoard has unique features in terms of quality, flexibility and usability; moreover it is one of the few Software of its category that permits the simultaneous playout of several graphic templates and the plugins.

1.3 Technical features

The technical features of LiveBoard are:

- Supported contents:
  - -Media: audio/video in the most common formats.
  - -Image: in the most common formats with alpha channel support.
  - -Graphics: graphic templates created with MoreCG, CastaliaCG and CoralCG.
  - -plugins: plugins integration and support.
  - -Group: for the synchronous execution of more contents.

- Features of the contents:
  - -Immediate execution by keyboard or mouse.
  - -Synchronous execution with audio files.
  - -Parallel execution of more contents.
  - -Exclusive mode execution.
  - -Organization in folders for a more simplified and advanced management.
  - -Preview image for each folder.
  - -Cut/Copy/Paste for each content.
  - -Preview image.
  - -Editing during the execution.
  - -Complete layout management with advanced positioning tools and real time preview.
  - -Visual priority management.

- Features of the graphic templates:
  - -Shared management of the resources for a maximum optimization of the memory usage.
  - -Immediate Take “frame” for the graphic templates.
  - -Programmable pauses.

-Frame based real-time preview.
-Control on the times and on the execution cycles
-Integration with CastaliaCG, MoreCG and CoralCG for editing the graphic templates.
-DataSource technology support that allows the complete management of external data sources such as Databases and TXT, CSV, XLS, RSS and HTML text files.
-Management of the projects' contents during the execution.

- SGL graphic acceleration support*.
- FrameLinkDriver technology support.
- No panic key.
- Project's saving and loading.
- Contextual video output preview.
- Programmable and customizable Interface.

* SGL - NVIDIA cards highly recommended, openGL

1.4 Conditions of sale, installation and support

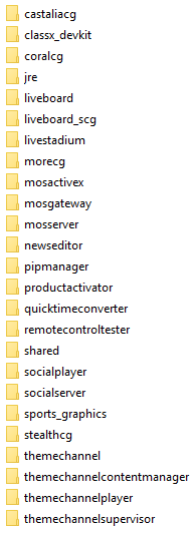
At the time of purchase, ClassX provides a detailed guide for installing and registering the software.
In addition, documents relating to the Annual Maintenance Agreement (CMA) and the general conditions of sale are also provided.
If you do not have these documents, you can request them by contacting ClassX directly at the official contact details indicated on the website or in the materials provided.

1.5 Software protection

The main risk to your and our investment is represented by software piracy. In many countries, unfortunately, illegal copying of the software is not even perceived as real theft.
Despite the efforts made to develop effective but non-invasive protection systems for LiveBoard, there are still attempts at unauthorized copying or use, despite the laws and related sanctions.
It is important to underline that the time spent designing and implementing new protection measures is time taken away from the development of new software features. Furthermore, it should be considered that each new protection inevitably involves some additional operations even for regularly registered users, who may have to perform less convenient procedures to protect their rights.
We therefore ask for your understanding: these measures are necessary to guarantee the value of the product and protect its integrity.

1.6 Organization of files and directories

After the installation process, as we have already seen, you will find a folder on your hard drive in C:\ called "ClassX_Applications" containing files and folders of the ClassX Distribution.



In detail, the folder "ClassX_Applications" contains the following directories:

Folder "LiveBoard" - contains the application and everything you need for LiveBoard.
Folder "jre" - contains the directories and libraries needed by the ClassX application.
"Shared" folder - contains all the resources shared by ClassX applications (Images, Video, DataSource, Scripts, Audio, 3D objects, preset themes, etc..).

The folder "LiveBoard" contains the main program, it supports files and any additional trays to complete the program. The following are the contents of the LiveBoard directory:
Folder **"doc"** - contains all the related documents, such as manuals.
Folder **"projects"** - contains all the projects that come with LiveBoard and is the directory where new projects will be saved unless the user defines a new path.
Folder **"export"** - contains all the animations and sequences of frames generated by LiveBoard.
Folder **"presets"** - contains all the presets that come with various versions of LiveBoard.

Warning: the folders "platform", "shared / plugins" contain files necessary to run LiveBoard. These folders and related files should neither be moved, nor renamed or altered.

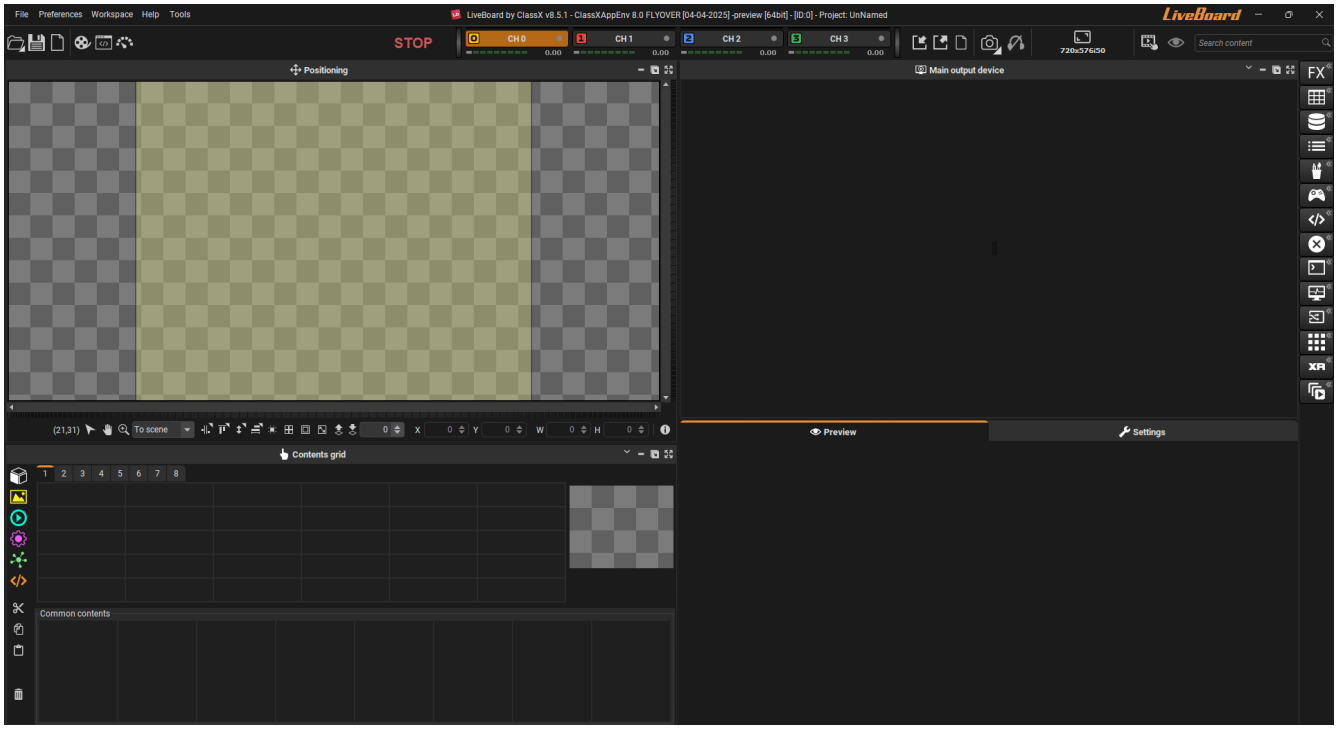
1.6.1 The ClassX conventions for naming files

To further facilitate the work of maintaining on-disk files, it is advisable to use appropriate names, chosen on the basis of small and clever everyday-rules that can greatly reduce searching times.
For example, if we were to save a LiveBoard project to a topic, it would be advisable to use the name "Argument.cst" and the corresponding sequence of frames with "Argument_xxx.jpg".
The name "topic" would immediately understand what it is, while suffixes ".cst" and ".jpg" serve to indicate the file type.
In general, therefore, the prefix should indicate the category of file (eg "Theme_", "Title_" etc..), While the suffix represents the type (ex ".gif", ". jpg", ". cst ", etc...).

Note: LiveBoard automatically adds the appropriate extensions when saving files.

2 LiveBoard Interface

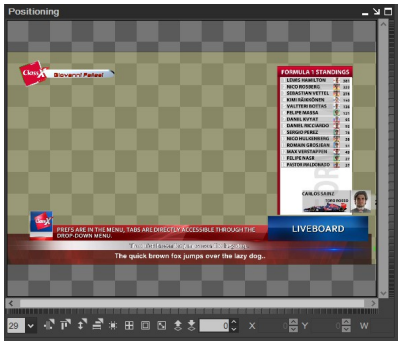
Learning how to use a new software often means to fit in or to “get used” to a new user interface, new menus and new features, sometimes very difficult to manage. LiveBoard, like every ClassX product, lies outside of this category, allowing to use any program features through an advanced user interface.
This kind of organization allows the user to exclusively concentrate on the project's creation, thus avoiding useless exercises of memory or weary searches in the manuals only for making simple things.
A great point of strength of LiveBoard is the possibility to work in real time during the whole creation and formulation of a project. This means that the program immediately responds to the actions of the user in real-time, without waiting times and always guaranteeing the maximum quality in the final result.



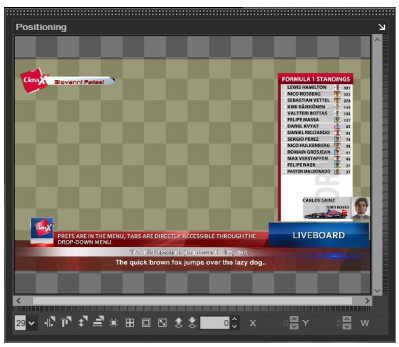
LiveBoard Interface

LiveBoard's user interface is flexible and customizable. Different environments are organized in panels that can be arranged (by docking/undocking and resizing as needed) as the user wishes and the newly made working environment can be saved.
The **Workspace** menu in the upper toolbar has two preset environments (Editing and Playout) and a Default one. Plus, any arrangement done custom by every single user can be saved in a “New” Workspace.

2.1 Docking/undocking panels



Panels are identified by a dark upper bar, the panel name is always on the left. On the right, there are three icons for minimizing, docking/undocking and full screen/popping back (see right).



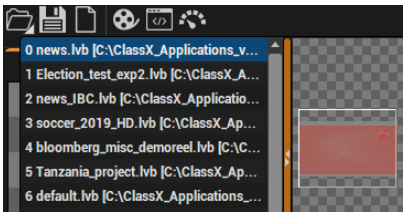
The undocked panel (see left) has a double dotted line on its upper side and only the arrow icon on its right, grab the double dotted line with the mouse in order to position it into a new place, or maybe onto another monitor. By dragging the undocked panel over the user interface, orange shapes will appear on the existing panels, suggesting new possible places where the undocked panel can be released, either as it is or as a tab of another panel. The **Contents Grid**, containing all buttons triggering the playout of contents, is as dockable panel as well and can thus be positioned and resized as the user wishes.

3 The Toolbar

Operations involving the management of a project are collected in the toolbar at the upper side of LiveBoard.



3.1 Loading a project



The first icon from the left opens a previously saved LiveBoard project. Loading a project can be done at any time.

Left click: opens the file selector to load a project from a specific path.

Right click: opens a small window containing a list of recently opened projects. The right part shows a preview of the selected project. Double clicking on the list will open the selected project directly. Left clicking on this icon will bring up the usual file browser, from which you can select your LiveBoard project from a folder in the file system.

3.2 Saving a project

The LiveBoard project consists of all objects, their attributes, timing, effects and details of the working environment.
When a project has been created or modified, it is usually necessary to permanently store it on a disk.
There are two functions that perform this task: one is the **"Save"** function, which you will find as a button or in the drop-down menu of the "file" and **"Save As"** menu. The latter allows you to open a file selector to set the name and path of the project before saving it. In a newly created project, clicking the "Save" button will automatically open the "Save As" saving procedure.

3.3 New project

The "**New Project**" button will clear the previously loaded project and all the resources it used from the User Interface. "New Project" also restores the application to its default condition, while keeping the customization on the Workspace unchanged.

3.4 Pack and Unpack a project

DEFINITIONS

A ClassX **Template** is an object generated by ClassX Template Editors such as MoreCG, CastaliaCG, CoralCG.

A ClassX **Project** is an object that groups resources or templates.

Resources are:

- Font
- Images
- Videos
- Audios
- Files
- URL
- Templates
- Projects

Absolute Resources means all those resources located outside the standard ClassX path. **Related Resources** means all those resources located within the standard ClassX path.

Overview

Project Packer is a set of APIs dedicated to collecting and distributing graphic templates and ClassX projects.
All this is integrated in all ClassX Broadcast applications with the aim of making the distribution and exchange of graphic projects and related resources much easier.
Project Packer is divided into Importer (pack project) and Exporter (unpack project).

Functionality

- Analyze a project or template generated with ClassX applications.
- Recursively collects references to all referenced resources.

At the end of the analysis, the resources are automatically excluded depending on the status of the flags of the "Preferences" panel. In the event that, during the analysis, datasources or URLs were found that could refer to external resources that cannot be relocated, it will be possible to display a "Warning" button which, once pressed, will display the warning list.

Relocation of absolute resources:

All absolute resources will be relocated under the standard ClassX path. They will be located under a folder that will take the name of the project.

Example:

Project name: "bloomberg.lvb"
Absolute resource path: "C:\Users\dar\Desktop\doc\clip.avi"
It will be relocated in: "C:\ClassX_Applications_v8\shared\video\bloomberg\clip.avi"

Relocation of related resources:

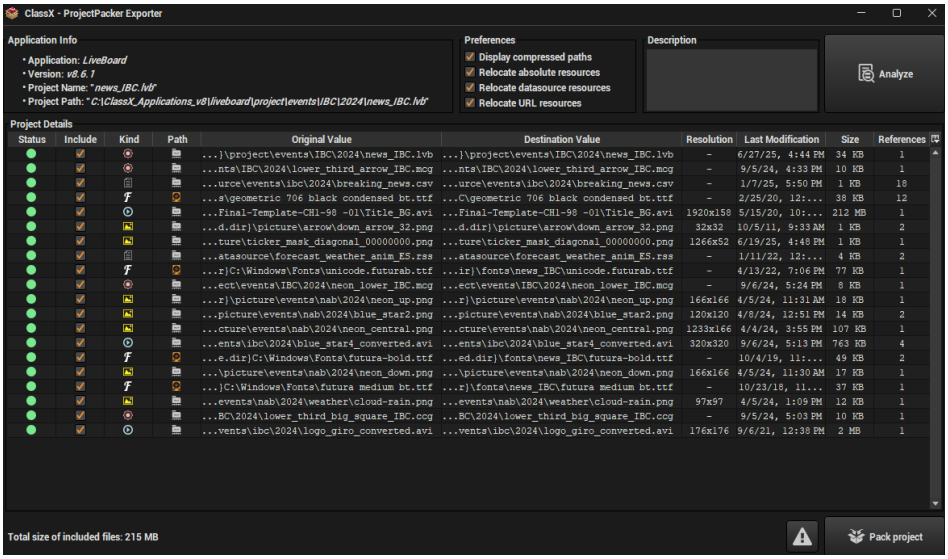
Relocation of the relative resources takes place by simply taking up the original path of the resource.

Example:

Relative resource source path: "C:\ClassX_Applications_v8\shared\video\clip.avi"
It will be relocated in: "C:\ClassX_Applications_v8\shared\video\clip.avi"

3.4.1 Pack Project (Export)

When you have selected the project to export, the following panel will open:



The top-left panel shows the current **Application Info**, including the project name.

The top-right panel shows the Project Packer file information. Here you can add the description.

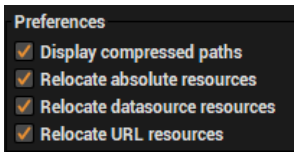
Resources such as picture or videos will be relocated in the “shared” folder inside the individual package folder.

Example:

Your “shared” folder is in C:/ClassX_Applications_v8/shared and you named the package “bloomberg”:

- all pictures will be relocated in C:/ClassX_Applications_v8/shared/picture/bloomberg folder
- all videos will be relocated in C:/ClassX_Applications_v8/shared/video/bloomberg
- all audios will be relocated in C:/ClassX_Applications_v8/shared/audio/bloomberg
- all 3dobjects will be relocated in C:/ClassX_Applications_v8/shared/3dobjects/bloomberg
- all alpha resources will be relocated in C:/ClassX_Applications_v8/shared/alpha_resources/
- all effects resources will be relocated in C:/ClassX_Applications_v8/shared/efx_resources
- all datasource will be relocated in C:/ClassX_Applications_v8/shared/fonts
- all fonts will be relocated in C:/ClassX_Applications_v8/shared/videos/bloomberg
- all LiveBoard projects will be relocated in C:/ClassX_Applications_v8/liveboard/projects/bloomberg
- all MoreCG projects will be relocated in C:/ClassX_Applications_v8/morecg/projects/bloomberg
- all CastaliaCG projects will be relocated in C:/ClassX_Applications_v8/castaliacg/projects/bloomberg
- all CoralCG projects will be relocated in C:/ClassX_Applications_v8/coralcg/projects/bloomberg

The "**Preferences**" panel contains four options:



- **Display compressed paths**:
  it sets how file or folder paths are displayed in the ProjectPacker list.
  Uncompressed form: C:\ClassX_Applications_v8\shared\picture\bloomberg
  Example compressed form of the same path:
  {shared.dir}\picture\bloomberg

- **Relocate absolute resources**:
  if disabled, Project Packer will not automatically include **absolute resources** during packaging.

- **Relocate datasource resources**:
  if disabled, Project Packer will not automatically include **datasources** in the packaging phase.

- **Relocate URL resources**:
  if disabled, Project Packer will not automatically include **URLs** in the packaging phase.

IMPORTANT NOTE
Special resources, such as **fonts** will be relocated in the C:/ClassX_Applications_v8/shared/fonts folder. These fonts will have to be installed manually by the user.

ANALYZE button: whenever you press the “Analyze” button, the analysis of the project will be re-executed.

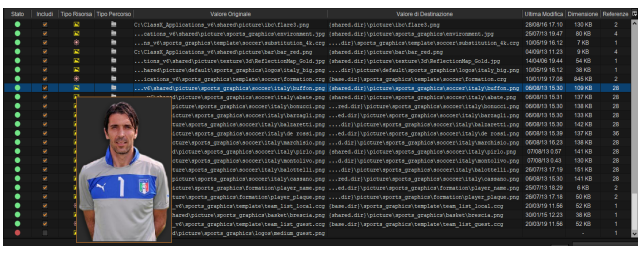
The center of the panel shows the table with resources that can be packed, resulting from the last analysis of the project.

This table shows the following information:

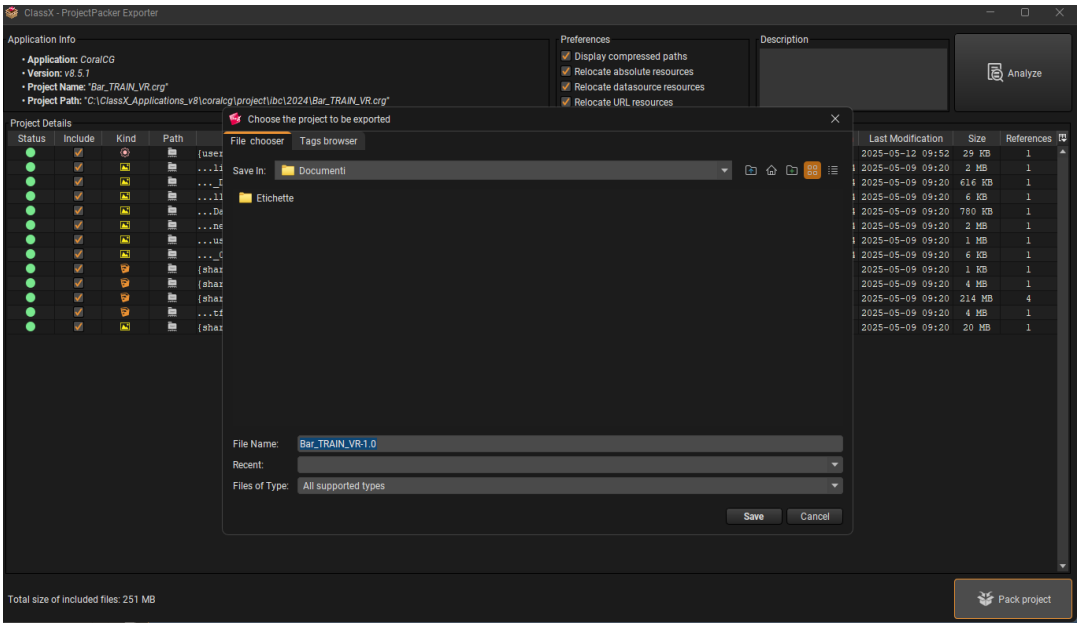
- **Status**: green if resource can be copied and red if resource is missing or not accessible
- **Include**: checked if you would like to include the resource in the package, otherwise unchecked
- **Resource kind**: the type of the resource: image, video, audio, obj3d file, mask image, efx resource, datasource file, URL, font, LiveBoard project and Template Editor project
- **Path kind**: relative to the installation folder (e.g. C:\ClassX_Applications_v8\shared) or absolute path (e.g. on Desktop or generic URL)
- **Original value**: it shows where resource is
- **Destination value**: it shows where resource will be relocated. If it is empty due to a resource not found, a dash "-" is automatically displayed
- **Last modification**: the date of last file editing
- **Size**: the size of the file
- **References**: the number of times that this resource is used in the current project

The bottom-left corner of the panel shows the total size of the resources that will be exported.

Moving the mouse over an image icon in the "Resource Kind" column, a small preview of the content will be displayed.

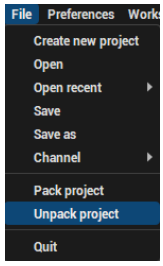


When you are ready to export, click to “**Pack project**” button in the lower right corner. This will open the system file chooser in order to choose where the package file will be exported. The default name pattern is <package.name>-<package.version>.clxpp . You can change the name as you prefer.

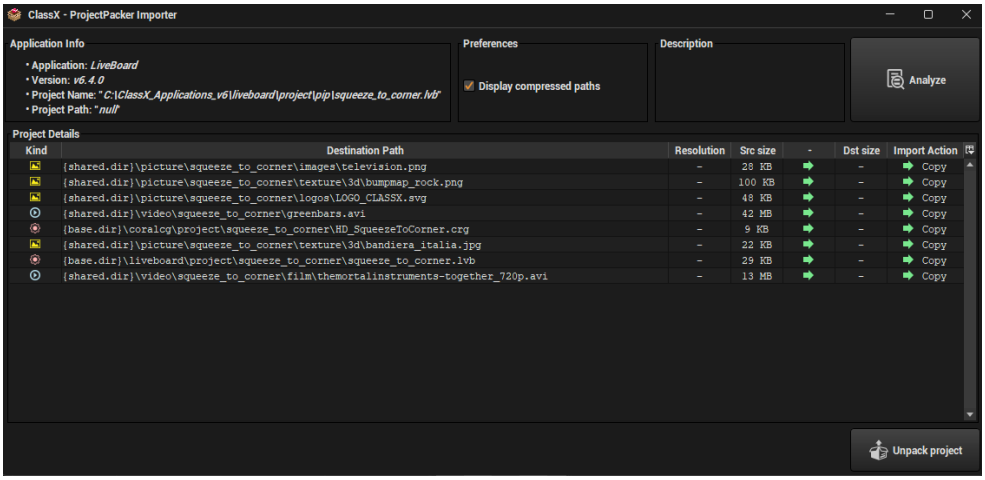


3.4.2 Unpack Project (Import)

The Project Packer files (clxpp files) can be imported directly from the ClassX Applications LiveBoard or the Template Editors MoreCG, CastaliaCG and CoralCG.



Select the clxpp file to unpack and the following panel will be opened:



In the upper side, the “**Application info**” and “**Package info**” show information created during the packing process.

In the center, “**Project Details**” shows the resources in the package.
The following resource information is shown:

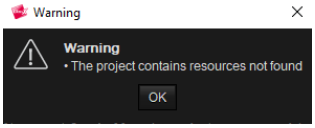
- **Resource kind**: it shows the icon associated to the resource type
- **Destination path**: the destination path where the resource will be relocated
- **Src Size**: the resource size
- **-** : shows the status of the importation
- **Dst Size**: shows the size of the resource in the destination folder. Before import, the value will be zero; once imported it will be the same as the "Src Size" if the process has been successful.
- **Import action**: it is the action to do during the importation:
  - Copy: copy the resource as it is
  - Replace: the destination folder already contains a resource with the same name. This action will replace the resource in the destination folder with the resource coming from the package
  - Skip: the resource will not be copied

Click the “**Unpack project**” button to unpack the project.

WARNING

In the event that there are problems or files not found, a "Warning" icon will appear at the bottom left of the "Pack" or "Unpack" project button. Clicking on it will display a pop-up with the list of problems encountered.





3.5 Menu “File”

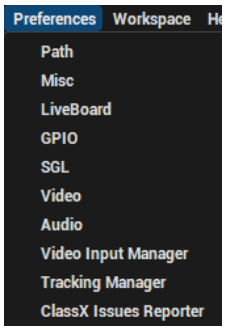
The File menu contains functions already described above, namely: Create New Project, Open, Open Recent and Save.

In addition, it includes two other functions:

- **Save As**: allows you to save the project with a different name;
- **Exit**: allows you to close the software.

3.6 Preferences

By clicking the "**Preferences**" menu in the upper toolbar, you open the panel with different tabs for the management of preferences, through which you can set the startup configuration of the program and control various using aspects.



The "**Default Page**" button restores the selected tab to "factory" settings. For example, if you are in the "**Paths**" tab and you press on "**Default Page**" button, you can restore the "factory" settings only for this selected tab.
The button "**Default All**" restores **all tabs** to "factory" settings.

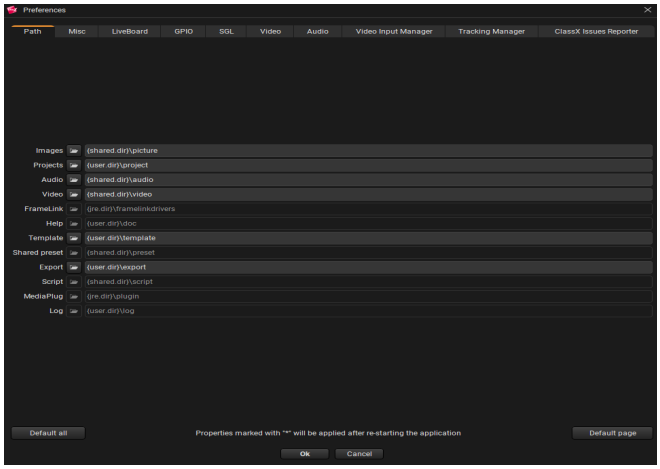
Before exiting from the "Preferences", you have to confirm or cancel the new settings using the buttons "**OK**" and "**Cancel**".

In LiveBoard you can directly access the desired tab of the Preferences Panel by selecting it from the dropdown-menu.

Let's see the single tabs of the **Prefs**:

3.6.1 Path

The “**Path**” tab shows the directories that the different contents or objects are stored in by default. You can change these paths by clicking onto the small folder icons.

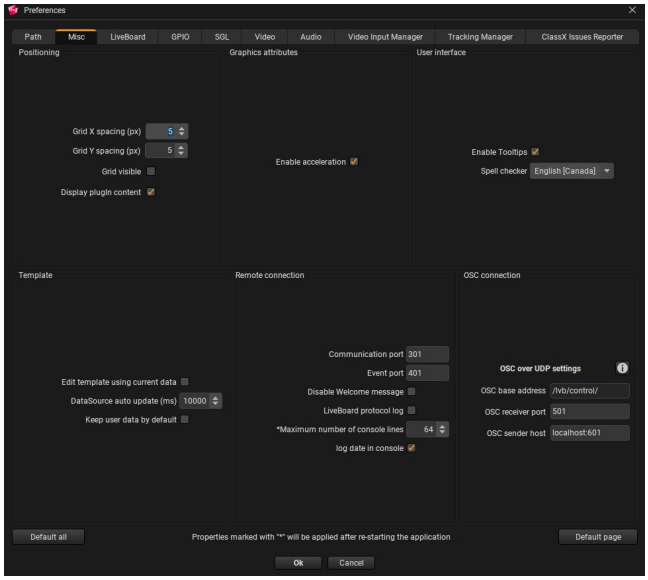


3.6.2 Misc

The "**Misc**" tab contains settings for the Positioning Panel and positioning templates.

3.6.2.1 Positioning

"Grid" allows you to define the number of pixels (grid value) used by this panel, to create a fictitious grid that helps you align and position the contents neatly. You can set the grid to be visible or not by checking/unchecking the “Grid visible” box.



The checkbox "**Display plugins content**" allows to display a full preview for each plugins in the Positioning Panel. For example, the Horizontal ticker plugins in its preview shows the phrase "the quick brown fox jumps over the lazy dog ..". With this option enabled, this plugins displays the inside of his real preview content uploaded by playlist default. NOTE: not all plugins support the full preview.

3.6.2.2 Template

The checkbox "**Edit last values**" enables a special mode that deals with the modification of a graphics template.
This feature is very useful when the user realizes that a text object in a template does not correctly display the desired value. By checking the box, any change done to the data values of the graphics template *through the LiveBoard interface* will be automatically saved in the LiveBoard project that you are working on.
Normally, the changing of a template data value is managed by the Editor application that generated it. The Editor will load a content (template) into LiveBoard ignoring all changes that have been previously made inside LiveBoard to the “editable” objects inside the graphics template, for instance when the LiveBoard project is loaded again after it has been closed, unless this checkbox is enabled.

3.6.2.3 Graphics attributes

You can choose to enable the hardware acceleration (by the vga card) of the graphics attributes by clicking onto this checkbox. Only certain LiveBoard plugins or contents have graphics attributes, such as Media Texts, Charts and Shapes. The aesthetic outcome of the accelerated and non-accelerated graphics attributes may be different, and it is up to the user to check or uncheck this feature, depending on the final effect that he wishes to obtain.

3.6.2.4 Remote connection

LiveBoard integrates TCP/IP and OSC protocols.
This section shows the communication port and event port values for connecting via TCP/IP or OSC. These information are important for the remote control of LiveBoard e.g. by an external application such as the SocialPlayer.
LiveBoard Protocol log is useful to store the logs generated by LiveBoard, for the user's reference or in case the user asks ClassX for assistance with an issue.

3.6.2.5 OSC protocol and integration with CuePilot

LiveBoard natively supports the OSC (Open Sound Control) protocol. Please consult:
https://en.wikipedia.org/wiki/Open_Sound_Control
https://opensoundcontrol.stanford.edu/spec-1_0.html

Through this protocol it is possible to control LiveBoard from external applications using the LBControl commands (see LBControl section).

Integration with CuePilot:

https://www.classxcg.com/cuepilot-integration-and-technological-partnership.htm
https://github.com/MikOfClassX/LiveBoard-CuePilot-schema

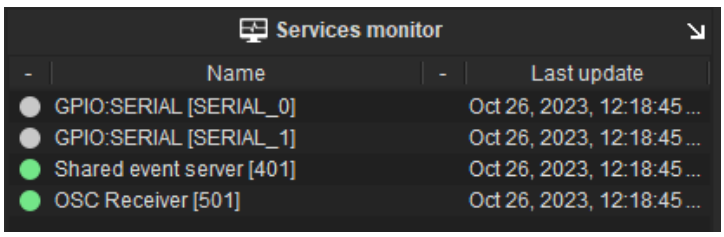
In the OSC-CuePilot protocol supported by LiveBoard there are 6 control commands:

Command	OSC address	Arguments	Descrizione
PLAY	/lvb/control/play	<content_name>	Play of the content
STOP	/lvb/control/stop	<content_name>	Stop of the content
PAUSE	/lvb/control/pause	<content_name>	Pause of the content
RESUME	/lvb/control/resume	<content_name>	Resume of the content
STOPALL	/lvb/control/stopall	-	Stop all content on air
LBControl	/lvb/control/lbcontrol	<command_lbcontrol>	Run a generic LBControl command

To set up an OSC connection you need to specify the connection port from the preferences in the “OSC Connection” section.

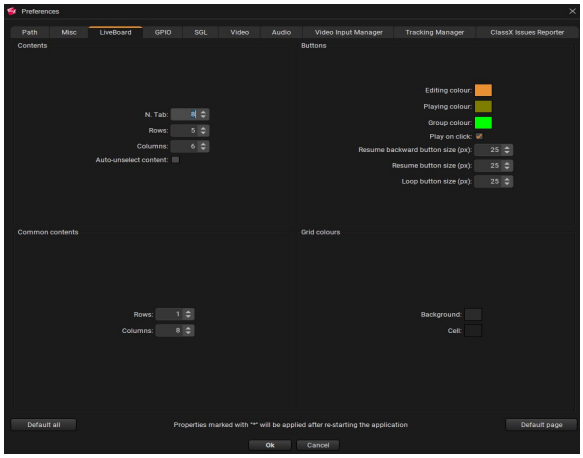


Once you specify the port, you can view the connection status from the “Service Monitor” dock.



3.6.3 LiveBoard

The “**LiveBoard**” tab manages the following features:



In **Contents**:

- Number of tabs
- Number of rows
- Number of columns

In **Common contents**:

- Number of rows
- Number of columns

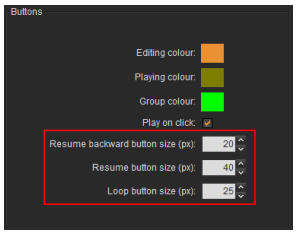
In **Button and Grid colours**:

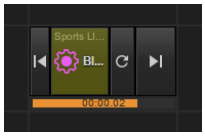
- Frame colour for the editing / playing and grouped statuses
- Background colour of the Contents grid

Every time we create a New project, the number of default folders and grid will be as defined in this panel. It is important to underline that the tabs configuration can be manually changed and saved inside the actual LiveBoard project, in order to be restored at any time this project is loaded in the future. In this case the default number of tabs, and grid values set in the Preferences, will be ignored.

3.6.3.1 Button size preferences (resume/loop)

In the "Buttons" section it is possible to customize the size in pixels of the buttons (resume/loop) of the content grid.





3.6.4 SGL

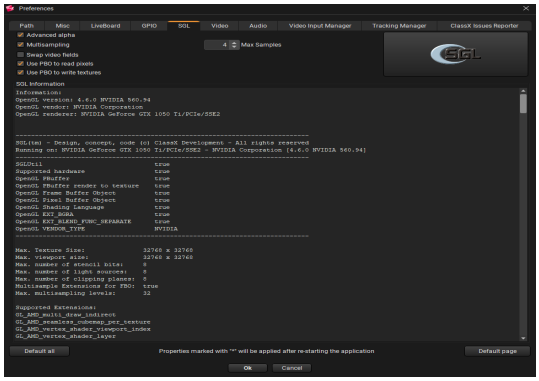
The “**SGL**” tab allows to enable the graphic acceleration by using the SGL library through the “**Enable**” component.
ClassX Software exploits the OpenGL acceleration of the VGA card through SGL. This component is disabled if the video card does not support OpenGL.

SGL is a proprietary technology developed and owned by ClassX.



The “**Test**” button opens a window showing an animation that verifies the actual performance of the video card.

The “**SGL Information**” box shows the technical features of the video card and the outcome of the acceleration test.



The “**Advanced Alpha**” checkbox enables an exceptionally proficient calculating mechanism for transparencies. More information in the SGL section.

The “**Multisampling**” checkbox enables advanced anti-aliasing used to improve image quality. The management of Multisampling requires more memory usage and the performance directly depends on the on board VGA card.

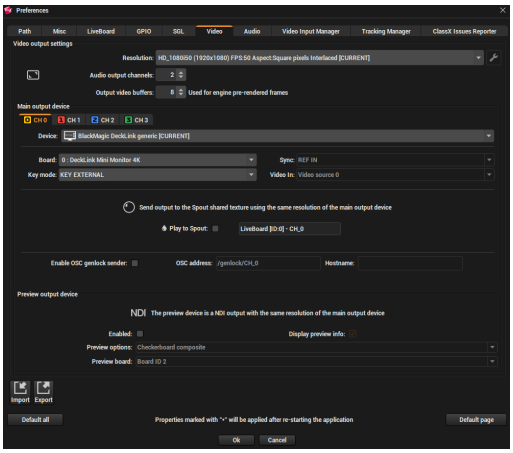
The “**Max Samples**” value improve or decrease the“ **Multisampling”** level. More the number is higher(the max value depends from VGA) than the result will be better.

The "**Swap video fields**" button is used only if after enabling SGL, the playout is drawn on the screen with incorrect ruling. The SGL library normally automatically detects dominant fields by interacting directly with the card, but on some VGA card models this automatic detection could not work due to the specific card drivers used. In these few cases, this checkbox must be checked.

PBO stands for “Pixel Buffer Object” and is a particular reading/writing protocol onto the VGA memory involving asynchronous pixel transfer operations. “**Use PBO to read pixels / write textures**” should be both checked for modern VGA cards; older ones though could need to selectively deactivate these features.

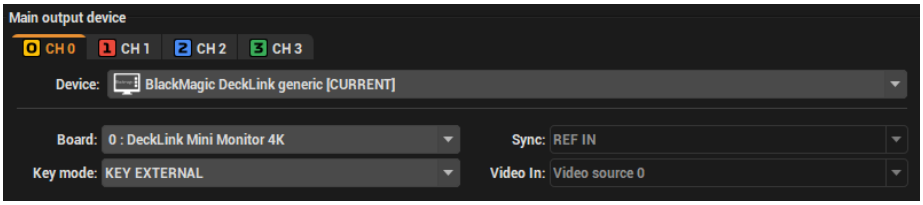
3.6.5 Video

To select the output device used by LiveBoard for content diffusion and to determine the resolution of the project, it is necessary to enter in the "**Video**" tab. It allows you to select the output device used by LiveBoard for airing contents and for determining the project resolution.



First of all, it's important to define the resolution for the playout (e.g. HD i50).

LiveBoard can play up to 4 channels simultaneously, but the resolution is common to all.
It's not possible to select different resolutions for each channel.



As shown in the figure above, for each channel you can choose the Device for the output. It's possible to select between NDI®, VGA, Spout, SDI and other option. There are three **SDI output devices available**: Blackmagic, Bluefish444 and AJA. For each selected device, you need to choose the BOARD ID to use depending on your video card configuration.

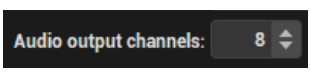
You can also select “**Play to SharedFrame**” to allow viewing in the homonymous plugins.

By double-click on the tab name, it's possible to change the name of the channel.



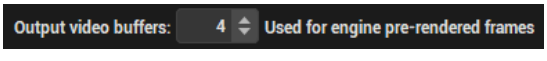
3.6.5.1 Audio output channels

Every channel in LiveBoard is able to play up to 8 different audio channels. According to the selected number, you will be able to see the sound bar in the panel “Main output device”.



3.6.5.2 Output video buffers

This setting that allows you to adjust the number of output video buffers used to store engine pre-rendered frames. Increasing the number of buffers can improve playback stability, especially under heavy processing loads, while reducing them may lower latency.



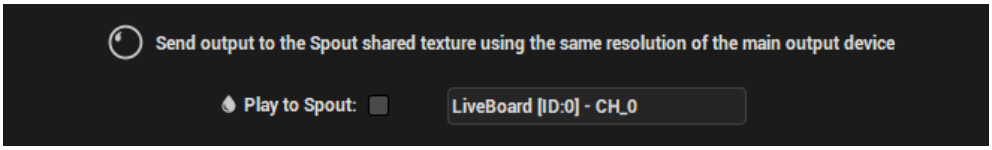
3.6.5.3 The Key mode

In case of SDI output, you need to select also the Key Mode of your signal:

KEY EXTERNAL: if you want to play out in fill&key
KEY INTERNAL: if you want to use your workstation as a pass-through
KEY OFF: if you want to have no key in your playout.

3.6.5.4 Spout output device

Simultaneously with the main output of LiveBoard via a video device (eg Blackmagic), it is also possible to enable the **output in Spout**, so as to be able to capture the signal also through any other platform that reads the Spout video signals.



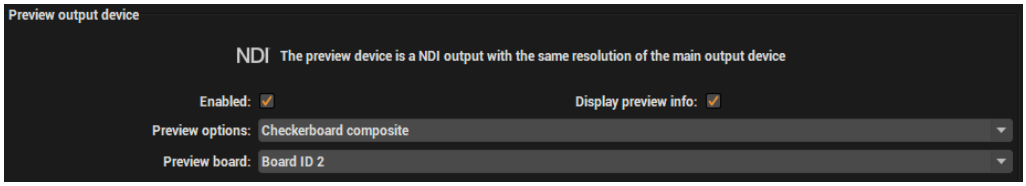
Spout is a technology that fully relies on the graphics card, so it doesn't burden the CPU.
It **does not currently support** audio.

For example, it is also possible to capture the signal directly from the **Video Input** in LiveBoard or from other ClassX and non-ClassX software that support Spout.

For further information, see the section dedicated to Video Input Spout (paragraph 7.19.1.6).

3.6.5.5 Preview output device

In the “Video” tab of LiveBoard there is also this section for managing the output device of the **Preview panel**.
The “preview output device” is a function that allows you to view the content played in the **Preview panel** on an external NDI® monitor.
In case you need to see for example an animation of a graphic template or a video in a different or larger screen than the LiveBoard Preview panel, you can very easily enable this function.



By selecting “**Display preview info**”, the name of the content we are viewing and the resolution will also appear in the monitor at the bottom of the scene.

E.g.: 

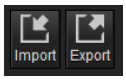
“Preview options” through a drop-down menu, gives the possibility to choose the background to be displayed in the monitor behind our graphic content:

- **Fill and key**: it displays the content with the selected background in the NDI® monitor settings
- **Checkerboard composite**: it sets the default background of the Preview panel as the monitor background
- **Black composite**: it sets a generic black background as the monitor background
- **White composite**: it sets a generic white background as the monitor background

“**Preview board**” allows to select the exit ID in NDI®.

Warning: if “NDI Network Device Interface” is selected as the LiveBoard output device, be sure to select two different “Board ID” for the two different outputs.

3.6.5.6 Video preferences import/export



The "export" function allows you to save a ".properties" file with all the settings inserted in the "Video" tab and export it to another workstation.
Instead, the "import" function allows you to import this file and retrieve the previously exported settings.

The preferences will be saved, by default, in the “Documents” folder with a default name and marked with a current timestamp: *CX_VideoOutputSettings_<date>_<time>.videooutput*

3.6.6 Audio

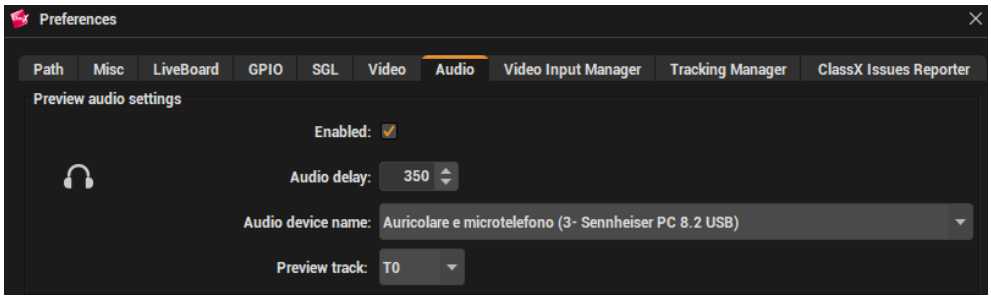
Within the **Audio** tab it is possible to activate the execution of the audio also on another external device in addition to the main output device chosen in the Video tab.

3.6.6.1 Preview audio settings

If enabled, it specifically activates the audio preview of the program.
It is also possible to set an audio delay in order to synchronize it with the program output audio.

In “**Audio device name**” you can of course choose the device to use for this function.

“**Preview Track**” allows you to choose which track to listen as preview.

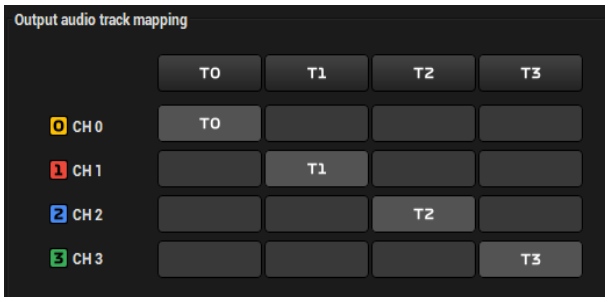


3.6.6.2 Output audio track mapping

This table allows you to map audio tracks to the various channels of LiveBoard.

To connect a track to a channel, simply drag and drop a track button into a cell. It is possible to combine all the audio tracks even in one single channel.

To delete a track from a cell, simply right-click on it.

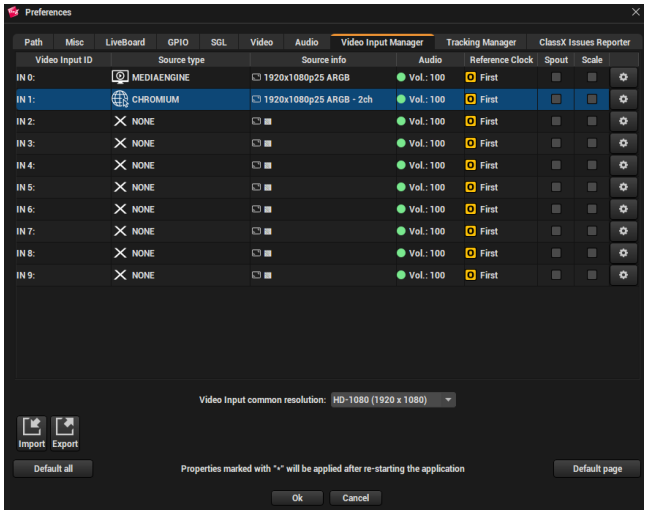




To set the audio mapping of the video input channels, you need to enter the appropriate panel via this icon directly inside the Video Input Monitor panel of each single video input.

3.6.7 Video Input Manager

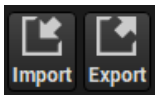
This tab allows to view and set the "Video Input" to use in LiveBoard.



- **Video Input ID**: channel name. With double-clicking it is possible to change the name of the source
- **Source type**: selected source type
- **Audio**: volume of the NDI® source
- **Source info**: it displays the input video information, such as resolution or the number of channels it has
- **Audio**: by clicking on the volume area, a slider will appear that will allow you to adjust the audio of the video input
- **Scale**: by checking the box, it allows to standardize in high quality the content to the resolution chosen in the list below "**Common Video Input Resolution**". This function is very useful when the original resolution of external sources does not coincide with that of the output.
- **Play to Spout**: it allows to play the resource at the same time even on a Spout channel.
- **Reference Clock**: like a genlock, the reference clock is the channel synchronization setting. The RC can be set to one of the available output channels and used to synchronize video inputs, audio inputs and tracking cameras to the specific channel. For smooth audio/video playout it's absolutely necessary to select the correct channel as reference. It is suggested to prefer channels with real hardware video output, as they guarantee the most accurate timing being themselves most probably connected to a real hardware genlock (e.g. blackburst or ext sync).

By clicking on the **gear** on the right of each resource, it is possible to select the source to be played by choosing from those ones listed by clicking on the drop-down menu located under each window.

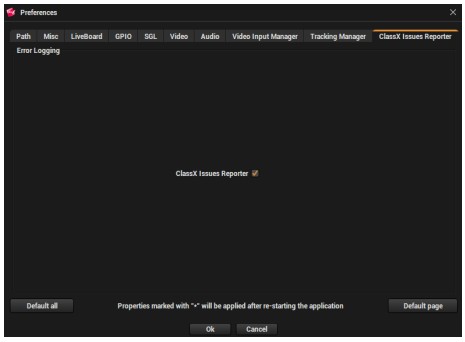
For details on the **various types of video inputs available** and how to configure them, refer to the chapter "**Video Input Source Types**".



The "export" function allows you to save a ".properties" file with all the settings and the various Video Input channels inserted in the "Video Input Manager" tab and export it to another workstation. Instead, the "import" function allows you to import this file and retrieve the previously exported settings.

The preferences will be saved, by default, in the "Documents" folder with a default name and marked with a current timestamp: CX_VideoInputSettings_<date>_<time>.videoinput

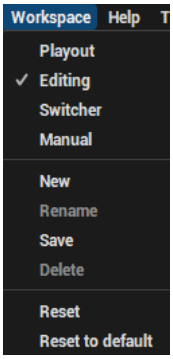
3.6.8 ClassX Issues Reporter



The last Prefs Tab gives the user the possibility to enable the Issues Reporter, that will send ClassX an error report, should an exception occur.

A dialogue will be shown on the Windows tray icons asking the user whether he wants to send ClassX an Error Report or not, if an exception should occur.
No hidden or automatic information will be sent to ClassX in any case, every single Issue sending will have to be approved by the user.

3.7 Menu Workspace



This menu allows you to set up, save and manage the workspace. There are some presets, designed for editing the LiveBoard project and for graphic playout.

Arrange the workspace as desired and then save the personal workspace:

- menu workspace → new → write the name of the new workspace → press ok

If you want the original workspace again, click "Restore default" in the workspace menu.
This is useful in case there are more operators working with the same software.
"Restore" restores the workspace to the last saved version, the one selected.
"Reset to default" will restore the factory settings.

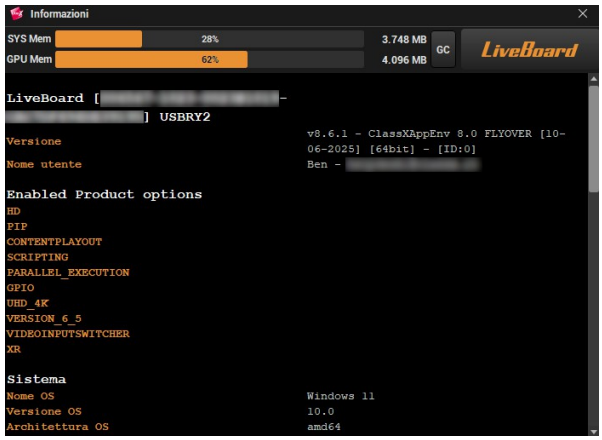
3.8 Menu Help

By clicking “Help”, the LiveBoard pdf manual will be automatically opened.

3.8.1 Info

The "**Info**" menu item opens a window that shows some useful information about the current installation and the status of the program. This is an extremely useful feature when you want to communicate the status of the program to ClassX with certainty, for example, in a bug report.
First of all, you can see the serial code assigned to the software license, the version you are using with the corresponding release date and, below, the product options activated with the license in use and various contacts to reach ClassX.

3.8.1.1 Memory management



LiveBoard has been programmed with the precise aim to optimize memory usage. At the program start-up, a certain amount of memory is allocated, enough to satisfy the initial needs of resources loading. LiveBoard exploits only a part of this memory, using the remaining one (“free memory”) in case of need while running the program. If the “free memory” is not enough, the program will allocate more.
The bar above the Information displays the state of the memory allocated by LiveBoard, showing the used, free and total amount. The orange bar in is the graphic representation of the memory: it represents the used memory, while the dark part represents the free memory. The bar is updated every time you press the “**GC**” button : you may verify the state of memory in real-time at any time.

3.8.1.2 The “GC” button

The term "GC" stands for garbage collection and allows you to perform a particular operation aimed at freeing portions of memory that are no longer in use.

LiveBoard uses all the resources made available by the operating system in a more optimized way to guarantee maximum performance. However, once the resources are released from LiveBoard, these are not available immediately but only after a certain period of time for the garbage collector action. The "GC" button starts the garbage collector which allows immediate recovery of the memory. However, this button should be used very carefully because once the garbage collector is in action, it requires a high use of resources and this could lead to drop frames. For this reason it is advisable to use it when there are no contents in execution.

3.9 Play to click



The "**Play to click on content**" check box controls the playback mode of the graphic contents. By default, when you press a content in the content grid, it goes on air directly; the icon on the interface is white.



You can choose not to activate this function; the icon on the LiveBoard interface will be in dark Gray. To disable this option, you need to remove the flag from "Play to Click" from the **Preferences** menu: *Preferences> LiveBoard> Play on click*

In this case, to play a live content from the content grid, you can choose between two ways:

- select "Play" in the drop down menu which appears by right-clicking on the content button
- key “shift” + right-click on content button.

3.10 No-panic key



The "**STOP**" key, also called "**no-panic key**", instantly interrupts all the contents in execution, producing an empty output on the current device. This function can also be called with the ESC key.
To stop the contents in all four channels at the same time, you need to do **CTRL + STOP**.

3.11 The Four channels



3.12 FPS meter



This very useful tool shows the number of frames produced at that time by the LiveBoard graphic server and sent to the output device; 25fps in the case of our example. Each LED represents a pre-calculated LED.

If all the LEDs are green it means that the graphics engine runs at full speed, while if the red LED lights up for more than a quarter of a second, drop-out may occur, resulting in a loss of frames.

Each channel has its FPS meter.

3.13 Import/Export/Clear channel and the resolution

These three functions are the same of the “File” menu → Channel.

 Import function: it allows you to import in the selected channel some contents previously exported from another one

 Export function: it allows you to export the contents of the selected channel to import them in another one

 Clear function: it allows you to clear all contents inside the selected channel



This icon shows the resolution of the project

3.14 Audio management of the preview

This button allows to enable or disable the audio in the preview panel or in the "Video Input Manager" tab in the Preferences menu.

 Audio enabled

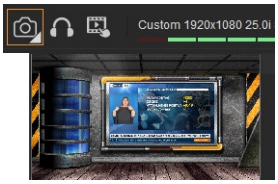
 Audio muted

3.15 Frame Capture

 By clicking on this icon it is possible to capture a snapshot of the content that is airing at the moment of the click. You can choose between 3 options:

- **Update preview icon**: it updates the “poster frame”, which is the preview image that is shown by the LiveBoard file-chooser when you want to choose a new project to open
- **Copy to clipboard**: the snapshot remains in the Windows temporary memory (it can be compared to the CTRL + C keyboard combination)
- **Save the image**: it allows to save the file in any folder on the workstation

By placing the mouse pointer directly over the icon without clicking on any function, the preview icon saved for that project will automatically be shown.



3.16 NDI preview status

The NDI preview device has the same resolution as the main output device and can be activated or deactivated from the Video tab of the Preferences menu (see paragraph 3.6.5.5).

 Preview output device activated

 Preview output device deactivated

3.17 Tools

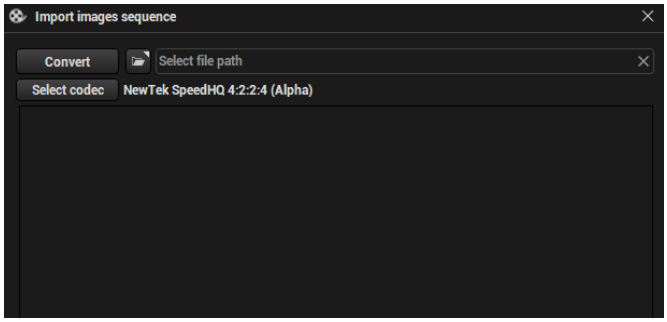
The tools in LiveBoard allow you to take advantage of some important functions for the success of your projects. The tools in LiveBoard are: “Import image sequence”, “Media Recorder” and “Media Converter”. Below are detailed explanations.

3.17.1 Import image sequence

This menu item allows you to import image sequences in TGA and PNG formats transforming them into animations with Alpha in the AVI format with Codec NewTek.

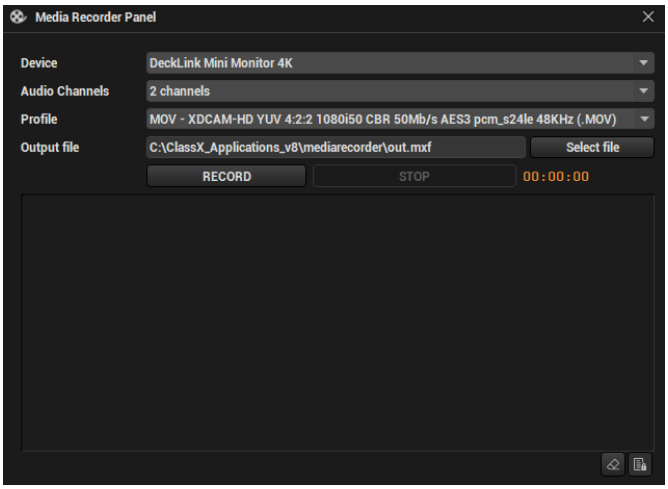
You can also choose the number of frames with which the animation will be created.

The recommended NewTek codec for import is “NewTek SpeedHQ 4:2:2:4 (Alpha).

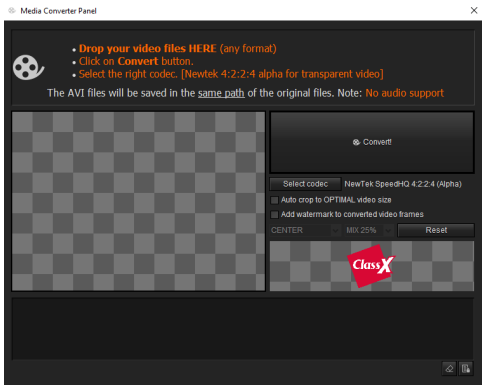


3.17.2 Media Recorder

This tool is very useful when you want to recorder an external SDI signal totally independent of the signals used in LiveBoard for input and output.
Through the small panel, you can select the SDI input device, the number of audio channels to recorder and the profile for the video imported (e.g. MOV, MXF or MP4). In the “Output file” section you can choose where to save your file.
This tool is completely independent of LiveBoard so it can record a video from a SDI input even if LiveBoard is playing out something on other channel.



3.17.3 Media Converter

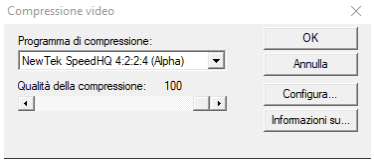


This software has been conceived and designed for those who needs to convert video files of various formats (eg .MOV, .MP4) into AVI videos by choosing between several codecs such as **NewTek**.
It transforms any video supported by **FFMPEG** in broadcast quality (with frame-accurate), supporting **alpha channel**.

3.17.3.1 Utilization

The user interface of **MediaConverter** is very simple and intuitive.

To convert a video, simply drag it anywhere within the interface and click on the "**Convert!**" button.
You can also add many videos at the same time via **multiselection.**



Automatically a little window will open. In the list in the drop-down menu, you can choose the codec you want.
The recommended NewTek codec for better performance and fluidity and for correct reading by ClassX software is the "**NewTek SpeedHQ 4:2:2:4 (Alpha)**".

Once you have selected the codec to use and after pressing "OK", the conversion process will begin.

In the central area of the interface, after conversion, "**conversion done**" will appear.

The new file will be saved directly in the same folder as the "original" video and will have the same name followed by "**_converted**".

Eg. *original → C:\Users\ClassX\Desktop\video1.mov*
converted → C:\Users\ClassX\Desktop\video1_converted.avi

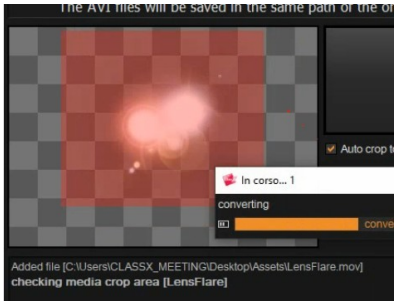
ATTENTION: the audio is not supported and consequently it is not converted.

3.17.3.2 Fill alpha channel

Some video formats do not support the "**Alpha**" channel (eg MP4); consequently it is not compiled and automatically displayed by some devices.
In these cases, during conversion, MediaConverter automatically compiles the alpha channel.

3.17.3.3 Auto Crop

Sometimes it can happen that you have videos with a larger size than the area that is actually filled with pixels. Example: in the case of a 1920x1080 video, with a lowerthird with real animated dimensions of 800x200, the space not occupied by the lowerthird is completely "empty", transparent.
For LiveBoard this "empty" space is actually a channel to be calculated for a correct rendering of the output. To optimize the playout of graphics calculated in real time, it is therefore advisable to upload videos on LiveBoard without unnecessary transparent spaces. For an exact crop of the video, the function "**Auto crop to OPTIMAL video size**" comes in handy.
Selecting this option, MediaConverter will analyze the entire video and return it cropped with the surface actually occupied by the pixels.



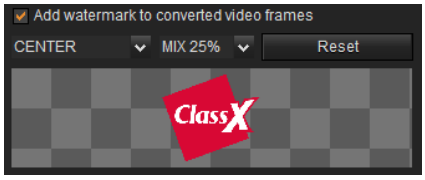
During the conversion, a red area will show the zone of the video actually covered by pixels and which will consequently be used for trimming.

N.B.: sometimes graphic technicians tend to create video in post production with the dimensions of the output resolution (SD, HD, UHD), even for very small contents, with the sole purpose of maintaining a precise positioning in the scene. In the case of a small object (logo, lowerthird, etc.) it is better to crop the video to its minimum size.

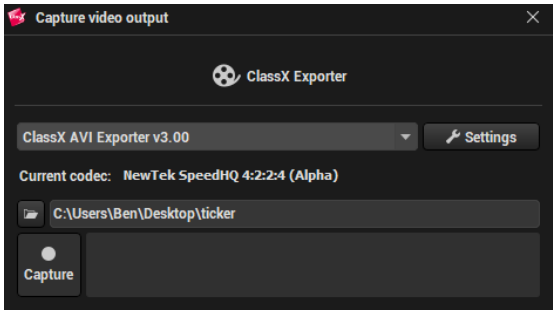
To maintain the original position in the LiveBoard output, simply set the same X and Y coordinates as the used ones for the content in the video editing program.
In the case of LiveBoard, the coordinates can be set in the positioning panel; in the case of graphic template editors, in the "Timeline settings" panel.
For further information, consult the dedicated user manuals.

3.17.3.4 Add Watermark

Through the Media Converter, it's also possible to insert a Watermark to the converted video(s). Just by cropping the Watermark image in the dedicated area and checking “Add watermark to converted video frames”, you can have a watermark on your video(s). You can also choose the position and the size.



3.18 Export



LiveBoard has an advanced feature that allows you to capture the video output directly to a video file with the full support of the transparency channel. You can access this feature either by clicking onto the **Export** icon or in File > Export.

By default, **AVI video NewTek SpeedHQ 4:2:2:4 (Alpha) codec** is selected as it is the recommended codec for obtaining high quality videos with full alpha channel preservation.

In the **settings**, however, it is possible to choose which codec to use for export using the "**Select codec**" button. You can also select the option to create "**Matte**".

By clicking on the icon with the folder, you can choose the destination path, while clicking on the text area you can change the name of the file. If no content is playing, the capture will automatically stop recording.
After the configuration is complete and choosing the export destination folder, simply click the "**Capture**" button to start recording the broadcast.

To change the export type, simply open the drop-down menu and select the one you want. The export types available besides AVI are **TGA, PNG, JPG, Jimi**. Also in these cases, the format and specific characteristics can be chosen from the settings.

WARNING: the export may not respect the frame-rate of content on the current time, such as timers and clocks.

3.18.1 Export to AVI with Alpha Matte

Almost all existing AVI codecs do not support transparency (alpha channel) necessary for proper overlay on a title video. Most video editing or playback systems, however, can overcome this problem by using Alpha Matte support or an AVI format mask animation to simulate alpha channel functionality.
For this reason we decided to implement this important function. To generate the matte settings, enable the "Create matte" check box in the settings. Automatically, the application will create two files: the animation file avi "Filename.avi" and the opaque alpha file "filename.matte.avi".



3.19 Console

LiveBoard has the ability to view the messages it receives from external software with which it operates simultaneously or from scripts created internally to the program. This console reports interchange messages or even communication errors.

3.20 Automatic Start-up

With the automatic start-up you can provide, through a command line (from CLI (Command Line Interface) such as DOS), the opening and execution of LiveBoard contents. The following parameters will guide you to the composition of the appropriate command:

- **FILE**: the path of the project to load;
- **ICONIFIED**: automatically reduces the LiveBoard window;
- **PLAYCONTENT**: defines the content to be run in the project;

Example:

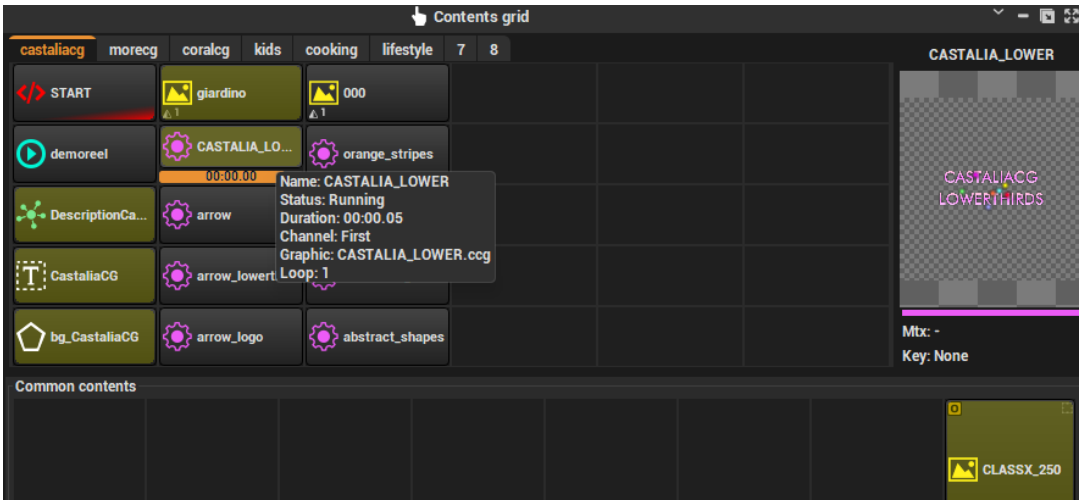
```
C:\ClassX_Applications\LiveBoard\liveboard.exe FILE=project\bloomberg.lvb,
PLAYCONTENT="Clocks", ICONIFIED
```

The path of project on an example is expressed as the relative path of the start-up directory (`C:\ClassX_Applications\LiveBoard\liveboard.exe`), but it is still possible to express it with an absolute path.

3.21 Closing LiveBoard

To quit the program you just need to click on the closing button (the “X”) of LiveBoard’s main window or select File > Quit.
At any request of program closure, LiveBoard controls if the current project has been saved. If not, it shows a message asking to confirm the operation. At the program closure, LiveBoard stores positions and size of the main editors and all of its panels, so that the user will find the same disposition at the next program start-up.

4 Contents grid

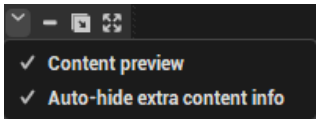


The content grid is the most important component of LiveBoard, allowing the user to manage and organize content with all the tools necessary for the most advanced features.
The LiveBoard operator will press the content buttons to send them live; this is the "on air" use of the contents in the Grid.

The grid has several tabs; a content can be moved from one part to another without restrictions, unless the operator is no longer on air. Moving a content from one folder to another can be done in two different ways:

- Through "cut / paste" operations
- Choosing the content with the mouse over the header label of the desired folder and waiting for its automatic opening after about a second. At this point the content can be inserted inside.

The grid size (rows x columns) is saved in the project so that it can be restored at the next reopening, ignoring the current settings.

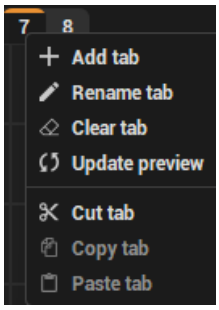


In the upper right corner of the Content Grid there are the usual three icons management icons (minimize, unlink, enlarge) and the small options icon.
By clicking on the latter, some options will be shown and they can be activated or deactivated.

- **Content preview**: allows you to show or hide the preview of the content on the grid
- **Hide additional information**: hide the extra information on the button; this can be useful when the operator wants to see a cleaner interface or if there are many buttons and the additional information of the buttons does not have a display space.

4.1 Management of the Tabs

The ability to sort contents in folders allows the user to better manage the transmission, inserting the contents according to a temporal or argumentative logic. The operations that can be performed on each folder are collected in a context menu opened with the right mouse button:



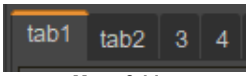
- **Add tab:** allows you to insert a new tab in the position following the selected one
- **Rename tab:** allows renaming the tab
- **Clear tab:** delete all contents from the selected tab
- **Update preview:** allows you to view / update the preview of the selected tab, which will be displayed in the preview grid simply by moving the mouse over the tab label. The preview of a card is calculated by simulating the simultaneous execution of all the contents contained in it.
- **Cut tab:** copy all the contents of the tab into memory and then delete the tab
- **Copy tab:** copy all contents of the tab into memory without deleting the tab
- **Paste tab:** inserts all the contents in memory into a new card

4.1.1 Move and rename a tab

You may rename tabs and order them by drag and drop. To rename a folder, simply double-click onto the tab label and edit the name in the text box.
Tabs can be organized through the operation of drag-n-drop: once you select the tab you want to reorder, just drag it and release the mouse in the desired position.
By saving the LiveBoard project, also the custom tab layout and naming will be maintained.
At the next load you will find the same tab configuration.



Rename folder



Move folder

4.1.2 Operations on contents

LiveBoard gathers in a contextual menu the main available operations on the content. According to the state of the content, some of these functions could not be available. To access to this menu, right-click onto the chosen button. The available operations are:

- **Play:** it plays out the content
- **Stop:** it stops the content that is being played out
- **Resume:** it resumes the pause of the content
- **Edit:** it selects the content in edit mode
- **Hide in pos:** it hides the content's preview in the positioning panel

LiveBoard allows you to access the state of the edit content also by drag – and - dropping the content onto any panel other that the Contents grid. Any Content that is being dragged, will turn the mouse pointer into the content icon. The dragging operation can be interrupted at any time by pressing ESC.

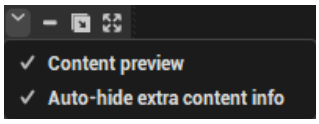
4.1.3 Content Preview in the Grid



Whenever the mouse is moved over the Contents grid, LiveBoard displays the content's preview in a special area on the right hand side of the buttons: the content's name is shown on top and underneath there is its preview.

The preview image is calculated and updated in different moments, according to the content. If the preview is not available, LiveBoard shows the icon of the content type as a preview.

You can hide the Content Preview by unchecking it in the Contents Grid extra menu (the small arrow showing downwards).



4.2 Contents



The concept of "content" represents the strength point of LiveBoard's graphic engine. A "Content" can be used to indicate separate graphics, video and audio. Generally, the content does not represent a specific clip, but an "abstract" element that, developed by LiveBoard's graphic engine, can be translated into a graphics, video animation, or more generally in a sequence of generic actions.
LiveBoard manages, with no distinction, several types of contents by identifying the following common operations:

Creation: procedure by which LiveBoard creates and manages, in the most appropriate way, the resources required for each content.

Destruction: all the resources generated during the creation are deleted.

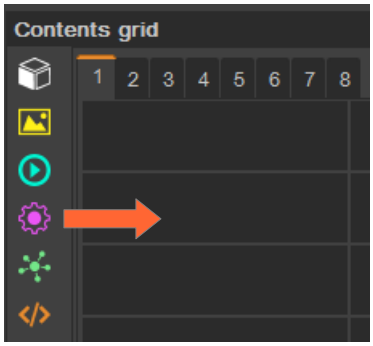
Execution: play / stop / pause are the three modes of execution that LiveBoard can handle according to the type of content.

Edit: it represents the phase where the user can modify each feature of a content, also during its execution.

Cut, **copy**, **paste**: the most common and simple operations that allow us to manage every content at its best.

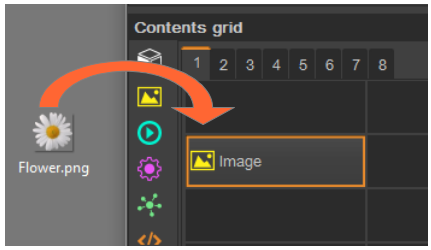
Most of these actions may be done using the mouse, through the "drag and drop" operation.
The concept of abstraction is also reflected in LiveBoard's graphic interface that associates and manages every content through a simple button.
The term abstraction, on which the concept of content is based, indicates the mental process through which a set of objects is replaced with a *concept*, more general, that describes the objects according to their common features.

4.2.1 Creating a content



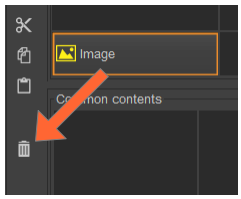
The available contents in LiveBoard are listed in the "Contents grid" on its left, they are represented by coloured icons. To create a content, it is enough to click on the desired button and drag it into the grid (see figure on the right) onto any empty button. At the beginning, this new button will be selected in "Edit" mode. LiveBoard's interface changes according to the type of content, all the other LiveBoard panels will contain information and parameters that refer to the new content.

4.2.2 Create a content from file system



You can create a new content with a simple drag-and-drop also directly from a windows file browser or your desktop: just drag the desired file directly inside the Contents grid onto an empty button. This type of operation is available only for files that are compatible and supported such as "Image", "Media" and "Graphics".
If there are more selected icons, LiveBoard will automatically import only the first element of the selection.
For all not supported file formats this type of operation is not possible.

4.2.3 Deleting a content – Intelligent Dustbin

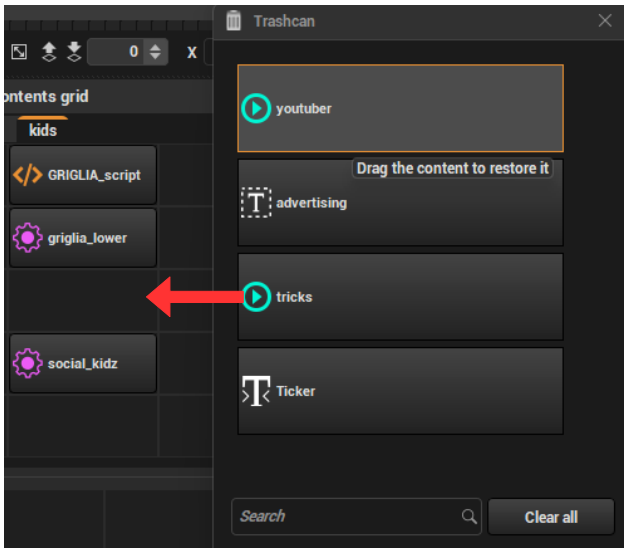


By removing the content from the grid, all the resources it required during its creation are freed. To delete a content, we just have to drag the undesired button onto the dustbin icon.

Deleted objects remain in the Trashcan, that will be emptied automatically upon the quitting of LiveBoard but will remain inside if new projects are loaded.

TRICK: use this intelligent tool to move a content from one project to another one.

You can restore the contents from the Trashcan to any tab and to any empty button at any time.



Note: a content cannot be deleted from the grid during its execution.

4.2.4 Cutting, copying and replicating a content

LiveBoard allows to perform the operations of cut, copy and paste on each content in order to give the user the maximum freedom in organizing and managing the contents.
The buttons in the “Operations” box manage three available operations:



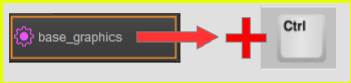
Cut: it copies the content into the memory and deletes it from the grid, freeing all the resources occupied during the creation. To perform this kind of operation, you just need to drag the button of the content over the related “Cut” button.

Copy: it copies the content into the memory without deleting it from the grid. To perform this kind of operation, you just need to drag the button of the content onto the related “Copy” button.

Paste: it inserts the object in memory into the grid. To perform this kind of operation, you just need to drag the “Paste” button into the grid on the desired position.

Note: The operations of cut, copy and paste are not available during the content’s execution.

TRICK: Copying a content can be achieved in a more convenient and faster way through the appropriate keyboard shortcut: once you select the content you want to copy, just drag it onto the desired location within the grid by holding down the CTRL key. The original content will not move, but the grid will automatically insert an exact copy of it into the chosen empty button.



4.2.5 State of the content

LiveBoard identifies and manages, for each content, 4 different states: Play, Stop, Pause, Edit.

Play : the content is being executed (running).

Stop: the content is not running (default state).

Pause: the content is paused during its execution.

Edit: the content is in edit mode and its features can be modified by the user.

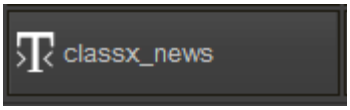
The states identify different situations, but some of them are intrinsically connected, as they describe the execution phases of a content. The passage from a state to another one happens following an easy logic:

Play <> Stop
Play <> Pause <> Stop

The first 3 states are mutually exclusive, while the Edit state describes a particular state that may occur in parallel to the other ones.

Note: the user can edit a content during its execution, when it is paused and when it is not running.

LiveBoard’s interface identifies and describes each state in a different way by automatically colouring and configuring the content’s button. Here are the different possible states:



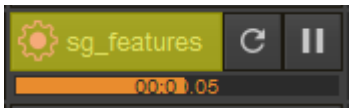
STOP



EDIT



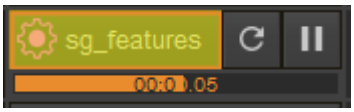
PLAY



PAUSE/LOOP



PLAY / EDIT

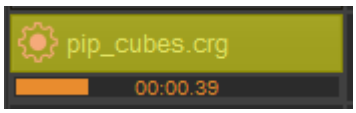


PAUSE/LOOP/ EDIT

Note: in Preferences > LiveBoard, it is possible to modify the colour that identifies the content's state.

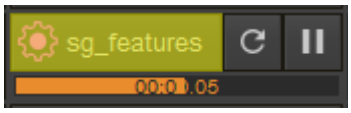
4.2.5.1 PLAY State

During the execution of a content, the button turns yellow (default colour). Moreover, if the content's execution can be expressed in terms of time, LiveBoard displays the temporal progress via an orange progress bar.



4.2.5.2 PAUSE/LOOP State

LiveBoard allows you to manage a particular execution state: the state of pause / loop. At the content's play, LiveBoard automatically displays the pause and / or loop buttons next to the content name. When content reaches the state, the button automatically activates the related "Resume" icon for pause and / or loop.

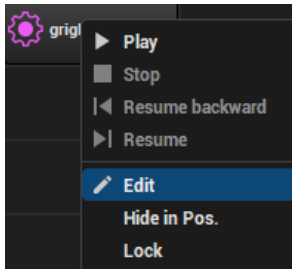


Note: the content that is in Pause / Loop is still "on air" and by pressing the Resume button, it will continue with its execution by executing the "state out" of pause / loop.

Note: the paused /loop state is available to LiveBoard only for contents that support it.

4.2.5.3 EDIT State

The editing of a content's properties happens through the edit state. Right-click onto any content that you wish to edit and select "Edit" from the small contextual menu that appears. The interface of LiveBoard will automatically adapt, showing the related panels.

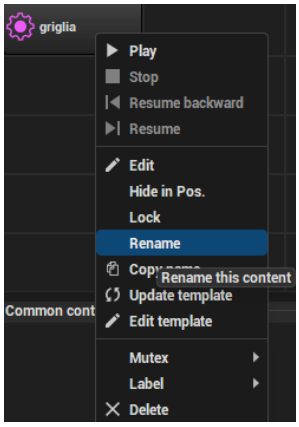


The properties of a content can be modified at any time with no limitation, also during the execution itself. As you can see in figure, the orange rectangle representing the content in edit, is visible during play, stop and pause phases.

Notes: the access to the edit state during the execution or the pause of a content is strictly connected to the implementation of the content itself. This means that for some contents the editing of the properties could not be available during the execution.

4.2.5.4 Rename a content

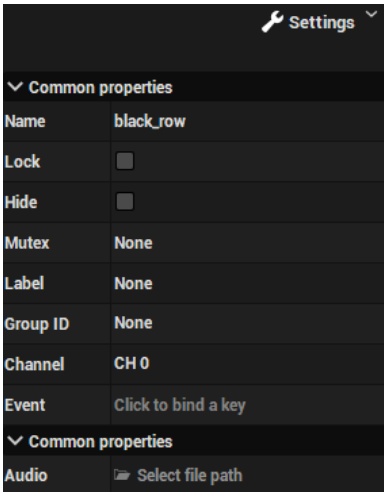
In addition to the common properties of the "Settings" panel, it is also possible to rename the buttons directly in the LiveBoard grid.
By right-clicking on the button, a list of actions will appear in the context menu where you need to select "Rename". In the text box, you can enter the new name to assign to the selected content. To confirm the change, type "Enter", to cancel it, type "Esc" from the keyboard.



4.2.6 Common properties

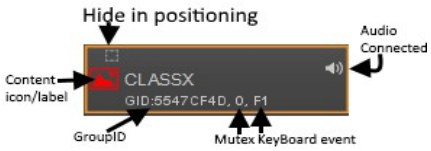
The “Common properties” contains all base content settings. This panel is always the same for each LiveBoard content.

The common properties are the following:



- **Name:** represents the name of the content in LiveBoard and by clicking on it, you can change it
- **Lock:** Lock the content in order to avoid editing, play or repositioning in Contents grid
- **Hide:** Hide the content in the positioning environment.
- **Mutex:** the index of mutual exclusion for the content.
- **Label:** assignee a color label to the content.
- **GroupID:** the index that identifies any group it belongs to.
- **Channel**: the channel the content belongs to
- **Event “**from keyboard”: allowing to setup a specific key to run or stop a content.
- **Audio:** bind an audio file to the content,it'll executed in sync with the content execution.

Here is how the content button displays the properties described above:



4.2.6.1 Icon and identification name

The icon and the identification name are very important because they allow to identify and distinguish very easily contents that are different or of the same type. For this reason we suggest you to always modify the default identification name.

4.2.6.2 Keyboard Event

The execution of a content can be managed also through a shortcut key. Clicking on “**Event**” row, a bind area will appear as in shown in figure and clicking inside it is possible to set a key for the content. Such (Event - Content) association allows the user to manage play/stop actions on a content without clicking on it into the grid: **play** / **stop** / **resume (in case of pause).**



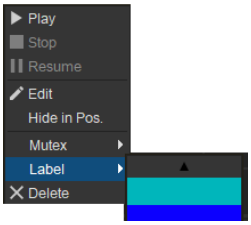
Notes: the keybind area will automatically display an error message if the shortcut key is already assigned to a new content.

4.2.6.3 Mutual exclusion

LiveBoard’s graphic engine manages the parallel execution of more contents without any limitation, but at the same time it allows to execute the contents in mutual exclusion. The term **mutex** (abbreviation of ***mut****ual* ***ex****clusion*) indicates a process of synchronization among contents, through it the program avoids that more parallel contents can be executed simultaneously.
Setting the same value of mutual exclusion on two or more different contents, the user will ensure that the execution of one excludes and stop the execution of the other.
The value of mutual exclusion fluctuates in a default range that is [1..20].

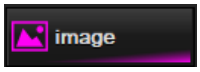
Note: the properties of mutual exclusion can be applied also to the Group content. In this special case all the contents belonging to the group automatically inherit this property.

4.2.6.4 Label



It's possible to assign different color for each button. Just by right-clicking on it and choosing “Label” in the list.

When the content doesn't play, it presents only a color shade on the button. Also the icon is colored:



When the content plays, the button is completely colored.



Example:



4.2.6.5 Execution in a group

All LiveBoard’s contents can be associated to a group by following a precise rule: a group can contain any content, except for the groups themselves.
The inclusion of a content within a group is done by defining the GroupID it belongs to. Once it is included in a group, the content can be performed individually anyway: the group is only for synchronizing the start or the stop of more elements. The Group content will be explained in detail later.

Notes: once the content has been assigned to a group, its value of mutual exclusion will be ignored because it will automatically inherit the one of the group it belongs to.

4.2.6.6 Hide in positioning

The positioning area displays the real content position on the screen, size and order visibility. In case of complex projects, this area can be chaotic because it shows all contents preview.
The “**Hide in positioning**” option, if enabled, allows the positioning area not to display the preview of such content. In this way the user can minimize to the essential the number of displayed contents, excluding the ones that are not essential. For example, in case of full screen video clips or graphic templates, this option can be very useful.

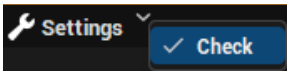
4.2.6.7 Associated audio file



LiveBoard allows to execute an audio file and a content at the same time, except for the Group content. The audio file execution happens in perfect sync with the associated content, and automatically stops when it ends its execution.

To associate an audio file to a content you just need to click on the “**Audio**” row and the audio management area will appear, then select the desired file using the “Folder” button. After this procedure the interface will display the path of the selected file.
The output volume can be managed using the commands on the right and it's possible enable the loop function using the “Loop” button.

4.2.6.8 Verifying a content



The verifying content operation allows to control the content health by checking the presence of any errors, which could compromise it execution. In case of errors, the content button displays it name in red, allowing the user to find the compromised content quickly. The “Verifying content” operation can be performed manually using the “Check” button inside the contextual menu in “Settings” tab.

Note: the verifying contents operation is automatically performed when a project is loaded, so the user can immediately verify the presence of any errors.

4.2.6.9 Advanced controls

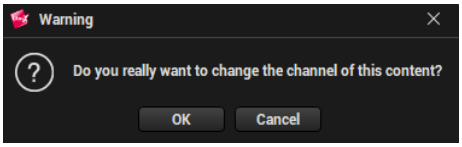
The “Advanced Controls” area put below the “Common properties” gives access to an advanced content settings. This area contains specified settings for each LiveBoard content, we'll analyze this area in LiveBoard contents later.

4.3 Common contents

In the "**Common contents**" grid you can insert those contents that you need to be displayed in all the "tabs" and in all the channels of the project.



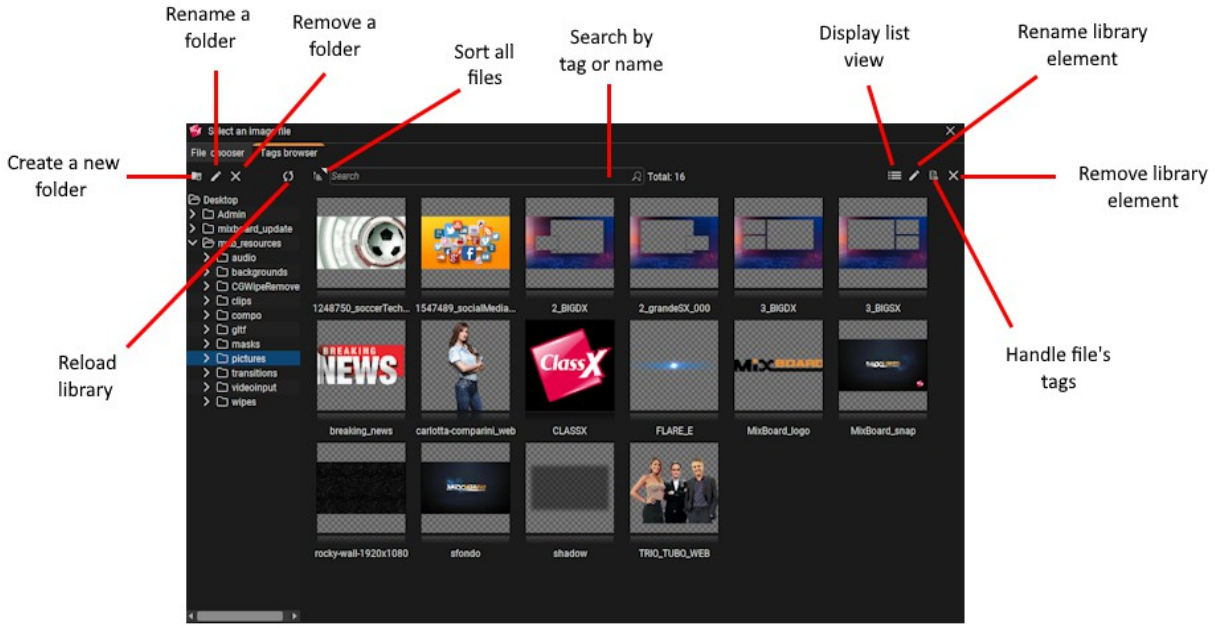
In this section each button displays a little number.
This indicates the channel it “belongs to”.
It is also possible to change the channel of a content. It's enough to drag-and-drop a content from “Common contents” grid to the contents grid of a different channel.



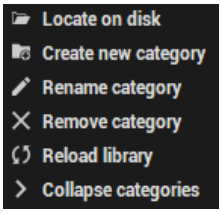
This message warns the operator if he really wants to change the channel to which content belongs.

4.3.1 File chooser and Tags browser

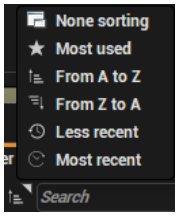
When you want to insert a new content on LiveBoard, a “file chooser” automatically appears on the screen. It allows to select a file to insert. When there are many files in the archive, searching for content is often more difficult.
To speed up this step, ClassX has introduced the possibility of inserting simple **TAGs** on each file to recall them quickly and intuitively. To insert new tags or to search by tags, simply enter in "**Tags browser**" tab of the file-chooser.



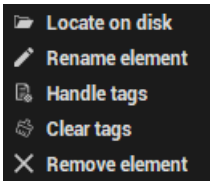
As visible in the image, on the left there is the list of files and folders within the path chosen in the "File chooser" tab. From the **Tag Browser** you can also locate on disk, rename, add or delete categories (folders). By right-clicking on each category, a small drop-down menu appears with the actions that can be performed.



On the right there is an area with all files inside each category and you can order them by clicking on the dedicated command. By right-clicking on the sort button, a drop-down menu will appear that will allow you to choose the criterion to be used for sorting.

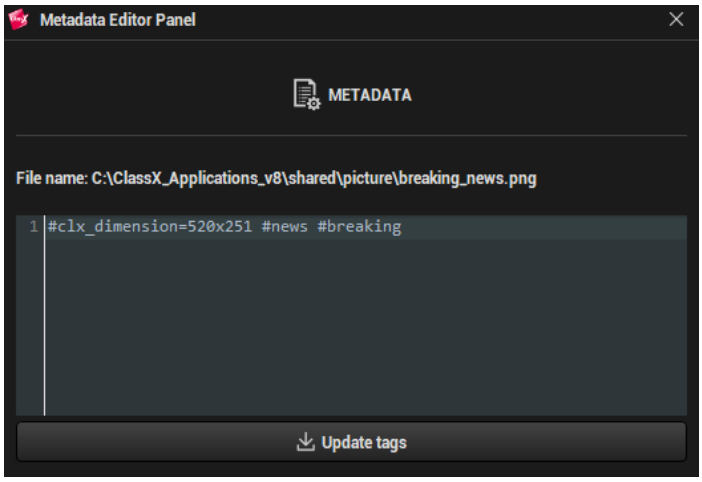


You can perform actions on a file (locate it on disk, rename it or add tags) by clicking on the dedicated buttons or, by right-clicking, directly above the selected file.
Specifically, by clicking directly on the file, it will also be possible to clean it of any previously inserted tags.



4.3.1.1 Tag insertion and search

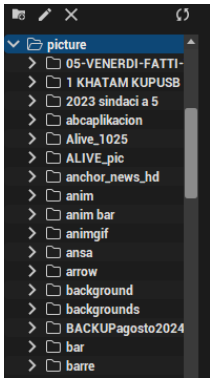
On the above-mentioned drop-down menu, you will find the command "**Handle tags**". If you click on it, a window will open where you can enter one or more words to be used as a tag for the selected file.



The "hash #" symbol that identifies a **HASHTAG** is created automatically by typing the various words separated by a space.
Once you have decided on the tags to insert, to correctly set them on the selected file, simply click on "**Update tags**" and close the **Metadata Editor Panel**.



In the search bar, by clicking on the magnifying glass, a list with all the hashtags used on the files will be displayed. Typing *#chosentags* will display all the files that contain **that specific** tag within the category selected in the left column, while typing #? will display all the files that contain **any** tag within the category selected.



Example: taking as a reference the list in the image on the right, as you can see, in this case "picture" is the parent category (or folder) that contains all the other various sub-categories.

If the search by tag includes all the categories within “picture”, it will be necessary to select "picture".
Otherwise if it is limited to a category ("buttons" for example), simply select the "buttons" folder.

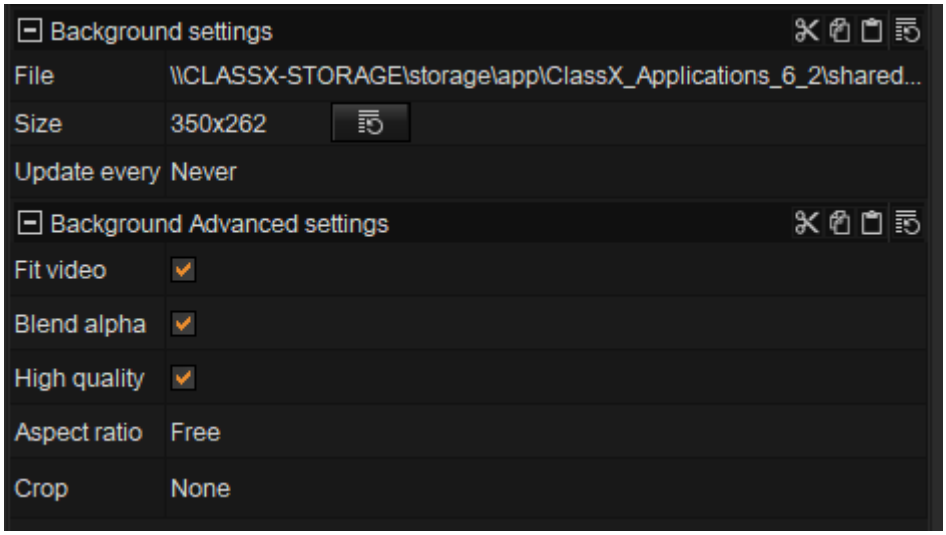
If you need to delete all the tags on a file, simply click with the right mouse button on the selected file and select "**Clear Tags**".

4.3.2 Image content

This kind of content manages and imports static images in the most common formats with full support of the Alpha Channel.
Once the desired image is loaded, it is displayed into the positioning area, through which it is possible to place and resize the content at our own will. To restore the original image size, we can reload it by pressing the “Enter” key into the field displaying the path on disk.



The “**Properties**” folder shows the following controls:

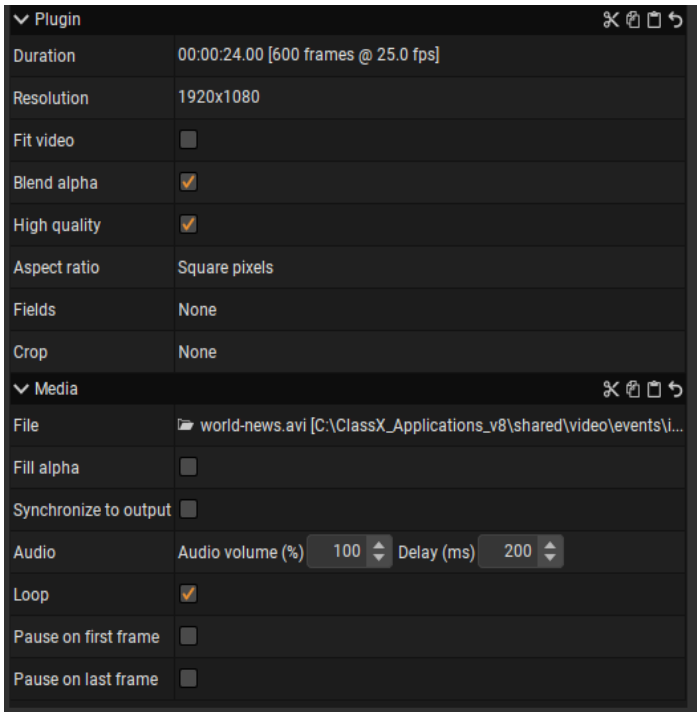


- **Update every**: this option allows the content to reload and automatically update the related image after a desired period. This option can be very useful if the content is referred to an image that is updated an external service or program.
- **Fit Video**: if this option is not enabled, each media file is executed in its original dimensions, ignoring the the content size. If the option is enabled, each file is automatically displayed and adapted to the current content size.
- **Blend Alpha**: it enables the management of the Alpha channel which, especially if it is large-sized, requires a greater time of calculation. For this reason, with images that don’t need the Alpha channel, such as, for example, backgrounds, bars, etc. it is advisable to disable this feature.
- **High quality**: it enables the high quality tracking of the image through bilinear interpolation. This function is very useful when a small-sized image has been enlarged, because the interpolation allows to eliminate most of the artifacts due to the resizing.
- **Aspect Ratio**: apply the proportions to the current media file.
- **Crop > Enable cropping**: it allows to crop the image on the edges, specifying, in pixels, the area that will be deleted.

4.3.3 Media content

This type of content manages and imports audio and video files in the most common formats.

The **"Properties**" folder contains the following controls:

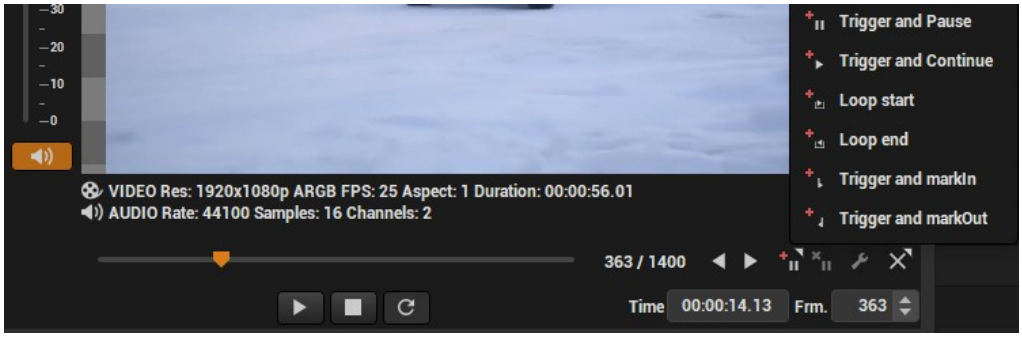


- **Duration**: it shows the media duration information.
- **Resolution**: it shows the media resolution (with x height).
- **Fit video:** if this option is not enabled, each media file is executed in its original dimensions, ignoring the the content size. If the option is enabled, each file is automatically displayed and adapted to the current content size.
- **Blend Alpha:** it enables the management of the Alpha channel for each video file.
- **High quality:** it adapts the video, using the highest quality.
- **Aspect ratio**: it applies the aspect ration to the current media file
- **Fields:** It applies the field direction the current media, for instance "Lower first(Even).
- **Crop**: it enables the media file cropping.
- **File**: it allows to change the media file.
- **Loop:** if enabled, it executes the content in loop.
- **Pause on first frame**: it pause the media at the first frame
- **Pause on last frame**: it pause the media at the last frame
- **Fill Alpha:** it fills the Alpha channel with the black color. This setting is useful for the MPEG files.
- **Synchronize to output:** it synchronizes the execution of the video files according to the display output chosen by the application. This option ensures a fluid execution, synchronized in time.
- **Audio:** it allows to set the play volume for the current media file, also it allows to apply an audio delay(in milliseconds) to synchronize the audio-video. In most cases the correct value is around 400 milliseconds, but we recommend in any case to perform the necessary checks. This value can be varied also during the execution of the content, in order to get an easier approach on it.

Warning: the Media content imports and manages both audio and video multimedia files according to the codec that are installed on the PC. If a video or audio file is not correctly executed, you should check the type of codec in use and install it correctly.
The performance of this content directly depends on the encoding the file has been created with, and on the hardware configuration. For this reason we suggest you to use high-performance codecs, preferring the ones directly supported by the video card, in order to share the work on the GPU rather than on the CPU.

4.3.3.1 MarkIn e MarkOut

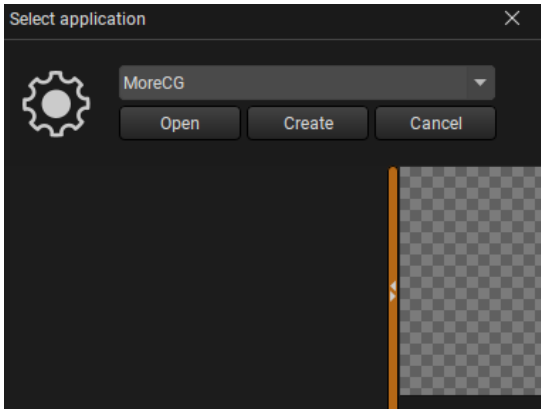
From the “**Preview**” panel, you can view a special interface to set the MarkIn and MarkOut points on the associated media file. These two points allow you to “cut” the video so as to broadcast only the part selected by the markers. A preview of the video file will allow a correct setting of the marker points.
The slider below allows access to each single frame of the media file and displays the MarkIn and MarkOut points with a special icon. The values associated with the two markers are shown in the lower part: these values can be modified manually by setting the time instant associated with the marker with extreme precision.



You can only set or only the markIn or markOut or both. The preview panel will be discussed further on.

4.3.4 Graphic Content

This section allows the integration into LiveBoard of graphic **templates** created with the MoreCG, CastaliaCG and CoralCG software.
The LiveBoard graphics engine allows the parallel execution of multiple graphic templates without any limitation. To do this LiveBoard uses advanced technology developed by ClassX, able to optimize and share the resources necessary for the execution and management of individual graphic projects.
Once the content is inserted in the grid, LiveBoard displays a panel used to select and load the graphic template, shown in the figure:

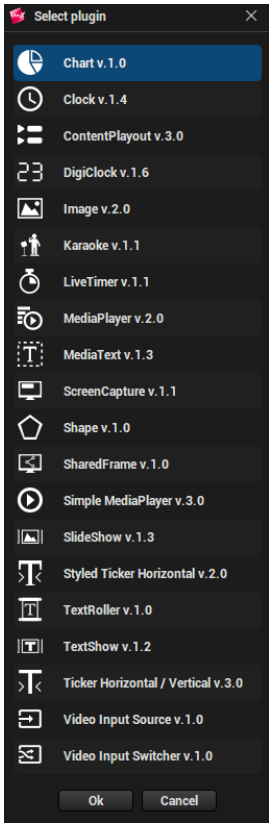


The "**Create**" button allows you to create a new graphic template using the selected titration software, requesting the user the name from the file to be saved. The "**Open**" button allows you to open an existing graphic template using a special chooser file.
At the bottom, the panel shows the list of recently uploaded graphic templates, filtered according to the selected titration software. Whenever you select an item from the list, the panel shows the preview on the right side. A double click of the mouse is sufficient to open a graphic template from the list.
In the "Positioning" panel it is possible **to relocate the template on the screen** in a new position respect to the original one set in the editor and also **scale its size**.

4.3.5 Plugins Content

This content allows the integration into LiveBoard of plugins, the suite of plugins developed by ClassX. Drag the white cube icon on the Content grid to the cell you want to edit. Once inserted into the grid, LiveBoard displays a list of all available plugins, as shown.
Select the desired plug-in and double-click on it or press "OK". The cell will contain the new content in Edit mode, so all the tabs and settings will be linked to it.

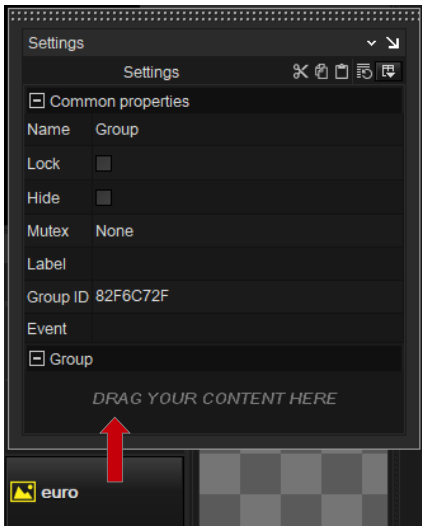
Some of the functions on plugins can be activated via the "plugins Controls" panel.



Notes: All the features and functionality of the individual plugins are described in the video tutorials within the ClassX Dashboard.

4.3.6 Group Content

All the contents of LiveBoard can be associated to a group following a precise rule: a group can contain any content, with the exception of the groups themselves. Each Group can be assigned a value called Group ID which is automatically created. This value is used by LiveBoard to identify the "**children**" **content** of each group: all the contents referring to the same Group ID are part of the same Group.



To create a group, drag the Group icon to any of the Content grid buttons.

The inclusion of any content within the group is done by dragging it to the dedicated area of the Settings tab ("Drag content here").

Even if it is included in a group, the content can still be done individually by clicking directly on it: the group only serves to synchronize the beginning or end of multiple elements.

Notes: once the content has been assigned to a group, its mutual exclusion value will be ignored because it will automatically inherit that of the group it belongs to.

4.3.7 Script Content

This content allows you to create executable code directly in the program with the "Script Editor" panel which we will analyze later.

5 The Panels

In addition to the panels already analyzed, the LiveBoard interface makes use of other separate panels for a more complete content management: **Preview, Positioning, Main output device, Data, Data Source, Services Monitor, Error Handling, Shared Console, Playlist, plugins control, Script Editor, Effects and Graphic Attributes**. As already described, the position of the above mentioned can be changed according to the user's wishes thanks to the docking / undocking settings.

At the top right of each panel or each section, there are small icons to apply actions on the modified values of the panel itself: Cut Values, Copy Values, Paste Values, Restore to Default, Open or Close a Session.

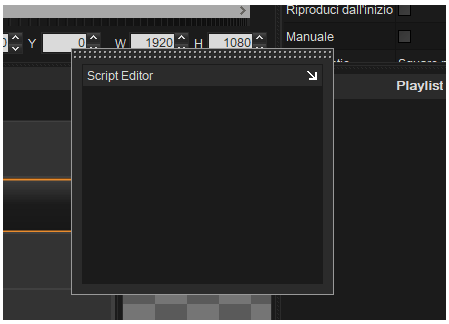


As already mentioned in the introduction to LiveBoard, the LiveBoard interface is customizable and the panels can be arranged as desired by the user.
By clicking on the panel name with the right mouse button, a window will appear allowing you to choose where to place the panel.

- **Minimize**: the panels are "closed" and placed to the right of the interface in a list of icons.



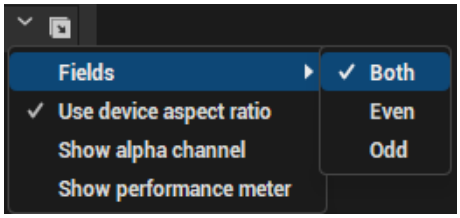
- **Detach**: the panel is detached from the others and can be placed in any position on the screen. With the down arrow it is possible to return it to its position from which it was detached.



- **Maximize**: the panels are extended and adapted according to the size of the screen. To return to the original settings, you need to click on the icon  at the top right.

5.1 Main output device Panel

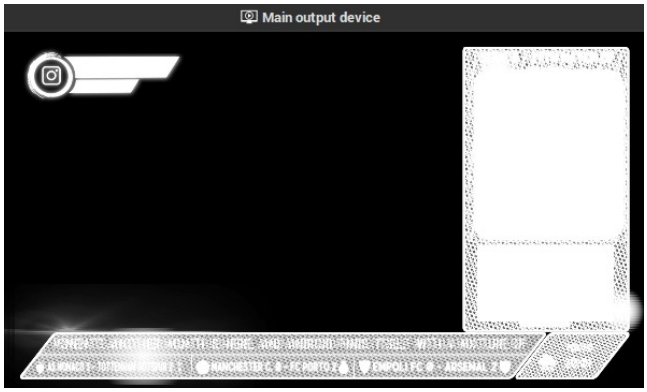
The main output device will always show any graphics that are played by LiveBoard. In the upper right corner, the down arrow opens a dedicated menu that allows you to set the fields, the aspect ratio, to show the alpha channel and finally to show the performance meter.
The quality of this preview does not reflect the actual output; it is only a quick preview rendering for the operator to know exactly what is being played in real time.



LiveBoard playout in real time

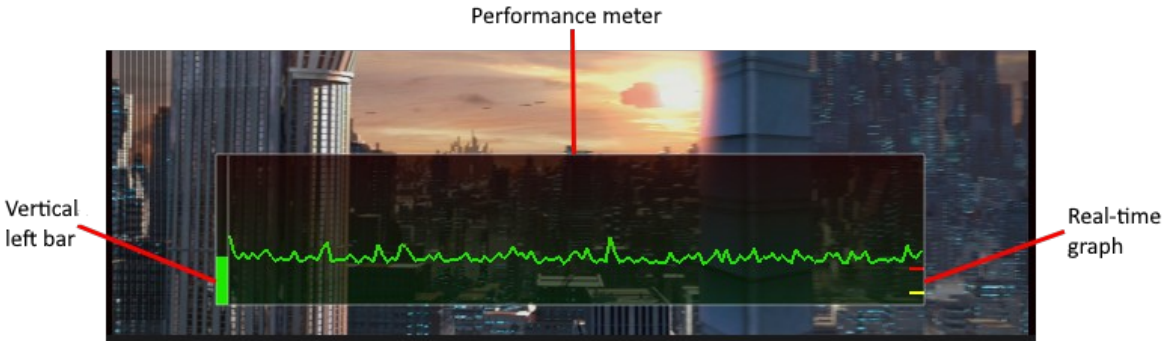


LiveBoard playout alpha channel in real time

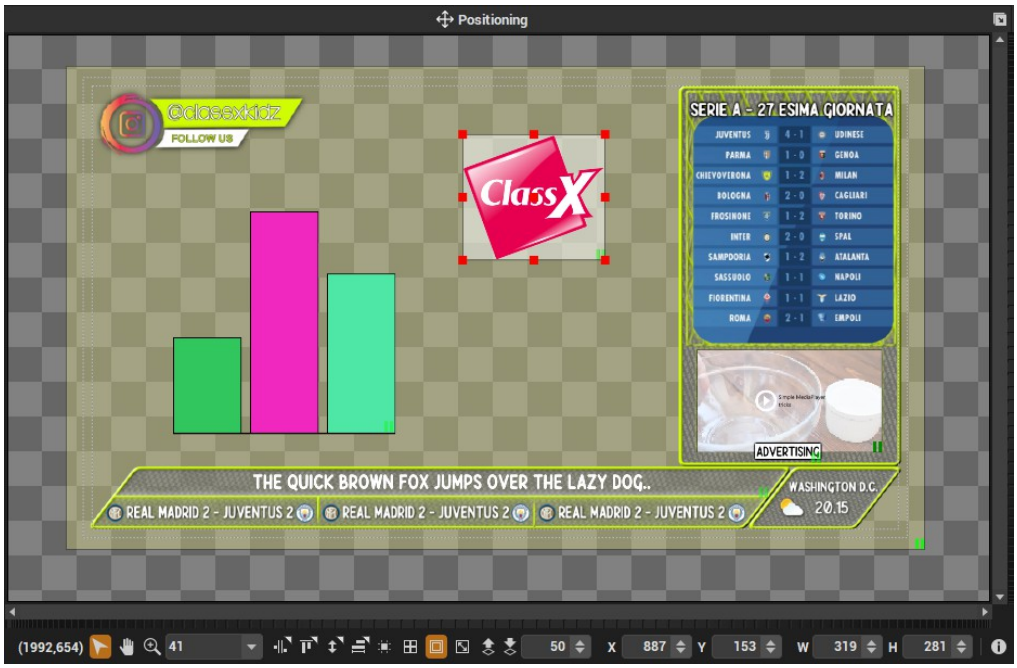


5.1.1 Performance meter

Clicking this icon opens a performance monitor directly on the **program** preview panel. The graph displays the engine load in real time, using a vertical bar on the left that ranges from 0% to 100%, colored from green (low load) to red (maximum load). When the engine reaches 100% load, audio/video frame drops may occur on the **program** outputs.
The vertical bar on the left shows the average engine load over the last 150 frames, while the graph on the right displays the real-time load along with a short history of recent frames.



5.2 Positioning Panel



The Positioning panel plays an important role in the use of our software. It is entrusted with the preview of the contents and the management of the graphic layout. For this reason, the positioning panel is designed to make editing and positioning extremely simple and fast.

The Positioning area represents the television screen and is represented by a light coloured rectangle on a chessboard pattern, representing transparency. The project resolution set in Preferences will define the aspect ratio and size of the Positioning area.

The Positioning area will display a preview of the plugins and allows its direct selection with a mouse-click. The selected content will be highlighted with a black frame and the appropriate red squares for resizing. In the Misc tab of LiveBoard's Preferences, the checkbox of “Display plugins content” has to be selected otherwise the plugins will have no graphic preview on the Positioning Panel.

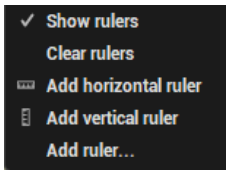
You can select multiple contents at the same time (multi-select) by holding down the **CTRL** key and clicking the individual elements with the mouse.
Alternatively, you can select multiple contents by drawing a selection area with the mouse.
At any time, you can cancel the current selection by right-clicking on an empty area.

By right-clicking onto a content, a drop-down menu with all functions of the Positioning Panel will open:



- **Hide**: it hides the content(s) from the positioning scene
- **Show all**: it shows the previously hidden content (only in Keyer editor)
- **Copy bounds**: it copies the size and position of the object
- **Paste bounds**: it pastes the size and position of the object previously copied
- **Zoom**: see the explanation below
- **Place objects:** see the explanation below
- **Align objects:** see the explanation below
- **Rulers:** alignment guides

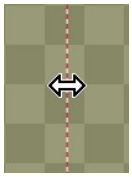
5.2.1 The rulers



By right-clicking within the positioning area, the context menu that appears includes, in addition to the options already described, a section for adding and managing rulers.
Alignment rulers are visual tools designed to help you position and align graphic elements in your project with precision. They enhance layout accuracy and improve the overall aesthetic of the design.
They are especially useful for aligning text, images, shapes and other objects in a clean and consistent manner.
Elements automatically snap to guides or rulers, making precise alignment quick and effortless.

As indicated by the menu, you can **add** multiple vertical or horizontal rulers.
By default, the "**Show rulers**" option is enabled; disabling it will hide all rulers currently placed in the area.
The "**Clear rulers**" command removes all rulers from the workspace.

"**Add ruler...**" is a dedicated function that allows you to insert a horizontal or vertical ruler by specifying its position in pixels from the edge of the scene.
The pixel scroll control lets you adjust the placement distance in real time.
Within this window, you can also choose the **color** of the ruler you are adding.

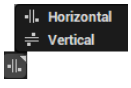
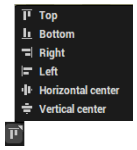
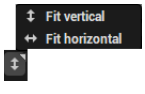
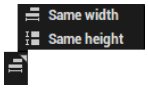


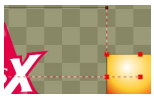
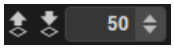
At any time, you can **reposition** the rulers within the area.
To move a ruler, simply hover your mouse cursor over it. As shown in the image on the right, a double-arrow cursor will appear, allowing you to drag and reposition the ruler easily.

5.2.2 Positioning Panel Functions

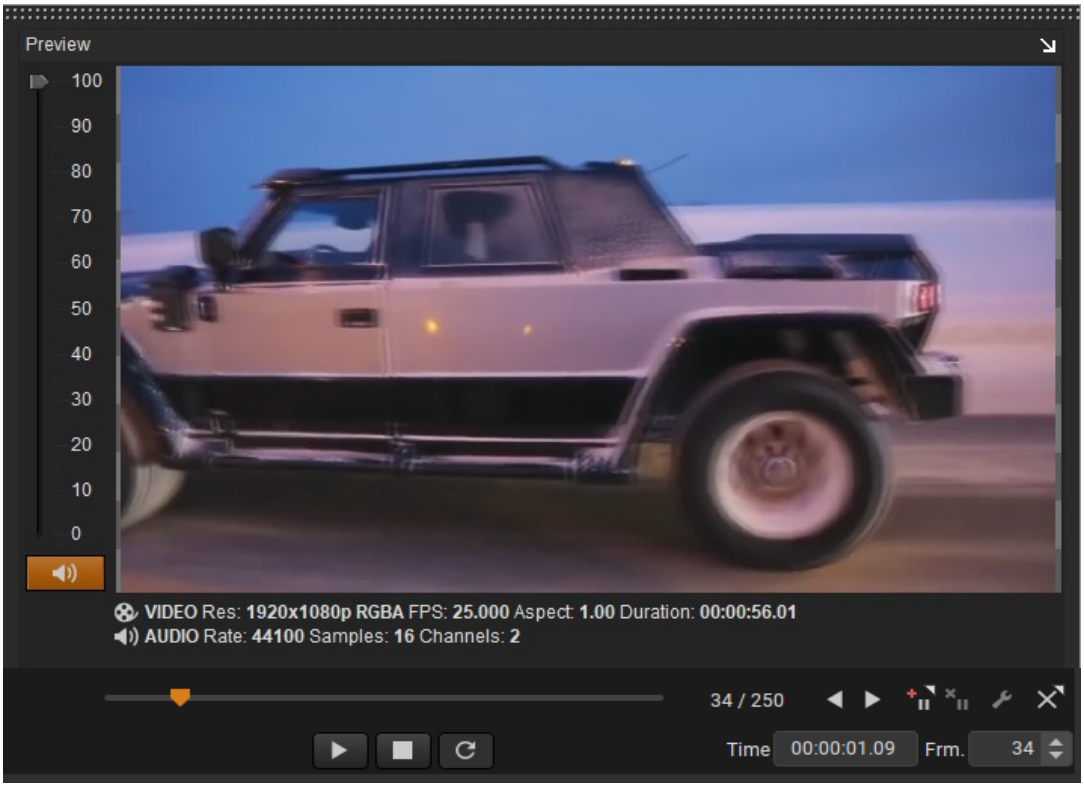
On the bottom bar of the Positioning Panel there are several tools that allow the correct creation of the project.
The following table explains the functions present on the bottom bar of the Positioning Panel.



	Real-time coordinates of the mouse pointer on the positioning area; the zero point is in the upper left corner of the Positioning area representing the television screen.
	The arrow button is the correct tool for selecting the desired content inside the Positioning area by directly clicking onto it and for resizing it.
	The hand button allows you to move the whole Positioning area with a mouse-click and drag.
	**Zoom and Display scale**: select “To Scene” to display the whole Positioning area as a best fit in the panel. By selecting the magnifier tool and dragging it over the Positioning area, you may select a rectangular area on the panel that will be magnified as soon as you drop it.
	**Place objects**: select three or more contents you wish to place and right-click onto this icon in order to vertically or horizontally put them at the same distance.
	**Align objects**: select the contents you wish to align and right-click onto this icon, then select the position of the virtual line that will be used as a base to align the selected contents.
	This function will stretch the selected content's size to fit vertically or horizontally to the project resolution.
	This function will set all selected contents to the same height or width. The first object to be selected will define the future height or width of the afterwards selected contents.

	**Snap to Objects**: Enables red-striped guidelines that help align content easily when moving or resizing elements. 
	**Positioning Grid:** Activates a virtual dotted grid that snaps content into place. Grid density and visibility can be adjusted in the "**Positioning**" tab of the Preferences menu. 
	**Safe Area button**: it displays a dashed double frame in the Positioning area to indicate the TV-safe zone. This helps ensure that titles and graphics remain fully visible on screen during broadcast. 
	**Enable Pixel Aspect Ratio**: it adjusts the positioning environment to match the project's aspect ratio, useful for 16:9 playback. To edit and resize images or video plugins accurately, enable the "free" aspect ratio in each plugin's advanced settings.
	The two buttons with the up and down pointing arrows are entrusted with the management of the visual priority. The current value is displayed in the field and can be directly edited to suit your needs. There is no limit to the number of contents you may add to the project. **TIP**: always set a different value of visual priority to each content, so you'll know at any time what the graphics playout will look like. Immediately set a value to each content you create in your project.
	The X and Y boxes show the coordinates of the selected content. You can move the content by typing in manually coordinates or by pressing the up and down arrows and dragging up and down the mouse.
	The W and H boxes show width and height of the selected content. You can change the content size by manually typing in size numbers or by pressing the up and down arrows and dragging up and down the mouse.
	If clicked, it displays the list of shortcuts available for the Positioning Panel.

5.3 Preview Panel

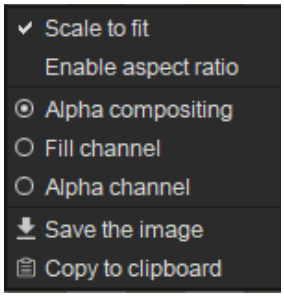


This panel is for:

- The real preview for the frame based graphic
- Frame Marker management
- ”Take to the current frame” function

The frame selection slider represents the project’s timeline and allows the complete preview of the graphic template by selecting its frames by “scrubbing” it.

By right-clicking over the area inside the preview, a small menu appears and you can select some actions that can be performed directly on each individual panel:



- **Scale to fit** and **Enable aspect ratio**: set the display proportion of the content. By removing the setting “Scale to fit”, the content will be shown in its real size;
- **Alpha compositing**: displays the video signal in fill and key;
- **Fill channel**: displays only the “Fill” channel;
- **Alpha channel**: displays only the “Alpha channel”;
- **Save the image**: save the frame of the visible content at the time of saving in png format;
- **Copy to clipboard**: copy, to the "clipboard" of the operating system, the frame of the visible content at the time of saving.

5.3.1 FrameMarker

Direct access to every single frame of the video allows LiveBoard to add one or more FrameMarkers (abbrev. **FM**) along the timeline. The FrameMarker is an action associated with a frame; once this frame is reached during the execution of the content, there are the the following options:



- **Pause** (“Trigger and pause”): the animation pauses and TemplateActions can be performed
- **Continue** (“Trigger and continue”): the animation does not undergo any change of reproduction and can run the TemplateAction
- **Loop Start/End:** they allow to loop only a certain interval within the animation and TemplateActions can be performed. In the edit panel it is possible to set the number of loop cycles to execute, even infinite (**set -1 for infinite**).

There are two different buttons ("**+**") ("**x**") used for inserting and removing a FrameMarker at the selected frame.

By right-clicking onto the icon, choose the desired FrameMarker from the list.

After adding it, this will be highlighted on the timeline through a dedicated icon: pressing onto it, LiveBoard automatically selects it along with the preview frame associated with it.

A single FrameMaker can be removed only if selected; press to remove the single selected FrameMarker.

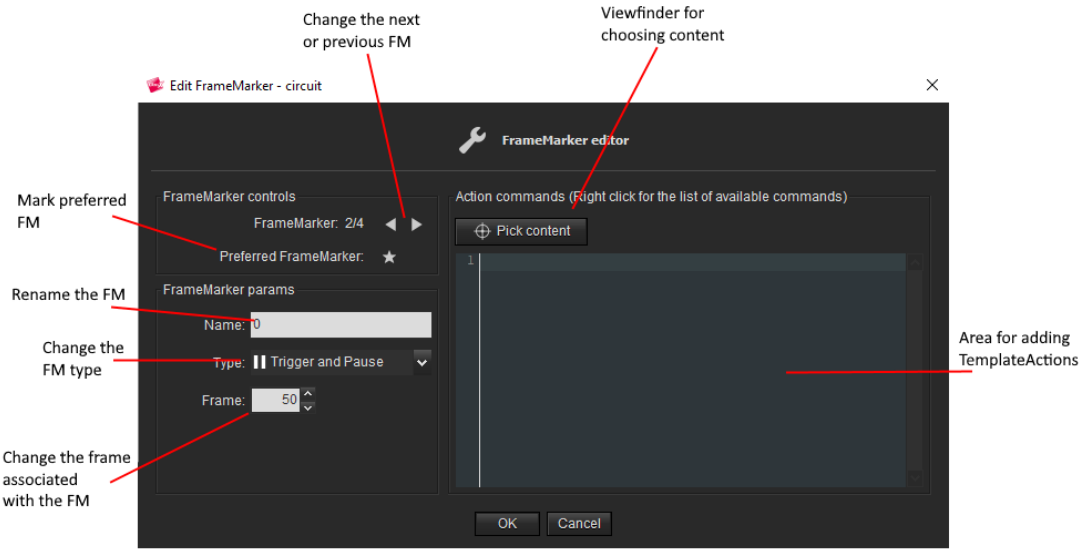
By pressing all FrameMarkers will be deleted in batch.

5.3.2 Edit FrameMarker

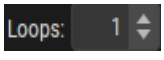
When the cursor is positioned over a FrameMarker, clicking on the icon will open the “Edit FrameMarker” panel.

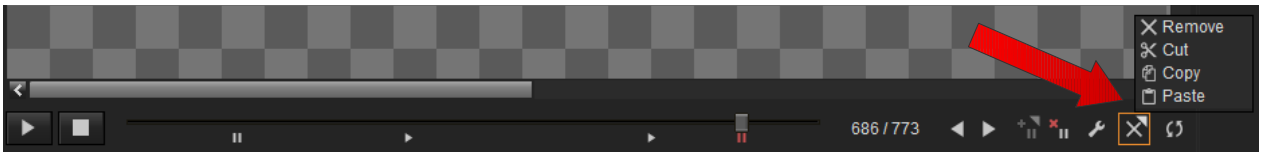
The "**Edit FrameMarker**" button opens another panel that allows you to:

- Change the type of FrameMarker
- Change the frame associated with the FrameMarker
- Select the next or previous FrameMarker
- Mark your preferred FrameMarker
- Rename the FrameMarker
- Define one or more TemplateActions



N.B.: In the case of the FM “Fine Loop”, the panel also contains the section for choosing the number of Loops to perform. In the case of an infinite loop, it is necessary to enter “-1”.



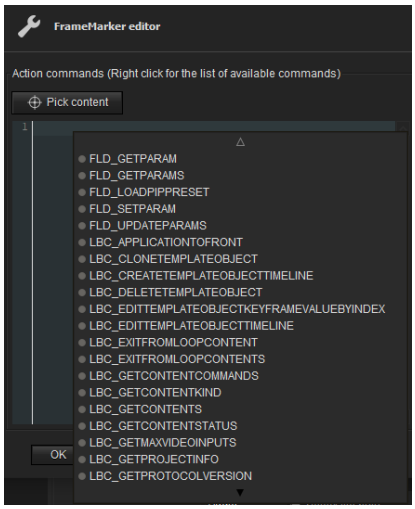


By right-click on this icon you can choose if remove, cut, copy or paste all FrameMarkers of a timeline.

5.3.2.1 TemplateActions

The TemplateActions are scripts; one or more commands executed in sequence from LiveBoard, which allow you to interact with other currently loaded contents. You can create and run a series of automations, synchronized in time. All actions or commands, are performed synchronously with the playout. Most TemplateActions interact with one or more contents, identified by their **name**.

The light grey area is used to insert TemplateActions; each action is placed on a separate line. The context menu, that opens with the right mouse click, contains all the available commands. Once you select a TemplateAction, it will automatically be inserted along with the full text of its syntax.



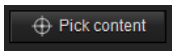
For example, if we wish to to playout the content named "End Titles", we select among the TemplateActions:

LBC_PLAYCONTENT CONTENTNAME="<value>"

and change `<value>` with the correct and precise content name.
Just double-click <value> to select the part to replace.
The correct syntax of command would therefore be:

LBC_PLAYCONTENT CONTENTNAME="End Titles"

NOTE: Quotes (") are necessary. Without them, the above command will not work



The viewfinder allows you to choose exactly the LiveBoard content whose name must be entered within a TemplateAction. Just click on the “Choose content” button, hold down the mouse click and drag it over the content of interest on the LiveBoard grid. Once you release the mouse button, the content name will appear inside the area. For example, if <value> is selected, the content name will be inserted in place of the content.

Warning: If there are multiple LiveBoard contents with the same name, applying a command may not have the desired result and the command will be applied only to the first content created; make sure all content has a different name.

5.4 Plugins controls Panel

This panel is activated only with specific plugins or when a template is selected on the LB grid.
The plugins that use this panel are **LiveTimer** and **Karaoke**. In the case of LiveTimer for example, it allows you to activate or stop the timer.

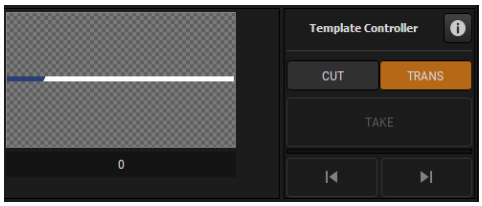
Note: For more information on PlugIns, see Chapter 7 below.

5.4.1 Template Controller

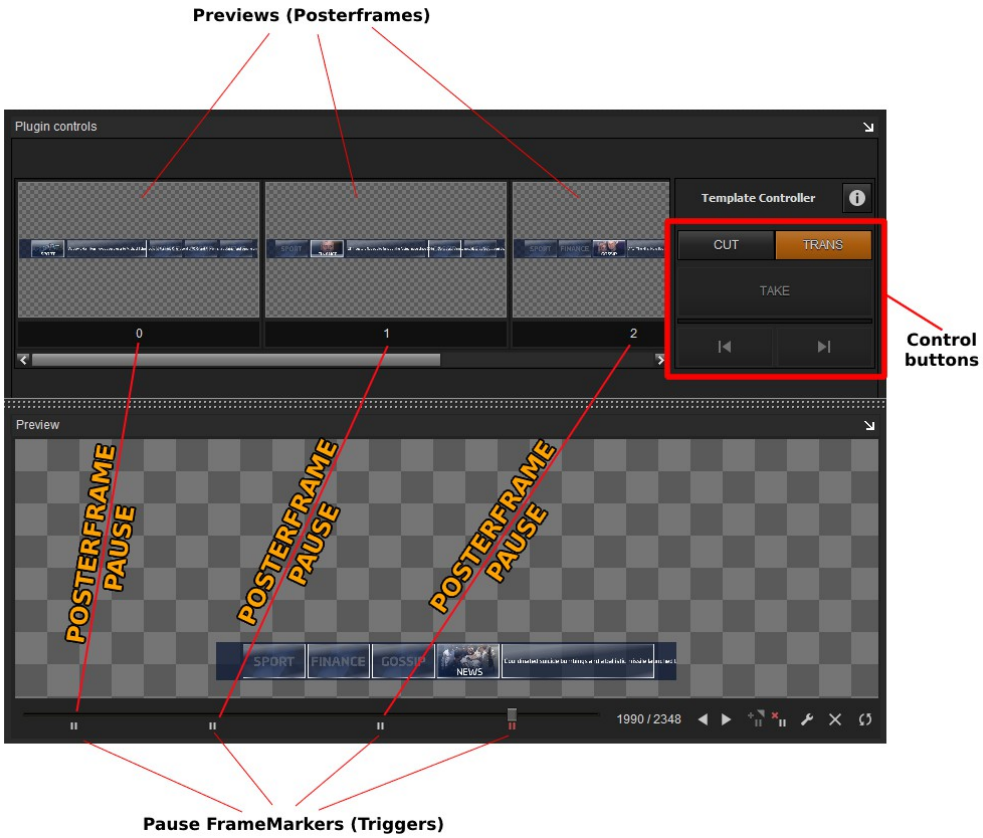
The "**Template Controller**" is a tool that is activated only when a template is selected in the LiveBoard grid, that is when a template is in "Edit status" (see paragraph "**5.2.5 State of the content**").

The Template Controller allows the controlled execution of templates via **pause FrameMarkers** (abbrev. **FM**), ie the pause FrameMarkers are used as the animation pivot point.

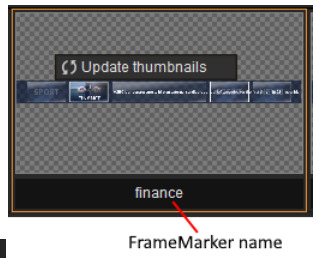
When the selected template has no pause FrameMarker, the panel is empty.



Whenever a pause FrameMarker is inserted on the template timeline in the "Preview" panel, a small preview (posterframe) of the corresponding frame is automatically inserted in the Template Controller.

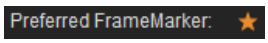


In case you need to move the position of the FrameMarker in the timeline and then update the preview of the frame, just right click on the preview and click on "**Update thumbnails**".



By double-clicking on a preview, automatically in the "Preview" panel, the cursor on the timeline will position itself over the corresponding FrameMarker.

To rename the FrameMarker, it is necessary to click on the "Preview" panel icon  (See chapter 6.3 for further information).



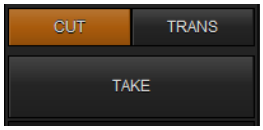
From the aforementioned panel it is also possible to mark a FrameMarker as a **"preferred"** by means of the star. The corresponding pause FM will automatically turn orange. Preferred FM is an animation starting point if you need to start from a specific frame.

5.4.1.1 Control buttons

Once the template is in the "play" state, thanks to the control buttons it is possible to move forward or backward within the template between the inserted pause FrameMarkers.



These two arrows allow you to move the template exactly to the two **adjacent** pause FrameMarkers to the right and left of the current one.



Instead, these controls allow you to switch to a FrameMarker that is **not adjacent** to the current one.
First of all it is necessary to select with the mouse the preview corresponding to the FrameMarker to "reach". Automatically the chosen preview will have an orange border. Thanks to the list of previews, you can select the one relating to the FrameMarker you want to view on air. After that it is necessary to choose the "switch" mode between **CUT** and **TRANS** and finally click on the "**TAKE**" button to finalize the operation.

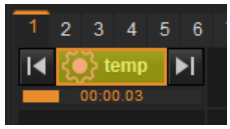
It is possible to choose between two types of modes:

- **CUT**: this is the "detach" mode, that is, it makes a clean cut from the current FM to the next
- **TRANS**: is the transition mode, that is it makes a transition from the current FM to the next FM.

N.B .: For both modes, any intermediate FM pause is ignored.

Shortcuts:
- **CTRL + left click:** to execute a TAKE in CUT mode
- **SHIFT + left click:** to execute a TAKE in TRANS mode

To switch back and forth between two adjacent pause FMs, you can also use the arrows directly on the button corresponding to the template on the LiveBoard grid. The two arrows on the button have exactly the same function as the arrows between the control buttons.

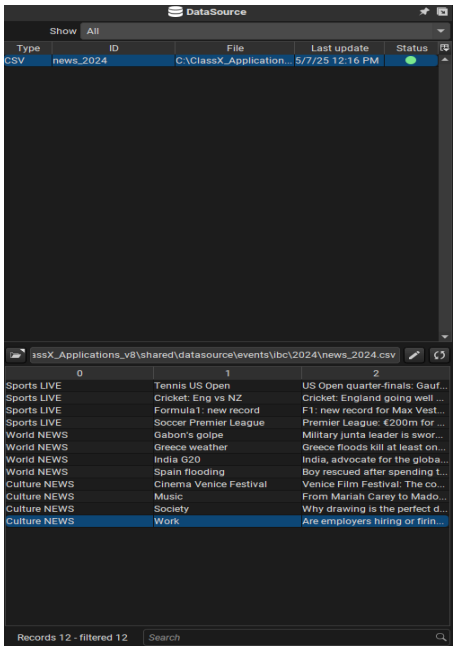


Warning: Any loops are always interpreted unidirectionally forwards: it is not possible to loop backwards. This means that if a loop is set on a template, it will only be executed in the FORWARD direction. In the BACKWARD direction the loop will be ignored.

5.5 DataSource Panel

This panel is used to manage the data sources connected to a graphic template. Data sources can be more than one and for this reason they are contained within a list. Each data source is graphically represented by a cell that describes its current status and last update.
The lower part displays all the records associated with the data source selected in the list. Depending on the selected record, all the values of the editable objects of a template associated with that source can be automatically changed.

The imported records can be in various formats depending on the need.



5.5.1 Status of a DataSource

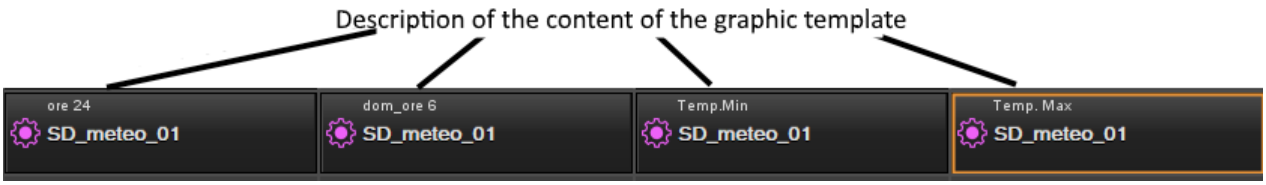
The status of a data source determines whether the loading of the associated file was successful or generated an error. In the first case, the list shows a green icon and the date of the last successful update, while in the second case the list displays a red icon.
The data source may be in an error state, for example if the file associated with it is opened by another application that does not allow it to be read.

The status of the data source is updated automatically after the "Open", "Edit" and "Reload" operations.

5.5.2 Selected Record

Based on the data source selected in the list, LiveBoard displays all the available records at the bottom, highlighting in particular the one selected. This list is updated automatically after the "Reload" operation in order to check its correct interpretation.
The selected record is shown in the "**Description**" area of the content button. In this way the user can create multiple copies of the same graphic content, directing each of them to different records whose value will be clearly visible in the aforementioned description area and can be selected from the list of sources.

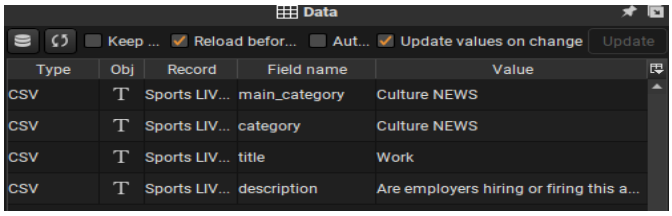
For example, if the source is a file with weather data in XLS format with multiple sheets in it depending on the data available, you can choose to attribute to each button a different "sheet" of our XLS file. You can then send information at different times depending on your needs.



5.5.3 Filter the available records

In cases where a data source contains a fairly large number of available records, selecting and then searching for the desired record may be difficult. In these cases, the tool that allows you to apply a search filter on records can be very useful as it reduces the list of available records to our liking.
To delete the filter applied, simply press the icon representing a “red X”.

5.6 Data Panel



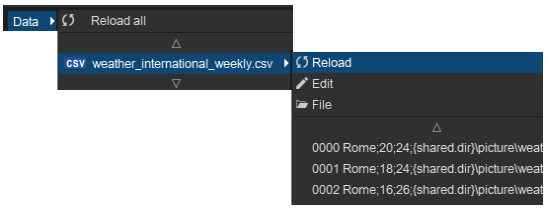
LiveBoard displays in a table all the objects of a graphic template connected to external data sources, offering the possibility to change their contents dynamically.

In this section we will limit ourselves to illustrating only the operation of the panel in the figure. For more information on data sources see the related appendix.

ClassX has developed DataSource, a technology that makes it possible to transform any graphic project (MoreCG, CastaliaCG, CoralCG) into a template, that is a scheme, in which the contents can be associated with external data sources. A data source contains within it one or more records based on its implementation. Selecting a data source record will automatically update all objects connected to it. There is therefore a well-defined link between the object's value and the current record of the data source.

The data panel summarizes all this information in five columns: type, object, record, field name and value. In each row the "**Obj**" column displays the icon of the record to which it refers identifying it as text, video or image. In the other fields, the identification name of the object and the selected record from which the object value is derived is displayed. The "**Value**" column shows the current value of the object and allows its modification either manually or via the list according to the template's **DataSource**™ settings.
Changing the value by hand in the table can be difficult in some cases as the available space is very small. For this reason, LiveBoard allows you to open an editor in the window by double clicking the mouse in the "Field name" column. In the case of text objects, LiveBoard will display a text editor; in the case of image or video, LiveBoard will open a file-chooser to select the desired file.
The modification of the values of an object can also occur during its execution: on the contrary, this possibility offers enormous advantages in case you want to use the same graphic to display more information in sequence while maintaining the same graphic template without any interruption.

By clicking on the icon  a context menu appears with the following characteristics:



- **View:** all data sources connected to the related icon.
- **Reload all:** updates all data sources in the list.
- **Reload:** updates the data source referred to by the menu. Action required after a data source has been modified externally.
- **Edit:** changes the data source using the default editor from the operating system.
- **File:** changes the source file of the data source.
- **See below:** all the data source records that can be selected.

Warning: Changing the data source via the default editor from the operating system actually starts a new application. The execution of the application could prove to be very expensive in terms of system resources and must be carried out with due caution, especially if there are contents on the air, as LiveBoard problems of fluency could occur.

Note: LiveBoard considers the contents referring to the same template as mutually exclusive, as it would make no sense to execute several instances of the same project in parallel.

5.6.1 DataSource Options

From the Data panel it is possible to enable some functions on updating the DataSource.

5.6.1.1 Auto update

This function allows you to automatically update the template DataSource following changes to the DataSource itself.

This option is managed in the “Data” panel of the template, with the “Auto update” checkbox.

The automatically updateable DataSource types are the following: XLS, TXT, CSV, JSON, XML.

Note: only the DataSource corresponding to physical files residing in local paths and on the network will be updateable. Resources referenced via URL will be ignored.

The “Auto update (ms)” setting in the “Misc” preferences allows you to set the auto-update frequency of the DataSource based on a set time in milliseconds.

The default setting (10 seconds) is the most commonly used and optimized for typical scenarios. Please note that the automatic update occurs in the background and does not affect normal LiveBoard operations.

Note: depending on the size of the DataSource file and/or its physical location (i.e. network), the update may take unpredictable times.

5.6.1.2 Update values on change

This function allows you to automatically update the template fields when the DataSource is modified or when the data is manually changed, regardless of whether the template is on air, paused or not running.

In any case, it will be possible to force the updating of the template fields manually using the "Update" button.

5.6.1.3 Keep user data

This option allows you to maintain the template data manually set by the user.

This option, if enabled, effectively prevents changes to the data source (i.e. updating the relevant file, reloading) from being applied directly to the template.

This implies that the data set by the user will be permanently saved on the template and consequently on the LiveBoard project.

If you want to update the template with the latest data from the data source, you will need to select the corresponding record in the "DataSource" panel. This way, the template will be updated with new data from the source.

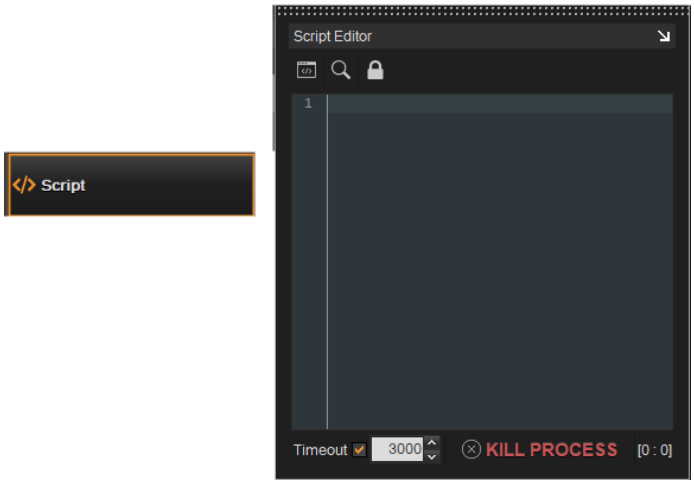
From the Preferences → Misc menu, you can enable this option by default.

Note: the “Reload before running” option will be ignored when we are in “Keep user data” mode. This is because the data is persistent to the template and independent of the data source.

5.7 Script Editor Panel

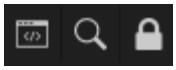
The term **Computer Script** identifies a sequence of instructions in a specific language that once interpreted gives rise to a result. The distinction between a normal program and a script is not clear, but that distinguishes the latter is the possibility to integrate perfectly in LiveBoard and to perform specific and very specific tasks. Thanks to this versatility, Scripts allow the user to create his own functions and for example to manage the score of a football match through a dedicated window.

In LiveBoard, to integrate a script you must first create a **Script content** in the Grid using the Drag-and-drop function from the icon as shown. At that point, via the Change status, you can write the desired algorithm directly from the Script Editor panel.



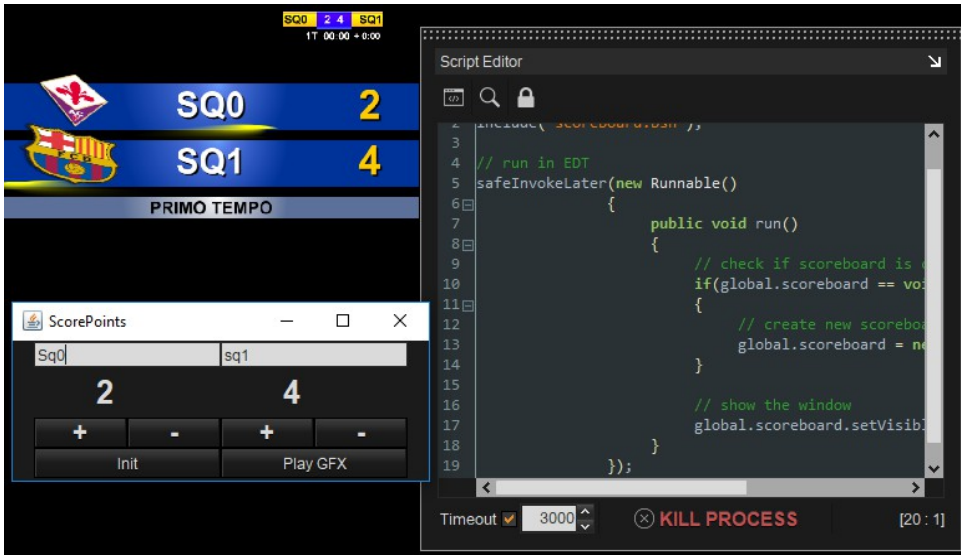
The engine that animates the scripts and runs the Java language is **BeanShell**: a powerful Interpreter that allows you to start a program in Virtual Machine without having to fill it out. In fact, it has the task of executing the instructions from the Java language, translating them from time to time into machine language instructions.

By clicking on the "**Output Console**" button, you open a console that returns any errors if there were any in the algorithm just written.



The magnifying glass allows you to easily search for strings or values within the code, while the lock stops the editing of the text to avoid any unwanted code changes.
In the next figure, an example of a script for the creation of a ScorePoints for the live assignment of the scores of two teams.

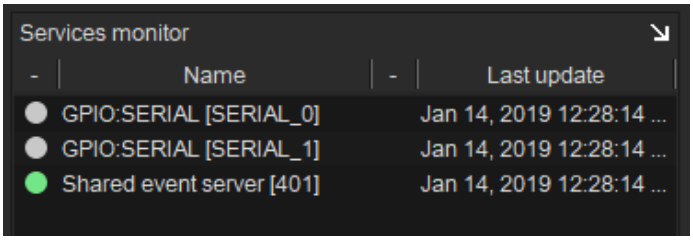
"**KILL PROCESS**" allows interrupting the process at any time.



5.8 Services Monitor Panel

This panel is a viewer of active services and connected via TCP / IP or GPIO SERIAL connections. Next to the active services, a different color indicator appears depending on the connection status:

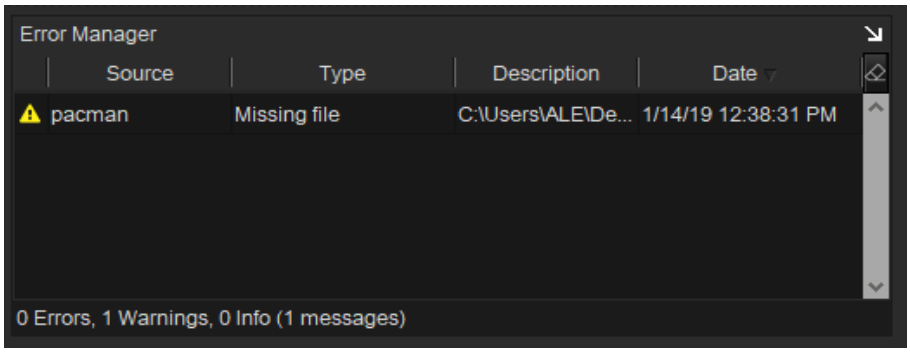
- **Gray**: input not available / not found
- **Red**: presence of errors in the connection
- **Yellow**: connection attempt
- **Green**: connection made correctly



If there is an activity in progress in one of the connected services, a small red light is shown to the left of the last update indicator.

5.9 Error Manager Panel

This console displays errors such as files not found. Since LiveBoard is a playout platform, it does not save multimedia files within created projects but opens them directly from external paths. If, in the meantime, a file is changed to a file between one save and the next, an error will be indicated at the next opening.



5.10 Shared Console Panel

The project created in LiveBoard could have within it the presence of one or more scripts. This console shows all possible errors in the scripts within the project.

The two icons on the bottom right allow you to clean the console or block the scroll.



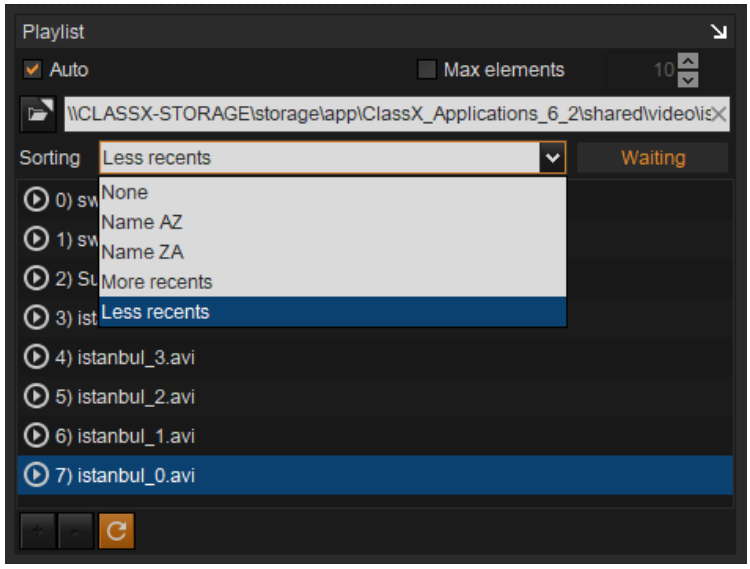
5.11 Playlist Panel

This panel is mainly connected to the use of some plugins content like “**ContentPlayout**”, “**MediaPlayer**”, “**SlideShow**”, “**Ticker**” and “**StyledTicker**”.
For more information on ContentPlayout and plugins, refer to the appropriate manuals.

For convenience, in this manual for the explanation of this section, we will take as an example "**MediaPlayer**" in which with simplicity, you can create a playlist of both audio and video content with the ability to transmit them automatically or in loop.

Files can be entered by various methods:

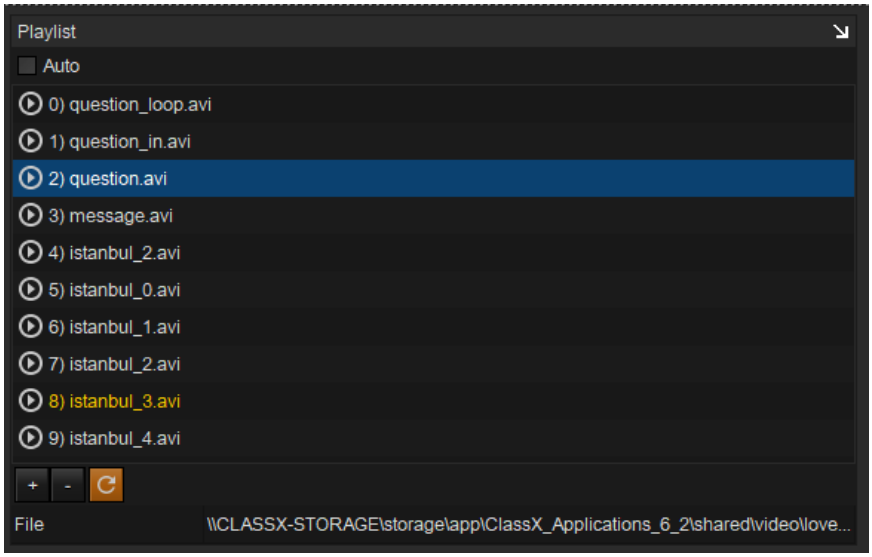
- They can be loaded in bulk from a folder, putting the flag on "Auto" and then selecting the folder of interest. In this way it is also possible to choose the order to be given for the transmission of files and to decide whether to indicate a maximum number of entries.



- It is possible to upload one or more files at a time by dragging them directly inside the panel from any folder or from the desktop, or by clicking on the lower left on the "+" and "-" keys.

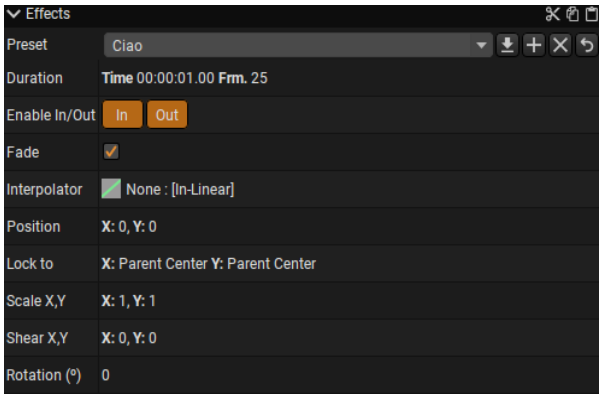
  In any case, the elements can be ordered manually by drag-and-drop.

  The icon  allows you to play the contents in loop.



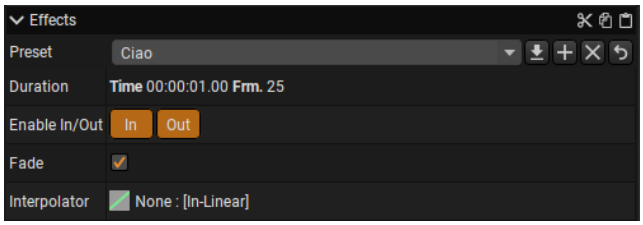
5.12 Effects Panel

The advanced graphics engine LiveBoard shows its flexibility and power in the administration and enforcement of the effects of content. Each effect can act on the properties of content such as scale, position, tilt and turn, changing its value over time through interpolation.



The interpolator is simply a mathematical function that returns vs. time and represents different values on a Cartesian plane by a spline.
The linear interpolator for example expresses a change in linear time. But if we take a spline consists of a curve and apply it to the property position, we can easily create a motion effect accelerated or decelerated.

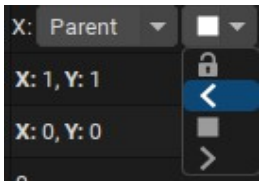
The effect is applied to content when it is executed (step input) and when it is interrupted (output phase). The duration of the effect can be customized to suit your needs, but it should be pointed out that once you pressed the button to stop the execution of the content, this will not disappear immediately, but will do so only after the effect of output. When applying the phasing-out, the button displays the contents of a special animated icon (see figure).

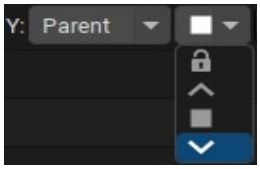


The **input phase** will be the interpolation of the Previous KeyFrame with the Parent KeyFrame.
The **output phase** will be the interpolation of the Parent KeyFrame with the Next KeyFrame.

The output and input phases are always opposite animations. You cannot choose to animate the input phase and the output phase differently.

Items on the Effects tab:





- **Preset**: you can choose a number of preset animations. Use the “+” and “x” buttons on the right to add a custom preset or to delete a preset
- **Duration**: set the effect duration in time or the number of frames
- **Enable In / Out**: enables the input and output phases
- **Fade**: fades out the plugins (remember to activate “blend alpha” for the plugins in its settings otherwise the fade effect will not be visible)
- **Interpolator**: defines the type of interpolation to use in the in/out animation
- **Position**: shows the coordinates of the Previous and Next KeyFrame. The coordinates can be changed here only if the X and/or Y positions are set to “Lock to > Parent > custom”
- **Lock to**: locks the position of X and Y coordinates of the first and last KeyFrame to a position related to the Parent KeyFrame.
- **Scale X, Y**: changes the size of the content by applying a scale factor for X and Y
- **Shear X, Y**: changes the inclination of the content on the axes X and Y during the animation
- **Rotation**: rotation of content during animation

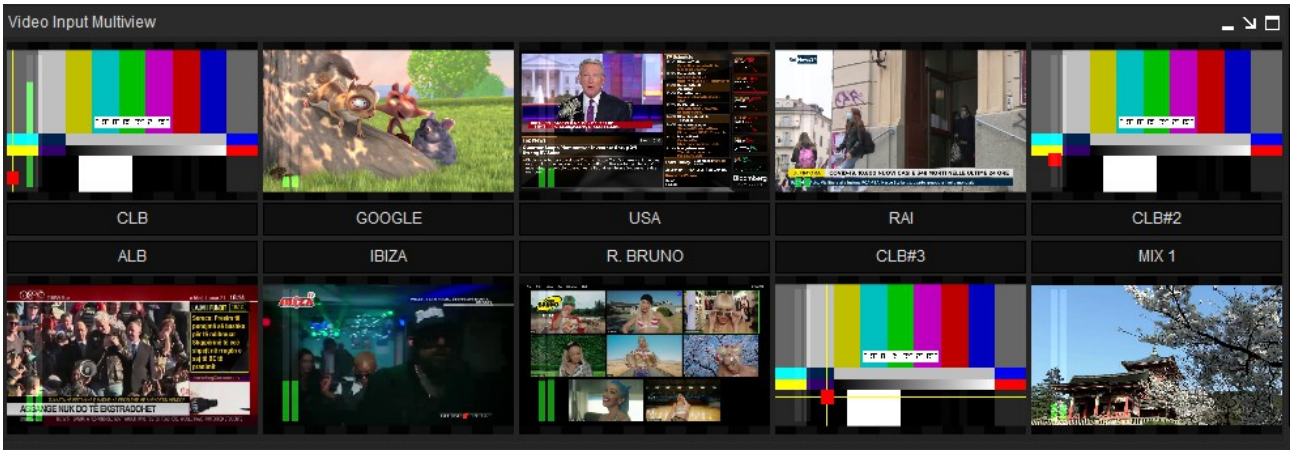
Find all the available operations collected in the tab header: cut, copy, paste and reset to default.

TRICK: by setting the **scale** value of either X or Z to **-1**, you can rotate the object once on its x or y axis during the input and output animation.

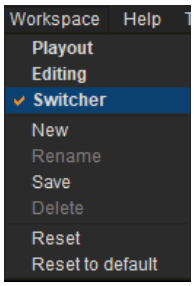
NOTE: Effects are orthogonally applicable to all contents except for the Group content

5.13 Video Input Multiview Panel

Video Input Multiview is a new LiveBoard panel that allows you to simultaneously view the 10 Video Input sources set by Video Input Manager.
It takes full advantage of the **dock / undock** benefits. It can be resized or detached from the main interface and used as an external window.
In each preview monitor there is a **vu-meter** that allows you to view the audio peaks of the source. Below each monitor there is a label dedicated to the name of the source set from the Video Input Manager menu.



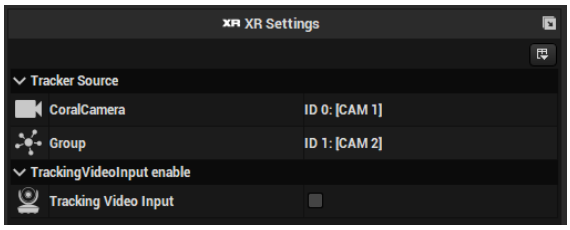
The new workspace "**Switcher**" has been added to the "**Workspace**" menu, which offers the user an accurate management of Video Input sources with a LiveBoard interface optimized for the use of the "**Video Input Switcher**" plugin.



5.14 XR Settings Panel

The “XR Settings” panel is dedicated to controlling the XR templates created with CoralCG. This panel allows for intuitive and fast management of the broadcasting of XR content and is divided into two sections:

- Tracker Source
- TrackingVideoInput enable



By "tracker" or "tracker source" we mean a device, a camera or in general a sensor system capable of generating positional information on the **FreeD** protocol.

The various trackers to be used in the project must be defined in the “**Tracking Manager**” tab of the Preferences menu. (See the “Tracking Manager” section).

Each Tracker can be associated with a name and can be connected to one or more targets (generally objects of a template that can manage tracking data) which applies the tracking information.
An example is the **CoralCamera** by CoralCG which, if connected to a tracker (PTZ, tracking devices with sensors, etc.), changes its position and rotation characteristics based on the data received from the tracker via the FreeD protocol.

The “Tracker Source” section allows you to assign a tracker to a specific target object.
As visible in the example image above, on the left there is the list of names of the target objects inserted in the XR template and on the right, the relevant assigned tracker.
By clicking on the tracking device (tracker), a combobox will appear where you can select the tracker to assign to the target.

TRICK: Two ways to perform the camera change (e.g. from CAM-1 to CAM-2):

1. Manual switch: assign the tracker related to CAM-1 or CAM-2 to the “CoralCamera” target via the tracker selector of the XR Settings panel.
2. Create a clone of the XR template from the LiveBoard content grid and assign the other camera tracker (CAM-2) to the CoralCamera. The execution of the two templates is mutually exclusive and each of them uses the tracking data of the selected camera (selected tracker).

In a CoralCG template it is possible to insert one or more "TrackingVideoInput" type objects that provide the image of the camera to which they are associated. They can be dynamically switched from one room to another.

In order to easily build a multi-layered XR environment (i.e. background, actor, foreground), LiveBoard allows you to run multiple XR templates at the same time.
These templates will be able to share the same camera trackers and each of them can display a TrackingVideoInput.

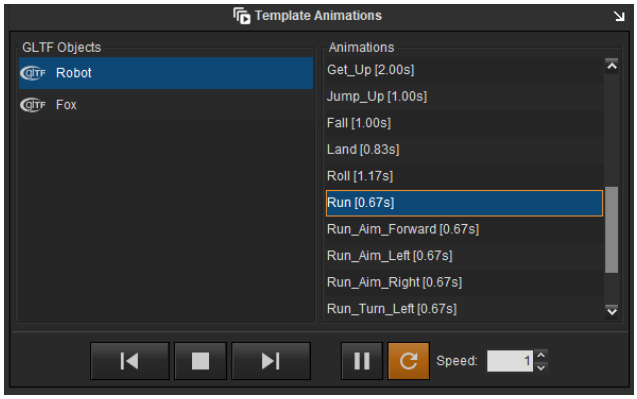
For implementation purposes, it may be necessary to disable one or more TrackingVideoInputs of a given template. In the “**TrackingVideoInput enable**” section, all the TrackingVideoInputs present in the selected template are listed and, via the checkboxes, it is possible to decide which to make visible or which to hide.

5.15 Template Animations Panel

The “Template Animations” panel is dedicated to controlling specific animations of templates and, in the case of CoralCG templates, it is possible to control the animations of glTF models.

The glTF animation control panel is divided into:

- “glTF Objects” section
- “Animations” section
- animation control section.



The “glTF Objects” panel contains the list of glTF models that present animations within the selected template. Once you have chosen the glTF model of interest, the “Animations” section will be populated with the list of names of the animations of the model itself.
From the list it is possible to run the selected animation by double clicking. The execution of the animation will follow the “Loop” and “Speed” settings below.

The control section has 5 buttons. In order from left:

- **Play backward**: Plays the selected animation backwards or resumes from pause in the backward direction
- **Stop**: stops the running animation returning the animation to its initial state
- **Play forward**: Plays the selected animation forward or resumes from pause in the forward direction
- **Pause**: pauses the animation or resumes execution if paused
- **Loop**: Enable or disable infinite animation loops
- **Speed**: sets the execution speed of the animation

The “Loop” and “Speed” settings are applied only when the animation runs.

5.16 Graphic attributes Panel

For attribute management, LiveBoard provides the "Graphics Attributes" panel; it has an advanced integrated graphics processing system that can act directly on objects.
The system acts in real time, the preview of the final result is calculated in real quality and what you see is what you get at the end of production.
In the presence of animated objects or video sequences, the graphic operators automatically act on each frame.
This is a very important feature and is explained by a simple example: if we wanted to add a simple shadow to an AnimGif Image object, we would first have to break it down into frames, then upload our photo editing software, cut out each frame, add the shadow via a filter, save the processed frames and reassemble them in a new AnimGif. On the contrary, LiveBoard does everything by itself: load the AnimGif, set the shadow and check the result immediately. This is just an example, in reality it is possible to do much more.

Available attributes are:

- **Transparency**: it makes the transparent object on the luminance or chrominance with the opportunity to interact with a tolerance value (from 0 to 200);
- **Solid**: makes the object of a single solid color programmable;
- **Background**: the object adds a solid background color programmable;
- **Color Correction**: Allows to change the hue, saturation and brightness of the object.
- **Flip:** the object rotates 180 degrees horizontally and / or vertically;
- **Buttonize**: It makes the object detected, thickness (from 0 to 10 points) and direction programmable;
- **Outline**: the edge of an object to color, thickness (from 0 to 10 points), opacity (0 to 255) and distribution (from 0 to 20) programmable;
- **Emboss**: it makes the padded object, thickness (from 0 to 10 points) and programmable directions;
- **Extrusion**: causes an object to three-dimensional border color, thickness (1 to 30 points) and direction programmable;
- **Antialiasing**: eliminates the annoying "jaggies" of an object;
- **Shadow**: Adds a shadow to the object color, distance (from 1 to 60 points), opacity (0 to 255), diffusion (from 0 to 20) and direction programmable.



The elaboration of the attributes is always done in real time. The computing time may vary depending on the performance of the machine, the type of attributes and the size of the object to be processed.

Referring to the panel "Attributes", shows a list of available attributes (identified by icons), from which, by clicking with the mouse, you can select what you want and change the parameters.

5.16.1 Operations on attributes

Operations cut, copy, paste and restores the default values, are collected in the upper-right panel of the attributes.

5.16.2 Transparency

The "**Transparency**" attribute transparency allows the processing of the selected object. The transparency is information that LiveBoard can inherit from the original, or recalculate based on the "**Chrominance**" or "**Luminance**" of the object.
The transparency on "Chrominance" allows you to make transparent a color image.

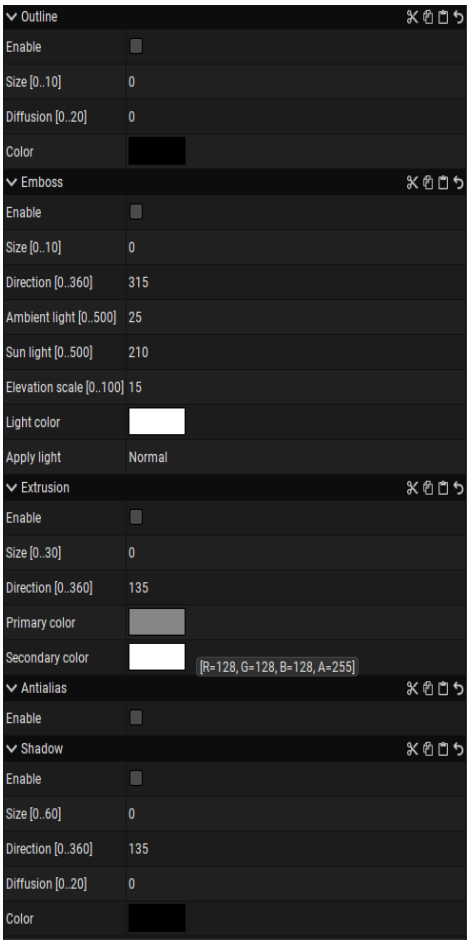

No Trasparency

The "**Tolerance**" introduces a tolerance to the selected color, the colors so that even the most "nearby", which fall in the value of tolerance, are considered to be transparent (the margins of tolerance range from a minimum of 0 to a maximum of 200).
Tolerance is useful to improve the transparency of graphic images or digitized where there are the "artifacts" (executed dirty) typical graphic formats "lossy" as the jpeg.
Tolerance, especially if used in conjunction with other attributes, can create excellent effects. The fact that tolerance, as well as other attributes, is calculated in real time, definitely contributes to a better realization of the effects.



Chrominance


Luminance



The "**Smooth**" introduces a gradient in semi-transparency on the edges of the image, to merge the object with the background perfectly. This feature allows you to further improve the graphic quality, eliminating the sharp cut that is created between the object and the background when you apply transparency chrominance (hue margins ranging from a minimum of 0 to a maximum of 200).

The checkbox "**Overlay**" allows you to add transparency to the original object that calculated from the attribute. This function is useful, you can make transparent a part of an object without losing the original transparency.

LiveBoard implements a second mode of transparency that works on the **luminance**. This mode makes it more or less transparent image so each point is inversely proportional to its brightness. In a nutshell: more light is less transparency to the limiting cases of the white (solid) and black (fully transparent).
Enabling the transparency of luminance mode, you get a great visual effects and, in combination with other attributes, you can make dramatic lighting effects. LiveBoard includes ("picture / mask") objects made specifically for the effects mentioned. The controls attribute is configured according to the mode selected transparency.

5.16.3 Solid



The "**Solid**" attribute has the function to apply a single color to all points non-transparent object. By selecting the checkbox "Solid", it activates the object in question becomes "solid" color selected. By clicking the colored button opens a color-chooser for the desired color calibration.

5.16.4 Background



The "**background**" attribute acting on the object with a given color by replacing the transparent object.
The parameters of "background" include the activation and deactivation of the attribute and setting the background color to match the object through the usual color-chooser.

5.16.5 Flip



The "**Flip**" attribute allows you to rotate the object 180 degrees horizontally and / or vertically. The panel is composed of two checkboxes "Flip Horizontal" and "Flip Vertical" with which the rotations are activated object horizontally and vertically, respectively.

5.16.6 Color correction

The attribute "**Color correction**" allows to change the color of the object, without the need to use an external image editing software.

5.16.7 Buttonize



The "**buttonize**" attribute serves to make the object "stuffed". It acts as if the object were actually subjected to a programmable light source according to various control parameters:

- "**Size** [0 .. 10]" allows you to adjust the width in pixels of the attribute.
- "**Elevation scale** [0 .. 100]" allows you to adjust the elevation of the board.
- "**Ambient light** [0 .. 500]" rule the incidence of ambient light (light background) object.
- "**Sunlight** [0 .. 500]" allows you to adjust the incidence of sunlight (direct) object.
- "**Application of light**" will determine the mode of application of light on the object.

Currently there are three modes of application:

- **Normal**: The light is uniformly applied, taking into account the edges of the object;
- **Luminance**: the light is applied taking into account the edges of the object and the brightness of the pixels that compose it;
- **Luminance reverse**: the light is applied taking into account the edges of the inverse of the brightness of the pixels that compose it;

Directional arrows determine the position of the light. Each time you select an arrow object will be illuminated by the new location.

5.16.8 Outline



The "**Outline**" attribute means adding a colored border around the object.
The edge can be customized according to different characteristics controllable by the user. The three numeric fields, in pixels width, opacity, and diffusion, respectively, to control the border width, opacity, which is superimposed on the edge of the background graphic, the spread with which it is drawn.
You can set the primary color of the border through the usual color-chooser, clicking on the colored button in the panel properties of the attribute.
The size of the edge varies from 0 (no border) to 10 (10 pixel border). If this value is set to 0, no border is not generated.
With great flexibility, you can create semi-transparent edges and / or disseminated on any object, even animated. The parameter of opacity ranges from 0 (fully transparent) to 255 (all opaque). The spread ranges from 0 (no diffusion) to 20 (maximum spread). The calculation of the diffusion takes place through a very accurate algorithm called "Gaussian Blur", which delivers excellent graphics performance ever.

5.16.9 Emboss



The attribute "**Emboss**" has the function of making the object "stuffed". It acts as if the object were actually subjected to a programmable light source according to various control parameters:

- "**Size** [0 .. 10]" allows you to adjust the width in pixels of the attribute.
- "**Elevation scale** [0 .. 100]" allows you to adjust the elevation of the board.
- "**Ambient light** [0 .. 500]" rule the incidence of ambient light (light background) object.
- "**Sunlight** [0 .. 500]" allows you to adjust the incidence of sunlight (direct) object.
- "**Application of light**" will determine the mode of application of light on the object.

Currently there are three modes of application:

- **Normal**: The light is uniformly applied, taking into account the edges of the object;
- **Luminance**: the light is applied taking into account the edges of the object and the brightness of the pixels that compose it;
- **Luminance reverse**: the light is applied taking into account the edges of the inverse of the brightness of the pixels that compose it;

Directional arrows determine the position of the light. Each time you select an arrow object will be illuminated by the new location.

5.16.10 Extrusion



The attribute "**Extrusion**" adds an edge to the object so that it appears three dimensional. The attribute can be customized according to different characteristics controllable by the user.
The numeric field "size [0 .. 30]" determines the thickness of the border, the two buttons allow you to calibrate two colors to accentuate the three-dimensional and directional arrows determine the direction of extrusion.
You can set the primary color and / or secondary extrusion through the usual color-chooser, clicking on the first and / or second colored button in the panel properties of the attribute.
The size of extrusion varies from 0 (no extrusion) to 30 (30 pixels extrusion). If this value is set to 0, no border is not generated.

5.16.11 Antialias



The attribute "**Antialias**" serves to smooth the jagged edges and the selected object.
The Anti-alias is a special procedure that allows graphics to eliminate unsightly jagged that normally can be seen working on objects in low resolution or very contrasting colors. The process is particularly useful as anti-aliasing can significantly reduce the flickering of the characters of a font when used for video production. Anti-aliasing is based on a rather simple principle which involves inserting one or more intermediate shades between two contrasting colors, so you move smoothly from one to another.
To enable anti-aliasing just select the checkbox "antialias".

5.16.12 Shadow



The attribute "**Shadow**" has the function of the object to appear as raised from the background, creating, under it, a shape of a solid color that represents him.
The shadow can be customized according to different characteristics controllable by the user. The three numeric fields, distance in pixels, opacity, and diffusion, respectively, to control the shadow's distance (from 0 to 60), the opacity with which the shadow is superimposed on the background graphic, with which it spread is drawn.
You can set the shadow color using the usual color-chooser, clicking on the colored button in the panel properties of the attribute.
With great flexibility, you can create semi-transparent shadows and / or disseminated on any object, even animated. The parameter of opacity ranges from 0 (fully transparent) to 255 (all opaque). The spread ranges from 0 (no diffusion) to 20 (maximum spread). The calculation of the diffusion takes place through a very accurate algorithm called "Gaussian Blur", which delivers excellent graphics performance ever.

5.16.13 Grab from Screen

By double clicking on **any color-chooser**, a window opens in which we can choose colors from various color tables with scales and gradations: sample colors, HSV, HSL, RGB, CMYK and **Grab from Screen**. Grab from Screen is an important function that allows the user to take color directly from the object. To use it, just click on the magnifying glass and drag it directly over the object from which we want to pick the color. Automatically the RGB and HTML codes of the desired color will be shown.

Notes: using the opacity and diffusion attributes of "Border" and "Shadow", you can create a neon-bright objects, or shadows spread (cast shadow) that blend in perfectly with the background

5.16.14 Hardware accelerated Attributes

The most recent technologies allowed us to implement and develop **GPU-based** algorithms for the computation of the attributes. The results, both in terms of speed, optimization and quality are extremely evident. We are talking of speeds over 100 times the **Software-based** counterpart.

The advantages of this technology are clearly indicated in cases where there is need to apply complex attributes to animated objects.

The differences in rendering results between Software-based and GPU-based attributes are minimal and are limited to basically Extrusion (more quality) and Emboss/Button (we can't use the Light application modes). We feel therefore to always recommend enabling this important function.

The acceleration is enabled by default, but you can unable this option in: **Preferences panel** → **Misc** → "**Enable acceleration**".

6 *The plugins*

The ClassX Plugin applications are the heart and allow for unlimited expandability. With Plugin you can implement new features in a very low, with a guarantee of compatibility and integration.

Almost all Plugin manage the automatic execution of playlists, freely customizable. This allows you to manage, highly flexible and precise contributions that will be displayed.

Every single Plugin can be resized, positioned by a suitable editor that instantly displays the result of each operation.

ClassX staff is available to develop ad hoc Plugin specific process control, meteorological measurements, alarm messages, viewing text messages, camera control, remote control devices and any other type of application.

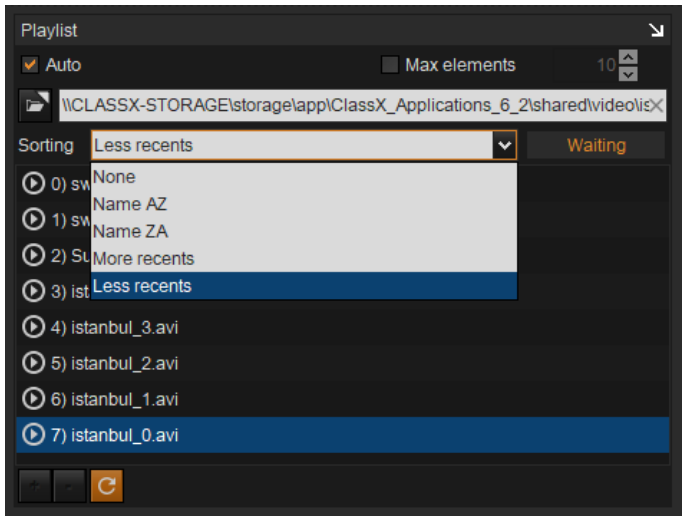
6.1 The playlist

Many plugins use a playlist to manage the content to display. For this reason, the operation of this component is shown in a separate section where all the plugin will refer later.

A playlist contains a set of files arranged in a specific order. As can be seen in the figure, these files are displayed in a list that implicitly determines the execution order. Each element is described by a number that identifies the location and associated files. The list order can be edited with a simple drag & drop.
The buttons lower left you to add and remove files in the list. By pressing the "+" button, it opens a file-chooser that lets you select the file or files to import into the playlist. If the format of an imported file is not compatible, you get an error message.
By pressing the button "**-**" you can remove the selected files after confirming the operation in the window. All operations on the playlist can be undone and repeated with the full support undo / redo called up by pressing "**Ctrl + Z**" and "**Ctrl + Y**".
The button shows an arrow on the right, enables the **"repeat"** mode of the playlist. This mode allows you to endlessly loop through the contents of the playlist.
Each time you select an object within the playlist, the file information is displayed at the bottom. The associated files can be changed at any time by changing the path directly in the field or by pressing the button "Open File" on the left.

Automatic Mode

The playlist has a mode called "**Auto**" which allows it to automatically define the elements within it. In this particular mode, the playlist loads and updates its entirety in a completely asynchronous to the execution of plugin that uses it, always guaranteeing the maximum fluidity of execution.
You can use this mode, a folder to use as a reservoir of data, in which to place the contributions that will be shown later in an order determined by the user.



In automatic mode you can define the following parameters:

- **Auto**: Enables automatic updating of the playlist.
- **Max Elements**: maximum number of items in the playlist.
- Source **file** or **folder** from which to derive the elements of the playlist.
- **Order** to be applied to individual elements: the most recent, older, AZ, ZA, no.

In automatic mode, the playlist is updated every 60 seconds. Update operations are highlighted with a simple progress indicator (progress).

6.1.1 Playlist from folder

In this mode, the playlist will be made up of all files of the correct type, contained within the selected folder. The subfolders will be ignored. We accept local paths (e.g. "C:\mik\images") and network (Ex:\\my_pc\images) but not remote (http).

6.1.2 Playlist from file

The elements of the playlist in this case are defined by a text file containing the list of files to upload. In particular, the file contains a list of ritch (both local and network and web (http)) separated by newline.

In contrast to the playlist folder, this mode allows you to insert files in the playlist is local but also remote files that are found for example on the web. The playlist file itself can reside on the web and be reached via http protocol.
Quite simply it is therefore possible to define for example, a playlist of videos or images to be displayed in the various plugin that automatically updates a list compiled by a third party or even copying the files to a shared folder.

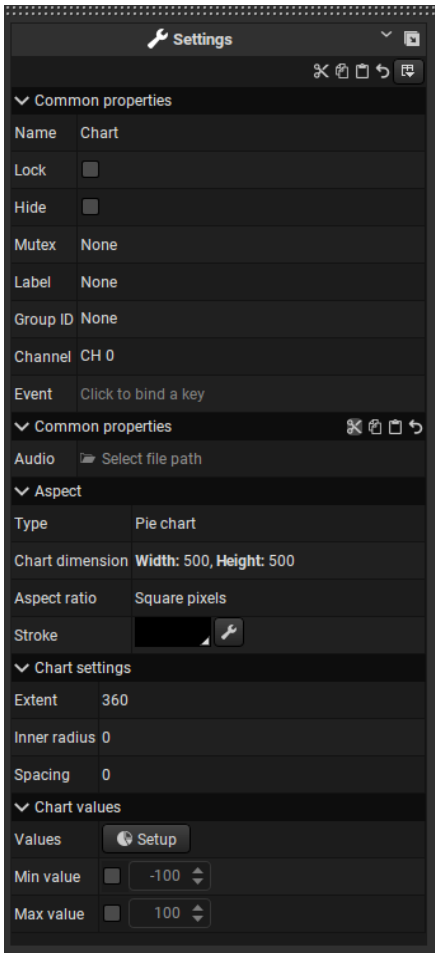
6.1.3 Advanced options for Ticker and StyledTicker

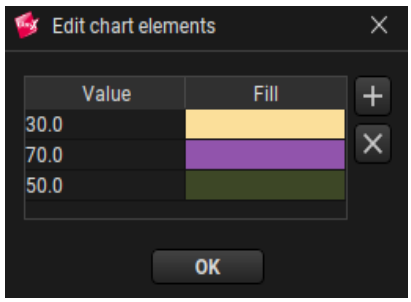
Generally speaking each playlist element refers to a local or remote file: each file is differently and specifically managed by a plugin. For instance, the Ticker and StyledTicker plugin interpret these files and extrapolate from them a sequence of information to be displayed: the field “Max imported records” defines the maximum number of the sequence elements. In this way, if the playlist contains a reference to an RSS file that is very long, it is possible to limit the number of the news displayed by the plugin. The drop-down menu on the right, indicates whether build the sequence from the beginning or end of the file.

6.2 Chart v.1.0

This plugin supports three different charts types: Pie, Histogram and line charts. These are different for their own properties but they are based on the same data model.

Pressing the button “Setup” with the pie charts icon, the panel shown below will appear. It contains a table that allows to edit the charts elements setting for each one the value and the fill paint.
The element fill is used to distinguish each chart part from each other. The paint editor is automatically opened by a double click on the related cell. How the editor works will be explained below.





6.2.1 Pie Chart

For this chart type the value may expressed in percentage or not. In the second case, each element converted and drawn as percentage of a total value derived by the sum of elements values. The user does not take care that the elements values sum is 100%.

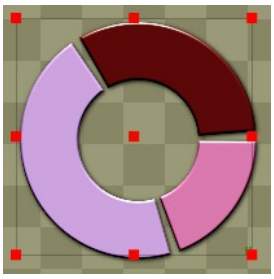
Example values:

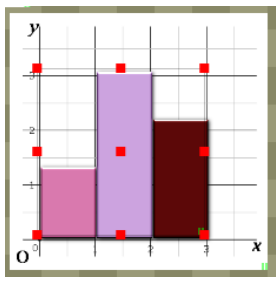
10, 40, 30, 20 the sum is 100%. Each element already defines the percentage of the relevant cake section.

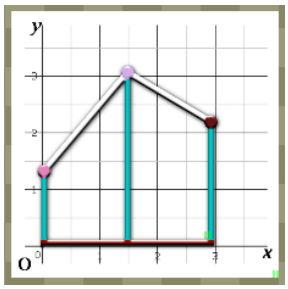
5, 7, 15, 34 whose sum is not 100%. Each value will be assigned a pie portion in percentage to the total of the values.

The settings for this chart type are:

- **Width**: the pie extent in degrees [0 – 360].
- **Internal radius**: used to cut out the pie inner part. The values is expressed in percentage.
- **Distance**: between the slices, expressed in degrees.



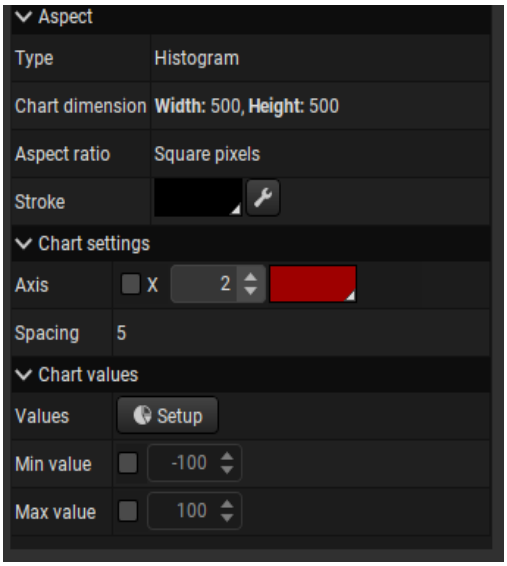




6.2.2 Histogram chart

The settings for this chart type are:

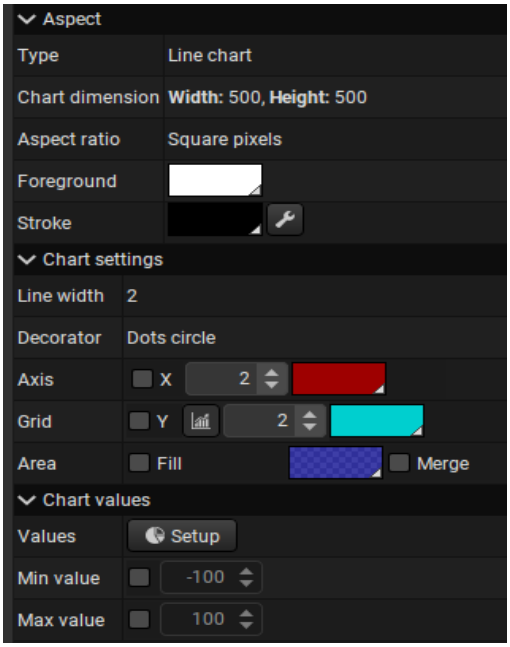
- The **minimum and maximum range of values** to display. All values out of this range will be displayed but will be limited to it.
- The **distance** in pixels between the chart elements.
- The **axis X** enables the drawing of the axis with the defined width and fill color.



6.2.3 Line chart

The settings for this chart type are:

- **Decorator**: used to mark each line element. It may be a circle, square shape or none.
- The **minimum and maximum range of values** to display. All values out of this range will be displayed but will be limited to it.
- **Width**: of the line in pixels.
- The point decorator as cycle, rectangle on none.
- The **axis X** enables the drawing of the axis with the defined width and fill color.
- The Y grid enable the drawing of vertical lines which may cover the whole cart or connect each point to the X axis.
- The Area enable the drawing of the area between the line and the X axis. If the merge option is enabled, all the areas generated by each point will be merged into a single element.



The fill of the shape is managed by the related button which you can edit the fill paint in the same way as already discussed for the text object.

6.2.4 Border and padding on the edge shape

The border of the shape and the border fill are handled by the appropriate buttons in the same way as already discussed for the text object.



6.3 ContentPlayout v.3.0

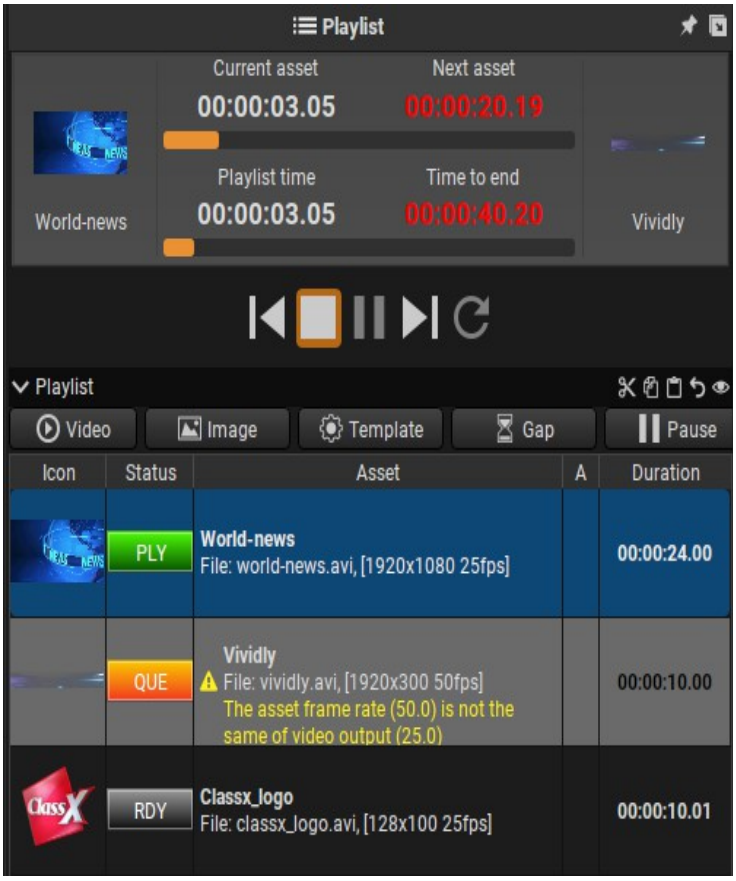
ContentPlayout is a LiveBoard plugIn that manages a playlist of events or assets which may be connected to a sequence of actions to run. Each asset has a duration that intrinsically defines a timeline where the actions are placed. It is a LiveBoard add-on with a separate license.

Thanks to this design, the ContentPlayout is able to manage with extreme simplicity the news airing: the audio and video are collected into playlist and by the actions, it is possible to run a graphics template in order to introduce each news clip. The final result is cool: a perfect execution where clips and graphics templates are in sync.

Looking the picture below, the ContentPlayout interface is subdivided into two parts: on the top it offers the playlist airing controls; below there is the playlist management.

6.3.1 Assets and playlist

The playlist is a sequence of assets of the following kinds: image, video, gap, template and pause. The assets are displayed from a table where they can be ordered by the drag&drop, that is, dragging each item into the desired position.



The meaning of the table columns are:

- **Icon**: contains a thumbnail of the asset. In case of video or image asset the icon contains a preview image.
- **Status**: the current asset status.
- **Asset**: displays the asset name and description. In case of image or video asset, this column shows the resolution and the source file.
- **A**: means actions. If it displays a gray icon means that the asset contains actions.
- **Duration**: the asset duration in HH:mm:ss.ff. The duration changing depends by the asset kind.

The asset can be put into the playlist simply dragging the related asset button into the playlist.
In case of image and video asset, ContentPlayout will ask you to select the source file before to inset it.
The keyboard delete button removes the selected assets and free all the allocated resource.
The right-click menu contains all the available operation as the actions cut/copy/paste, the actions edit and the asset sorting A → Z or Z → A.

6.3.1.1 The asset status

The asset status may be:

- **RDY** (Ready): the asset is ready for the airing.
- **ERR** (Error): the asset creation has failed. The asset is anyway into the playlist but it will not put on air. For instance if the source file of asset cannot be loaded the asset status will set to error. In general this status depends by the asset kind.
- **PLY** (Play): the asset is on air.
- **QUE** (Queue): the asset is in the airing queue and is next to be put on air.
- **PAU** (Pause): the asset is on air and paused.

The play and queue status are really important because they point out that the asset is on air on next to it. For this reason the assets with these status are locked and cannot be modified when the playlist is running. During the playlist execution the assets can be added, removed or sorted without affect its execution.

6.3.1.2 Video asset

This asset loads and puts on air a multimedia audio/video file. ContentPlayout allows to import single or multiple files into the playlist: drag and drop them from the file system or select them into the windows opened by the “import” menu. For each file which the loading does not fail, the ContentPlayout will automatically create a new playlist element where the name is got from the related file.
Before insert a new file into the playlist the ContentPlayout loads it and checks its correctness. If these operations fail the insertion will be denied. In this case we suggest to check if the audio/video codec of the multimedia file are installed on the machine.

6.3.1.3 Gap asset

The asset Gap has not graphics representation and may be considered as a blank video. In fact it defines a time period in which ContentPlayout will execute the actions joined to the asset. Unlike video asset whose timeline is derived from the duration of the media file, in this type of asset duration is set manually. By this way it is possible to different Gap assets useful to automate and synchronize the playlist with external elements as remote camera devices.

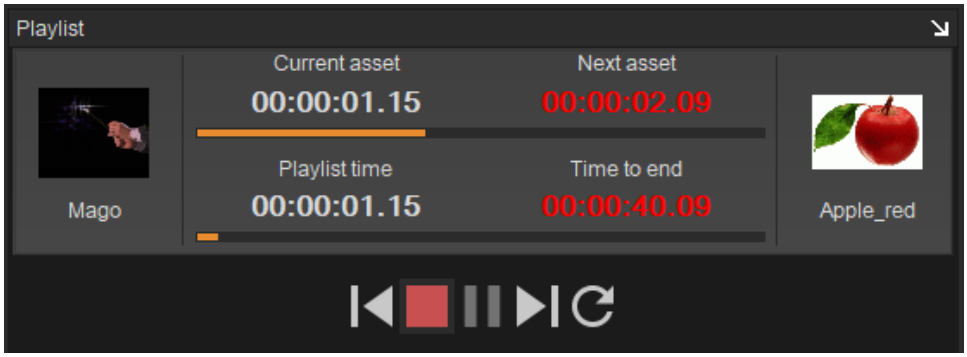
6.3.1.4 Image asset

This asset loads and put on air an image file. Its duration can be set manually as the Gap asset. ContentPlayout allows to import single or multiple image files into the playlist: drag and drop them from the file system or select them into the windows opened by the “import” menu..

6.3.1.5 Pause asset

This Asset defines one or more than one pause time into the playlist. This asset is really useful to control the playlist execution. The asset duration is fixed and set to 3 frames: the middle frame is the pause frame.
For instance in a news playlist, this asset can be put between clips. So every time a clip ends the playlist will be automatically paused, waiting to play the next clip.

6.3.2 Playlist execution



The playlist controls are placed into the middle of the panel above and allow to:

	Go to the previous asset
	Stop playlist
	Resume playlist
	Go to the next asset
	Enable the playlist loop

While the playlist is running the playlist panel shows on the left the name and the thumbnail of the running asset and on the right the name and the thumbnail of the next. The panels contains some progress bars and timecode labels: on the top they refer to the current playing asset and the red timecode label shows a countdown to the next asset. While on the bottom they refer to the playlist execution displaying its current time and total duration.

6.3.3 The Actions

As told before, the ContentPlayout allows to insert into the asset timeline some actions to run. This actions allow to interact with the LiveBoard contents: it is possible play/stop a content of for instance set values into the template and then play it. The action types are: EDIT, PLAY and STOP. While the first one can be applied only to the template, the others work with all LiveBoard contents.
The actions are stored into a FrameMarker that is placed into the asset timeline. In fact the FrameMarker is linked to a timeline frame, and its actions will be executed when the timeline reaches it. The playlist table shows a gray icon into the Action column when the asset timeline contains actions.

6.3.3.1 Actions editor

The action editor manages the actions on the timeline of an asset. Access to the editor can be done either by double clicking on the icon of the desired asset or by using the classic context menu under the "Actions" → "Set" items.

6.3.3.2 FrameMarker

A FrameMarker stores actions as is linked to a timeline frame. The FrameMarker types are:

MarkIn	Defines the media start time only for Video Asset. Only one MarkIn can be linked to the timeline.
MarkOut	Defines the media end time only for Video Asset. Only one MarkOut can be linked to the timeline.
Trigger Continue	Run the stored actions without interrupt the asset playing
Trigger Pause	Run the actions and pause the asset

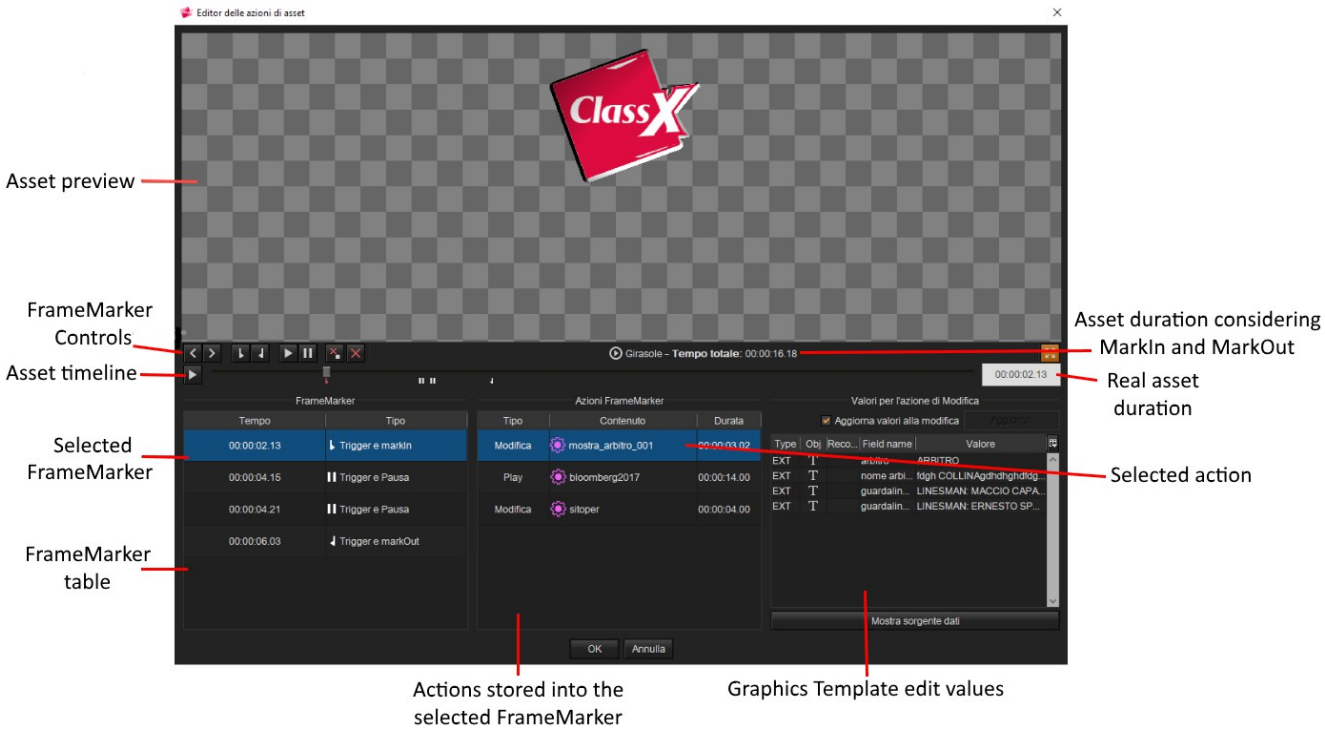
Generally when the Asset reaches a FrameMarker, executes its actions. If the FrameMarker type is “Trigger Pause” the assets execution is paused automatically. It will be resumed by the “resume button” into playlist control panel.

6.3.3.3 Playlist Import/Export

ContentPlayout allows to import and export the playlist. The export operation saves all playlist elements into a file. This file can be imported and all the stored playlist elements will be inserted to the end of the current playlist.

6.3.3.4 Timeline and FrameMarker

The actions editor manages the actions linked to the asset timeline. The editor can be opened by a double-click on the asset icon or by the right-click menu.



The actions editor allows to create, remove and edit FrameMarkers and its linked actions. The timeline is represented by a slider that allows to select each single frames. When the current frame changes, the preview panel places above, will be automatically update to check the correctness of the actions at this frame.
The timeline display the FrameMarkers by their icon. So the user can easily select it and verify the FrameMarkers linked to the timeline. A red icon will show the selected FrameMarker.
The controls located above the timeline collect all the FrameMarker operations. The first two buttons allows to

navigate through the FrameMarker, selecting the next of the previous. The other four button allows to insert the FrameMarker into the timeline at the selected frame. The last two button allow to remove the selected FrameMarker or remove all of them ignoring the selection.
Every time a FrameMarker is removed, added or selected the related FrameMarker table is automatically updated.

As shown in the figure, the panel is divided into two parts: the upper part is used to preview the asset and to manage the FrameMarker on the timeline. The bottom part contains three tables: "FrameMarker", "FrameMarker Actions" and "Edit Action Values".

6.3.3.5 The FrameMarker Table

This table shows all the FrameMarkers and display their linked frame using a comfortable timecode. The table allows to edit the FrameMarker time and its type. Every time a FrameMarker changes the timeline slider will be updated automatically.
The FrameMarker type column may show a “warning” icon when the FrameMarker will be unreachable. It happens when a FrameMarker is placed before a MarkIn or after a MarkOut FrameMarker.
These two types of FrameMarkers are only available for the Video asset: in this case if they are present on the timeline, the video clip starts at the MarkIn and ends at the MarkOut. As a result, all the FrameMarkers that lie outside the MarkIn and MarkOut will not be processed.

6.3.3.6 FrameMarker Actions table

This table contains the actions stored into the selected FrameMarker. For each action this table shows: type and linked LiveBoard content. As told before the action types are EDIT, PLAY and STOP. Each action is applied to a LiveBoard content that can be selected by a drop down menu. This menu contains only the contents contained into the current LiveBoard project and to make the selection ease it shows the content name and icon.
The duration column display the content duration but it is not always available and directly depends by the content type. In case of graphics template the duration can be useful to avoid the overlap of the different FrameMarker actions.

6.3.3.7 Edit action values

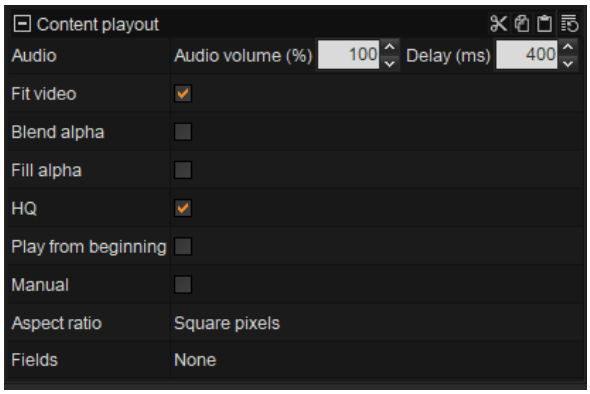
This table display the list of the graphics template field that can be changed. Its content changes automatically according with the selected action. If the template is connected to external DataSource, the filling of the template values can be facilitated by the DataSource panel. This panel is opened pressing the button “Show DataSource”. Here the selection of the DataSource record will fill the template values automatically.

6.3.3.8 Asset preview

This panel show a real time preview of the Asset that is drawn into the ContentPlayout area. So for a better understanding of the ContentPlayout on air position, this panel shows a yellow rectangle which size is set to the current output device. The white rectangle represent the ContentPlayout size and position on screen. Each playlist asset will be drawn into the center of the ContentPlayout area keeping its original aspect.
The preview panel has been designed to display and check with extremely precision either the asset preview either the graphics templates linked to the asset. By the selected frame into the timeline slider, the ContentPlayout get the nearest FrameMarker and draw its actions preview. If the actions are EDIT or PLAY of a graphics template, this preview panel will draw the graphics template filled with the edit values. This advanced function allows in case of a news playlist, to check the edit values of each template. If the actions work with more than one templates the preview will take care of their visual priority drawing them.

6.3.4 Asset video settings

The panel in the figure contains the Asset Video Settings.



- **Audio**: sets the playback volume for all files in the playlist
- **Delay (ms)**: allows you to set a delay to be applied to audio in milliseconds. The correct value to associate with this control varies depending on the display device chosen by the application that manages this content. In most cases the more correct value is about 400 milliseconds, but in any case it is advisable to carry out the necessary checks. This value can be changed during the execution of the content so as to make it easier to set.
- **Fit video**: If this option is not enabled each video file in the playlist is executed in its original size, ignoring the size of the content. Conversely, if enabled, each file is automatically displayed and adapted to the current size of the content.
- **Blend Alpha**: Alpha channel management skills for each video file.
- **Fill Alpha**: Alpha channel fills with the color black. This setting is useful for the file in MPEG format.
- **HQ**: fit the video using the highest quality
- **Play from beginning**: allows to play always the first item in playlist
- **Manual**: allows to play manual the items in the playlist
- **Aspect Ratio**: setting the Pixel Aspect Ratio of your video..
- **Fields**: Sets the polarity of the fields of the video running, choosing between deinterlace, lower first or upper first..

6.4 Image v.2.0

This type of care and manages plugin still images in popular formats with full alpha channel support. Once loaded the image you want you can resize and position them as desired. To restore the original size is sufficient to charge it by pressing "Enter" in the field that displays the location on disk.

Equivalent to the content of the content grid. They therefore have the same characteristics of use and it is possible to change their settings and properties from the "Settings" panel.



The sections “**Properties**” and “**Settings**” shows the following controls:

- **Auto update every** allows you to refresh the image selected, each 'x' minutes. This feature can be used whenever you need to display an image that is changed periodically to the source. This is the case, for example, a web cam located in a tourist place, or in front of a business, which periodically updates the image you are framing and saves it to a specific address (http://www.aptabetone.it/abetone/pics/lat001.jpg) pointed to our plugin.

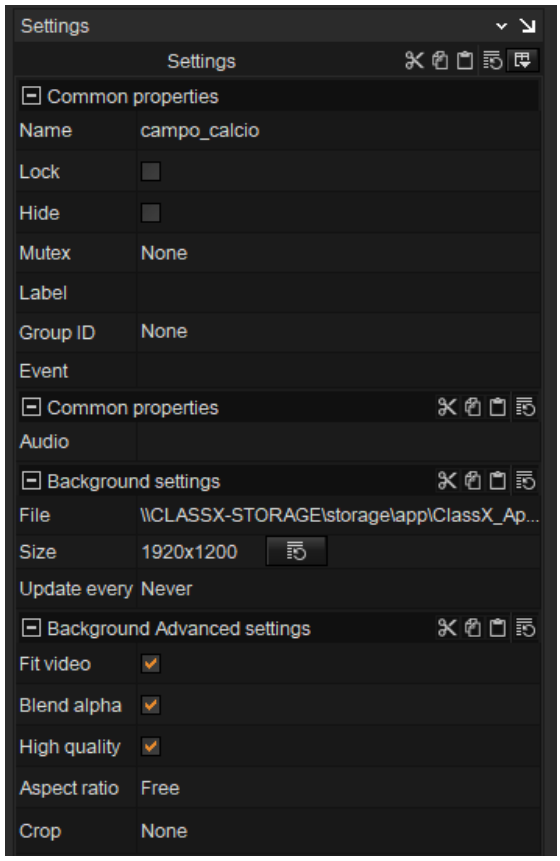
- **Min**. Sets how long to wait before reloading the selected image. It is enabled only when "Auto refresh every" is selected. If you set the value 0 minutes the plugin will automatically update its content every 5 seconds.

- **Fit Video**: if this option is not enabled, each media file is executed in its original dimensions, ignoring the the content size. If the option is enabled, each file is automatically displayed and adapted to the current content size.

- **Blend Alpha** enables management of the transparency channel. With SGL disabled the Alpha channel management may require more computing time. For this reason, in the presence of sources that do not require transparency channel is best to disable this feature.

- **High Quality** enable tracing through high-quality bilinear interpolation. This function is very useful when a small image has been magnified because the interpolation can eliminate much of all the artifacts due to downsizing.

- **Crop > Enable Cropping** allows you to crop the image edges, in pixels specifying the area to be deleted.

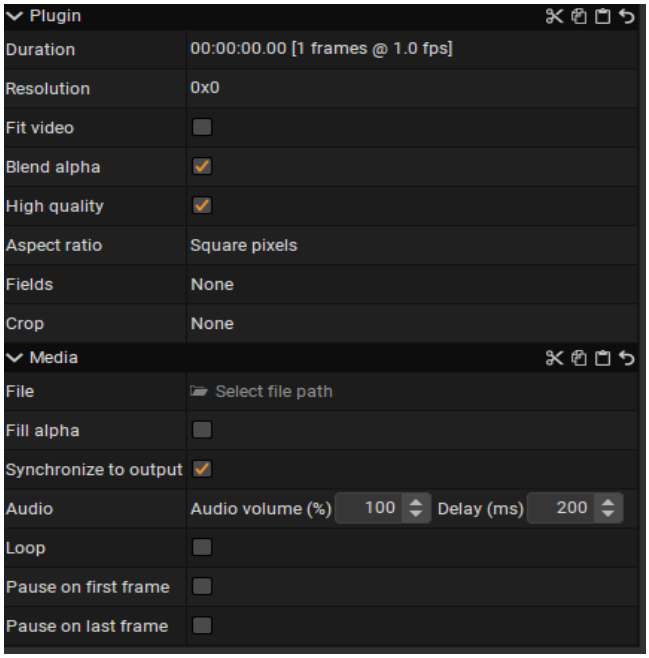


6.5 Simple MediaPlayer v.3.0

This type of plugin manages and imports videos in the most common formats with the full support of the Alpha channel. Once the desired video is loaded, you can position it and resize it as you like. To restore the original size of the video simply reload it by pressing the "Enter" key in the field that displays the path to disk.

Equivalent to the content of the content grid. They therefore have the same characteristics of use and it is possible to change their settings and properties from the "Settings" panel.

The **"Properties"** folder contains the following controls:



- **Duration**: it shows the media duration information.
- **Resolution**: it shows the media resolution(with x height).
- **Fit video:** if this option is not enabled, each media file is executed in its original dimensions, ignoring the the content size. If the option is enabled, each file is automatically displayed and adapted to the current content size.
- **Blend Alpha:** it enables the management of the Alpha channel for each video file.
- **High quality:** it adapts the video, using the highest quality.
- **Aspect ratio**: it applies the aspect ration to the current media file
- **Fields:** It applies the field direction the current media, for instance "Lower first(Even).
- **Crop**: it enables the media file cropping.
- **File**: it allows to change the media file.
- **Loop:** if enabled, it executes the content in loop.
- **Pause on first frame**: it pause the media at the first frame
- **Pause on last frame**: it pause the media at the last frame
- **Fill Alpha:** it fills the Alpha channel with the black color. This setting is useful for the MPEG files.
- **Synchronize to output:** it synchronizes the execution of the video files according to the display output chosen by the application. This option ensures a fluid execution, synchronized in time.
- **Audio:** it allows to set the play volume for the current media file, also it allows to apply an audio delay(in milliseconds) to synchronize the audio-video. In most cases the correct value is around 400 milliseconds, but we recommend in any case to perform the necessary checks. This value can be varied also during the execution of the content, in order to get an easier approach on it.

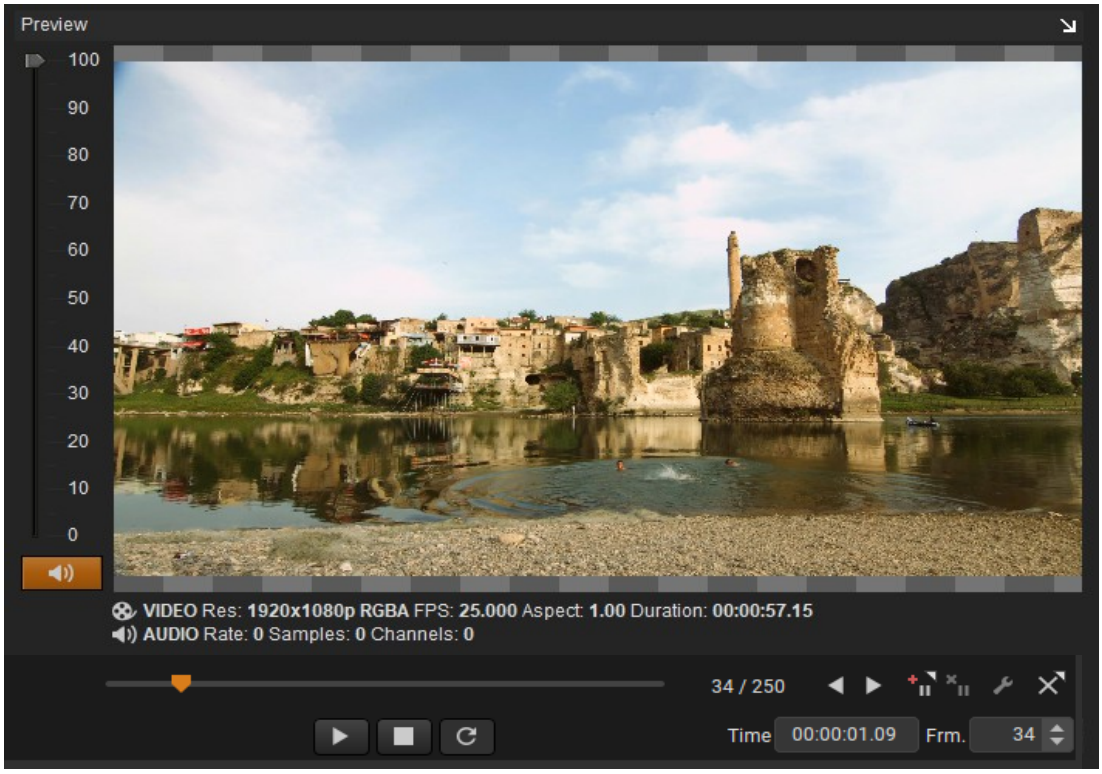
Warning: the Media content imports and manages both audio and video multimedia files according to the codec that are installed on the Pc. If a video or audio file is not correctly executed, you should check the type of codec in use and install it correctly.
The performance of this content directly depends on the encoding the file has been created with, and on the hardware configuration. For this reason we suggest you to use high-performance codecs, preferring the ones directly supported by the video card, in order to share the work on the GPU rather than on the CPU.

For more information on how to use the best audio and video files, you should consult the guide "How to use the best video / audio files."

6.5.1.1 MarkIn e MarkOut

The content Media offers the possibility to crop the video while it's running.
By pressing “Setup” button opens a special panel where can decide the MarkIn and MarkOut cropping point.

The slider above gives you access to every single media frame. By pressing the MarkIn/Out button you can decide to put a marker at the selected frame. By clicking on the markIn/Out icon it'll go directly and displays with an icon the MarkIn/Out points. In the lower values are associated with two markers: these values can be changed manually by setting precisely the point in time associated with the marker.



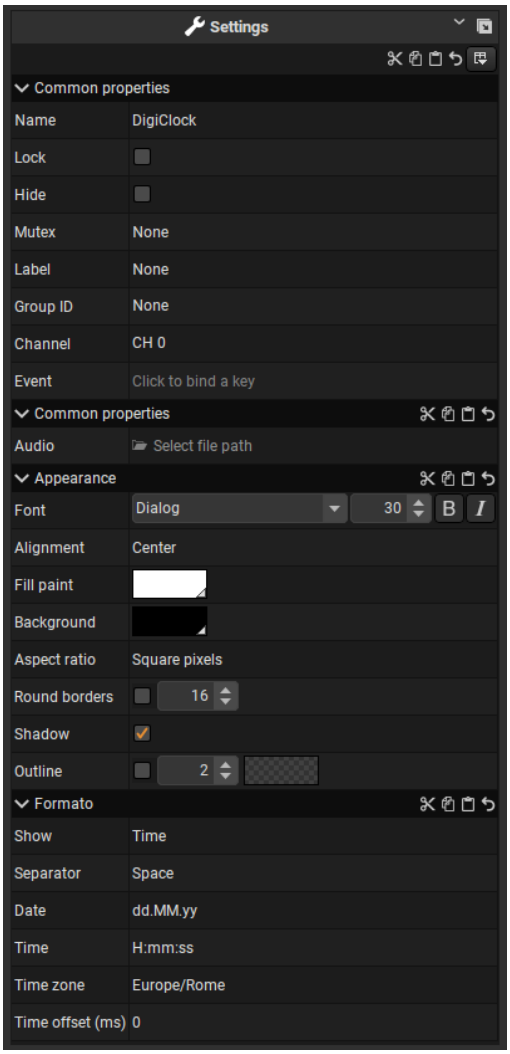
You can only set or only the MarkOut or MarkIn, or both. The preview panel will be discussed further on.

6.6 DigiClock v.1.6

The plugin "**DigiClock**" is designed to manage and display a digital clock. Using this plugin, because of its simplicity, it is very immediate.
The plugin "DigiClock" within a defined area displays time in digital and / or the current date. Through the panel shown in the figure, you can change at will all the properties of this plugin.



Through the **Settings panel**, you can change its features:



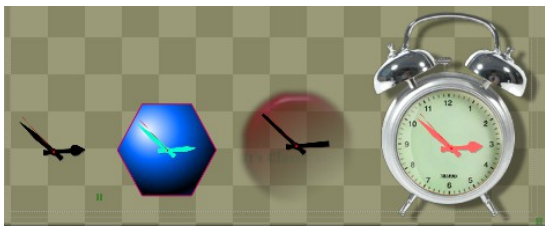
- **Appearance** allows you to customize the look of the plugin by setting fonts, style, alignment, padding, background, shadow and rounded corners.

- **Format** allows you to customize the time format and set the date and time zone. The combobox at the top allows you to choose which information to display: only the time, only the date, time and date, time and date. In the latter two modes, you can separate the time and date from a "head" by pressing the button next to the right.

- **Time Zone** allows you to select the time zone reference.

- **Time Offset (ms)** sets a value of positive or negative offset that is added to the current clock time. Time offset, in milliseconds, is used primarily to adjust delays between the system clock and the real one.

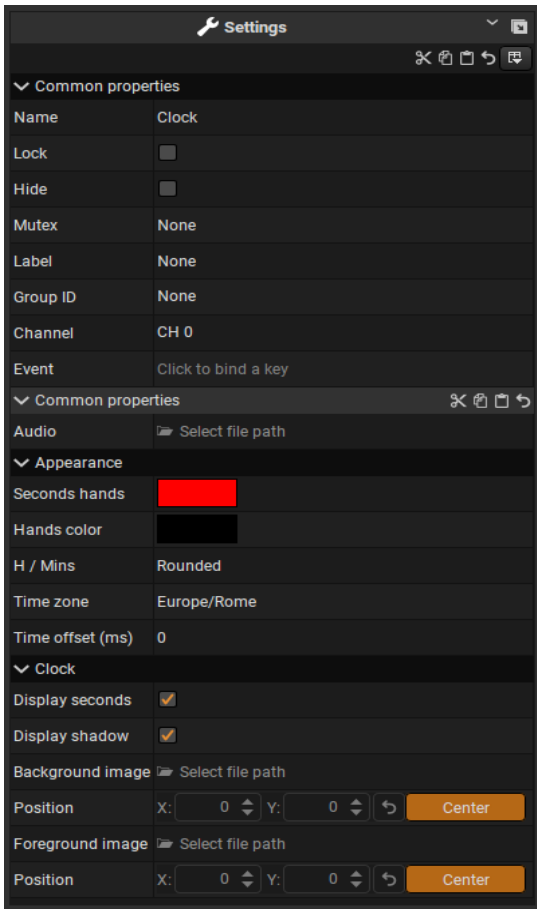
6.7 Clock v. 1.4



The plugin "**Clock**" is dedicated to the management of an analog clock. The analog clock is divided into 3 parts: background image, foreground image and hands.

The **settings panel** allows you to define its characteristics.

The folder "**Appearance**" allows you to customize the color, the shape of the clock and timezone. A second background image is used, these settings need to be modified to ensure proper display.
The combobox "**Form**" allows you to set the shape of the minute and second hands. As the digital clock, the combobox "**Time Zone**" allows you to select the time zone.
The "**Time Offset (ms)**" sets a value of positive or negative offset that is added to the current clock time. Time offset, in milliseconds, is used primarily to adjust delays between the system clock and the real one.
The settings box lets you choose which image to use as background or as foreground, and to determine its position relative to the hands.

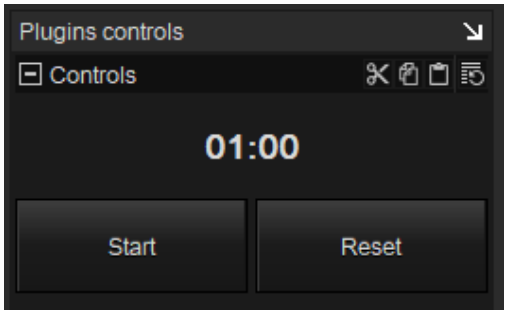


The checkbox "**At the center**" allows you to automatically place images in the center of the clock. Disabling this option can be defined for each image to a specific location. This feature allows you to create virtually any type of watch from the simplest to the most elaborate. The hands are automatically adapted to the size of the plugin. In this way it is possible to arrange the hands on the background image simply by resizing the plugin.
N.B .: in this way, by enlarging the plugin, only the hands will be enlarged. The background will remain of its original size to avoid images with grainy pixels.

It is also possible to manually insert any background image under the hands, overlaying them with any image already inserted in our project.

The other two checkboxes in the content pane settings allow you to display the second hand ("**Display seconds**"), and shade for all hands ("**Display shadow**").

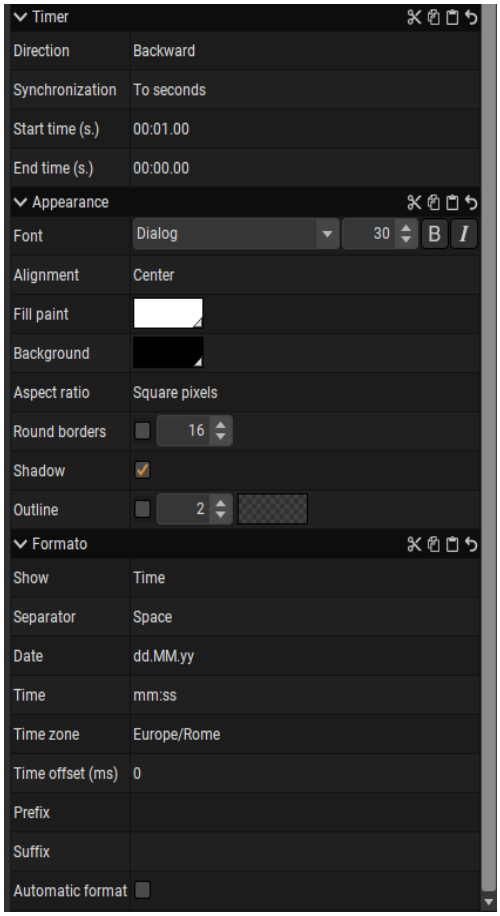
6.8 LiveTimer v.1.1



The plugin LiveTimer provides all the tools needed to manage all aspects of an up / down counter. The “**plugins Controls**” panel allows to start and stop the timer (button **Start / Stop**) and restores the current state of the timer with the initial one, for a new use (**Reset button**).

From the "**Settings**" panel it is possible to manage all the controls used to manage and customize the various properties of this plugin.

The panel shows all the controls used to manage and customize various properties of this plugin.



As can be seen in the figure, the panel is divided into several sections:

- **"Timer",** contains the main settings of the timer: **start**, **end**, **direction** and level of **synchronization**. Down towards setting the timer will serve as a countdown timer, and setting up the timer will towards the simple stopwatch function. The synchronization level has the purpose of synchronizing more timers. If in the scene, in fact, there is more than one timer, it is possible to make the seconds or the tenths of second change all together in each of them;

- "**Format**", you can setup the timer format as mm:ss, the delay and a prefix or a suffix to be added either before or after the timer;

- "**Appearance**", you can define fonts, size, style, text alignment, background color and shadow.

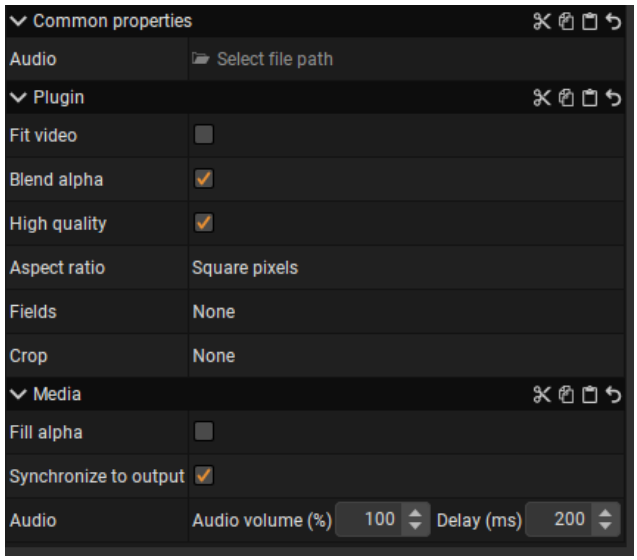
From the "**Effects**" panel you can apply DVE effects to the timer during input and output.

6.9 MediaPlayer v.2.0

The plugin "**MediaPlayer**" allows you to import and view both audio and video media files or import and view an external video source for example from webcam, cameras and video capture cards.

To change the features, you need to enter the **Settings** panel:

The first panel provides the following controls:



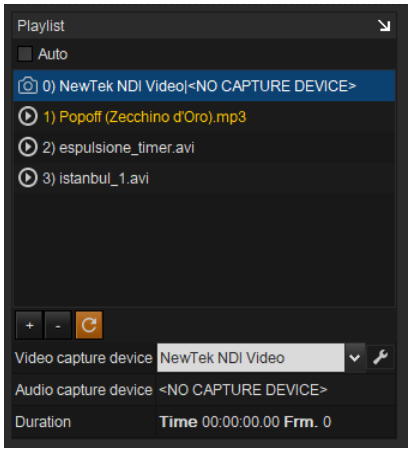
- **Fit video**: If this option is not enabled each video file in the playlist is executed in its original size, ignoring the size of the content. Conversely, if enabled, each file is automatically displayed and adapted to the current size of the content.
- **Blend Alpha: Alpha channel** management skills for each video file.
- **High Quality**: fit the video using the highest quality.
- **Aspect Ratio**: set the Pixel Aspect Ratio of the video.
- **Fields**: allows you to set the polarity of the fields of the video being played, to allow a correct display. Deinterlacing can also be entered.
- **Crop**: allows you to crop on the video
- **Fill Alpha:** Alpha channel fills with the color black. This setting is useful for the file in MPEG format.
- **Synchronize with video output**: run the video files synchronizing the execution according to the display device chosen by the application (see FrameLinkDriver™), which manages this content. This option ensures the smooth functioning and synchronized in time by forcing the speed run to the middle of the video refresh rate of the output device. Any audio tracks may not be synchronized using this mode.
- **Volume**: Sets the playback volume for all files in the playlist
- **Audio Delay**: Allows you to set a delay to be applied to audio in milliseconds. The correct value to associate with this control varies depending on the display device chosen by the application that manages this content. In most cases the more correct value is about 400 milliseconds, but in any case it is advisable to carry out the necessary checks. This value can be changed during the execution of the content so as to make it easier to set.

From the **Playlist** Panel you can enter the desired contents. The management of the playlist has already been illustrated in the previous chapters.

Warning: MediaPlayer matter and manages media files both audio and video codecs installed on the basis of the machine. If a video or audio file is not executed properly it is advisable to check the type of codec used and to perform a successful installation.
The performance of this plugin directly dependent on the encoding with which the file was created and the hardware configuration. For this reason, we recommend the use of high performance codec preference they are directly supported by the video card in order to distribute the load on the GPU rather than CPU.

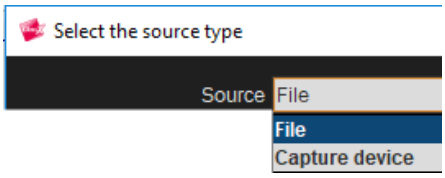
Supported File Types

MediaPlayer supports video files with the extension AVI, MPG, M1V, DIVX, WMV, DVD and MP3 and WAV audio files. It is also possible to decode files in web streaming and open URL links.



MediaPlayer plugin offers the possibility to crop the video is running and that coming through the button below. Pressing this button opens a special panel where you can set the number of pixels to be cropped on all four sides.
The panel of the playlist allows you to define a sequence of video from files or capture devices. As you can see in the picture elements that refer to files in playlists or media capture devices are represented by different icons. Each element contains a brief description as the file name or the name of the associated capture device.

Pressing the "+" button at the bottom will open a window where you can decide which type of item to insert in the combobox by selecting "**Source**".



It allows you to import and display an external video source eg from webcam, cameras and video capture cards. It interfaces directly with all video input devices (WDM compatible) connected to the machine.
When you add an item to select playlists in the combobox should be "**Source**" option "*capture device*". After adding the item into the playlist at the bottom you can set the "capture time" with this option you can switch video sources into a playlist from files with those from capture device.

To change the device settings you must access the configuration panel using the button. 

To control the capture operations, the button allows access to the configuration panel of the selected device. Each device uses a different control system based on its characteristics.

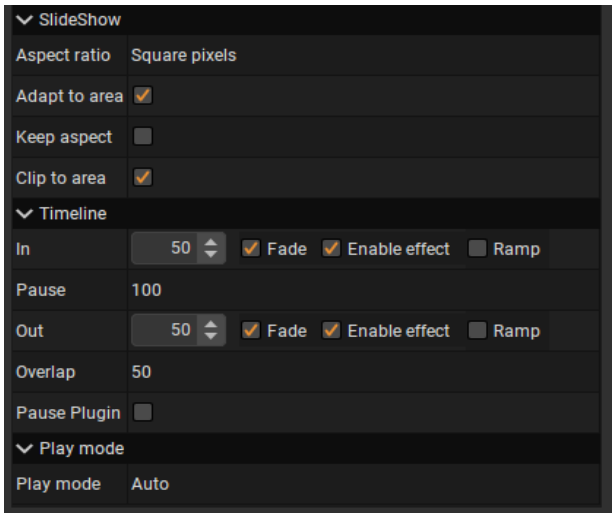
Warning: It is important to note that usually the same input device can be managed only by a single *MediaCapture* at a time. This means that you can not connect two different *MediaCapture* the same source video or audio.

Each time an input device is used, the operating system allocates resources to ensure proper display. The resources allocated directly depend on the type of device such as a webcam may require significantly less resources than a DV camera. For this reason you should always evaluate the resources used on a case by case basis and opt for a more powerful device.

If during the execution of a project, the video device is disconnected, the corresponding MediaCapture will be automatically removed without generating errors. In this case in order to redisplay the MediaCapture must stop running the current project, connect the device to your computer and rerun the plugin.

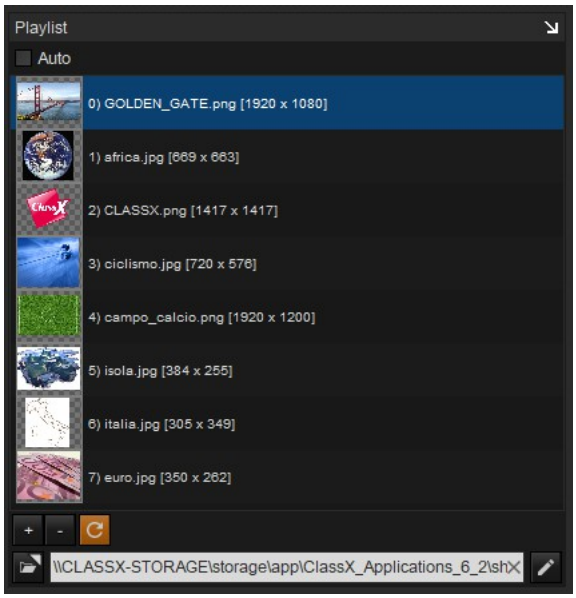
Warning: *MediaCapture* has been deprecated and replaced with Video Input. This feature will be completely removed from LiveBoard in a next version.

6.10 SlideShow v.1.3



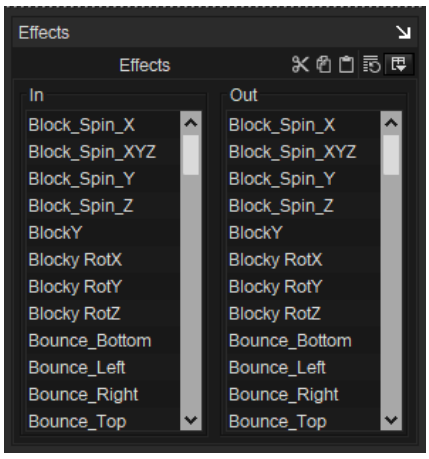
The plugin "**SlideShow**", as you can guess from the name suggests, is dedicated to the management and viewing a list of default images. In particular, this plugin allows you to display these images using 2D DVE effects, 3D. The images to be displayed are managed within a playlist; so it is possible to add them in the playlist panel.

The plugin "**SlideShow**" once added to the list, can be positioned and resized by positioning the panel. The size of the plugin automatically defines the area in which images are displayed.
From the “Settings” panel, in the "**SlideShow**" section, the checkbox "**Clip to area**" allows you to view all your pictures by cropping the excess. It is important to note that the displayed images do not resize to the size of the plugin, but are shown in its original size.
The checkbox "**Adapt in the area**", instructs the plugin so that your images are automatically scaled to the size of the plugin itself, maintaining the appearance that the feature is enabled.
The sidebar "**Timeline**”, defines the animation to apply to each individual image. The timeline of each image can simply be broken down into 3 phases: input, pause, exit. Through its stepping-field you can define the number of frames for each phase.
For the phase of input and output, you can enable the following properties:



Ramp: on the effect that causes the selected effect is executed following a **ramp-up** (during output) or **deceleration** (during input) seamlessly and automatically, giving the effect that touch of professionalism that never fails.
It's possible to enable **FadeIn** or **FadeOut** depending on the stage.
"**Overlap**" allows you to define the frames of overlap between the output and input of the next image. If this value is zero, the detachment between the images is clear. On the contrary if this value is exactly equal to the output frame images overlap perfectly. This way you can easily create images that transition between the classic cross-fade.

> **Notes:** in all cases of overlapping frames can never exceed that of output.



The "**Effects**" panel allows you to select the effect to be applied both in the dve of input and output. Some effects can be quite challenging and is sensitive to the user to choose the most appropriate and commensurate with the performance of the machine.

> The folder "**Playlist**" contains a list of plugin that will display images. The operation of the playlist has been previously described. In this case, however, every single item displays the playlist and the name of the file, even its preview and its size.

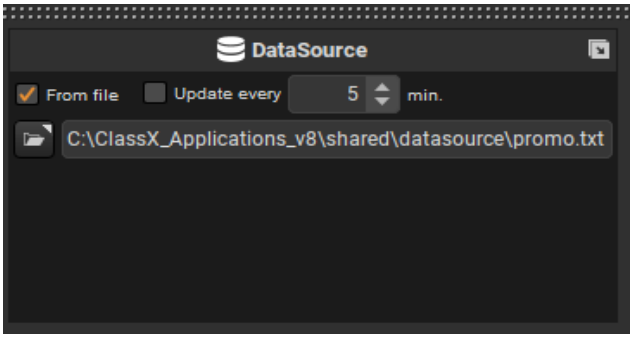
6.11 MediaText v.1.3

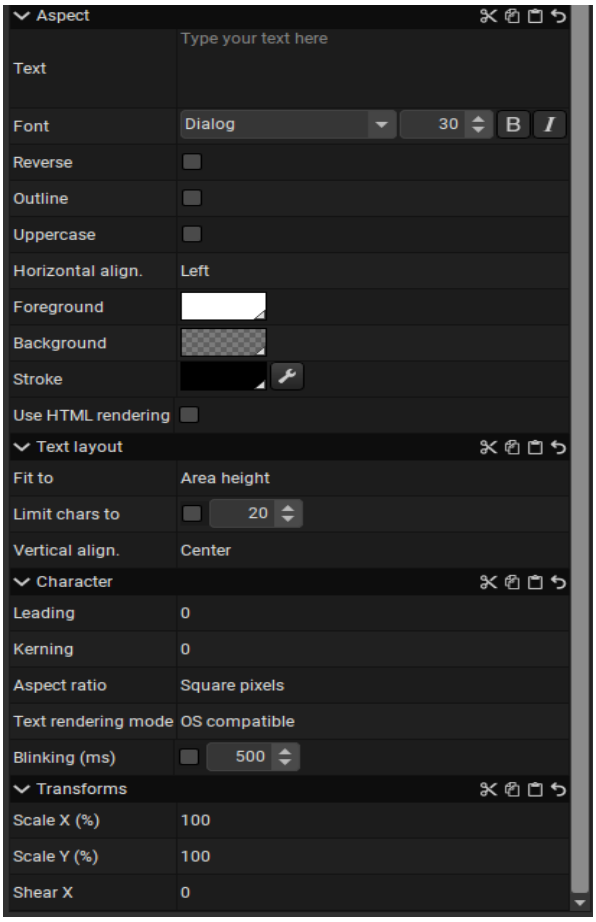
The plugin "**MediaText**" displays a text within a specified area. The management and implementation of this plugin is entrusted to the “The functions on the text and the "Text-area" chapter.

In the Settings panel you can define the characteristics and manually enter a text by double clicking next to the "Text" field.

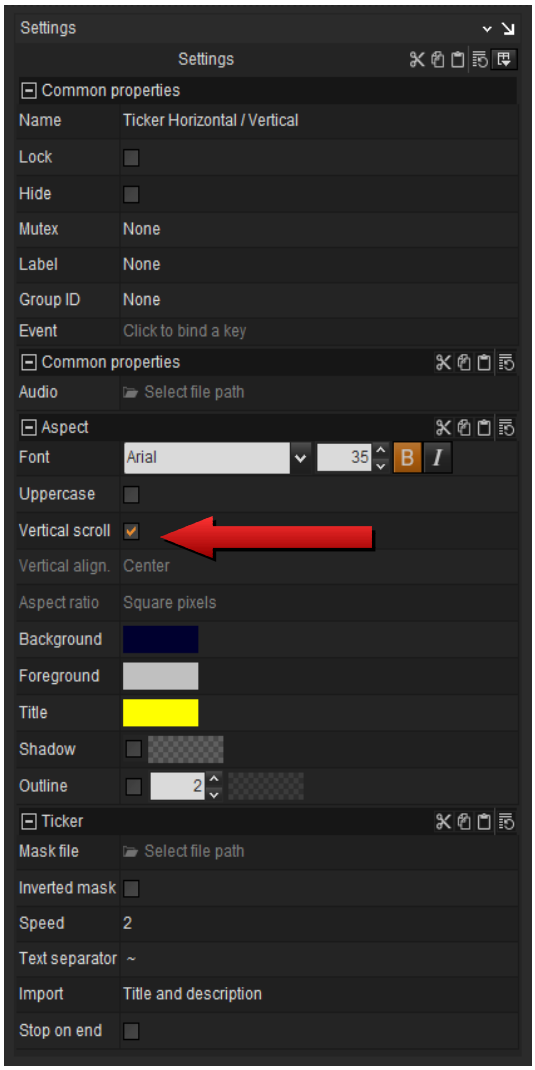
From the "Data source" panel it is possible to insert a text file from an external file and it will automatically be inserted in the Text field.

This mode can be enabled using the "From file" option and selecting the corresponding file.
The option "**Auto update every**" allows plugin to update its contents by reloading the file according to an interval of time in minutes. If this value is set to 0, the plugin will automatically update every 5 seconds. The "Attributes" allow quick and complete customization of the text.





6.12 Ticker Horizontal / Vertical v.3.0



The plugin "**Ticker horizontal**" was designed to display information in a series of horizontal scrolling within a predefined area. This plugin is able to collect information from both local files via a web technology by offering full support for RSS / XML.

For text files (*. TXT) to the ticker will automatically format the content, combining any lines of text on one line using the tilde "~" (tilde) as a symbol of the junction between lines.
With full support of technology RSS / XML Ticker is able to individually interpret the title and body of each news that the tags defined by this standard. In this way, the ticker can display the title and body with different colors chosen by the user.

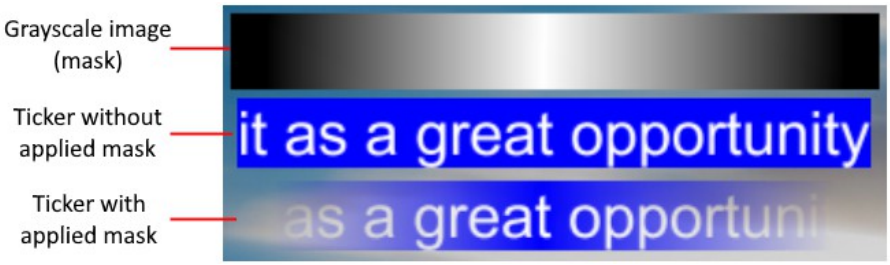
The ticker displays all the information in a predefined area, whose position and size is set by the panel positioning.

From the Settings Panel you can define its characteristics:

Appearance allows extremely easy to customize the look of the font with bold, italics, forcing the text in all caps, the **aspect ratio** and vertical alignment. In the same box you can also find the controls to set the **color of the title** (only in case of texts in XML / RSS) and **text**, **color** and thickness of the **edge** with the **shadow** and its **color**, and finally the color **background**.
By selecting the "**Vertical Scroll**" option and resizing and moving the ticker area as desired within the Positioning panel, you can give the ticker a vertical trend.
Ticker allows you to define the speed of the text, and set the separator character from a text to another, to choose what to import from the RSS and finally to decide whether to stop scrolling with the end of the text or not.

From the **Playlist Panel** you can enter the desired contents. The management of the playlist has already been illustrated in the previous chapters.

In the Ticker Horizontal/Vertical it is also possible to add a **mask**. In "Mask file", by adding a black and white (or grayscale) image, the effect is calculated by the luminance of the image pixels. Black creates complete transparency. It is also possible to reverse the effect, thus bringing white to transparency. Just tick "**Inverted mask**".



Playlist panel shows the amounts used by the Ticker in a playlist whose operation is similar to that described above. The playlist can contain both files without distinction of type TXT and RSS both local and on the web.
The loading and/or update texts and contributions RSS is done automatically during the execution of the ticker. The button in the lower right corner depicting a pen, you can open and edit the currently selected file.

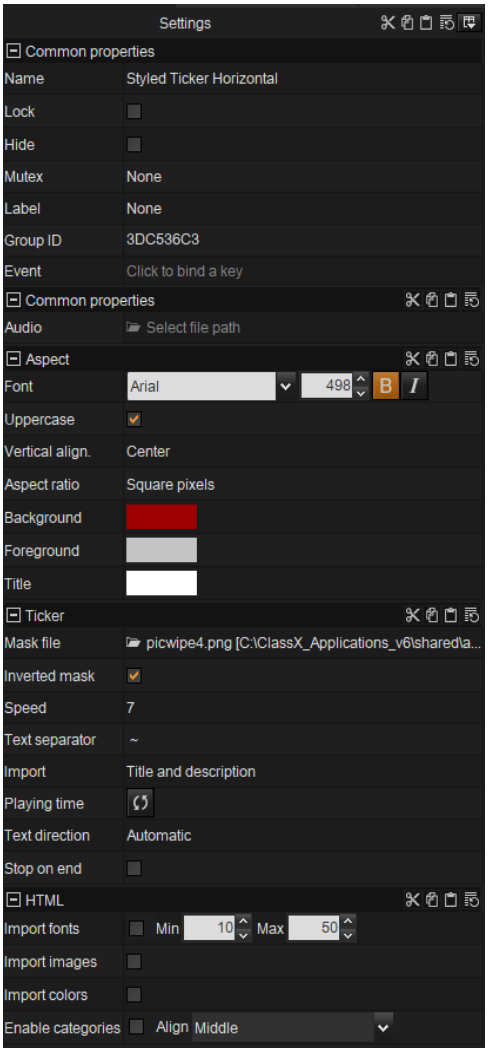
6.13 Styled Ticker Horizontal v.2.0

The "**StyledTicker**" plugin was designed to display information in a series of horizontal scrolling within a predefined area. This plugin is able to collect information from both local files via a web technology by offering full support for RSS/XML.

It represents an evolution of the previous ticker plugin (which nevertheless remains in the official distribution) significantly extending the functionality, performance and flexibility.

The StyledTicker allows you to:

- Importing a text file with automatic filtering of unwanted characters, and any conjunction of lines separated by newline.
- Import files of type **RSS-XML** tags with automatic **title** <rss title>, **description** <rss description> and **category** <rss category> for each line of the RSS file. You can also select to display the "Title" and "Description" or both of each news item.
- Handle full **HTML** formatting for all text, supporting font tags, color, alignment, image, style, superscript, subscript, strikethrough, and more.
- Insert **pictures** and **animations** including remote network address or URL.
- Insert separator between news and news text as an image or animation based on simple control sequences.
- Out any connection, download images and text-formatting completely in the **background**, during the execution of the playlist.



In addition, the StyledTicker fully **unicode** supports and automatically reverses the direction of scrolling texts according to their coding. This makes it possible to use the same plugin to run a mixed playlist of texts in various languages.

With full support of the RSS-XML technology, **StyledTicker** is able to interpret the individual title and the body of each news that the tags defined by this standard. This way you can display the title and body with different colors chosen by the user.

Through the **Playlist panel** it is possible to connect sources or files to insert.

In the **Settings panel** you can define its characteristics.

In the "**Aspect**" section it is possible to sets the default **font** and vertical **alignment** of text within the scrolling of plugin. In the same panel, you can set the **aspect ratio**, the **scroll speed** and color of text, background and title that will be used by StyledTicker. Using the "**Import**" menu for select to display the "Title" and "Description" or both for each line of the RSS file.

The **separator** represents a sequence of characters that is used as a separator between news and the next. Although it is set to "**~**" (tilde), you can use appropriate control characters to insert static or animated images along with the ordinary text.
<C:\ClassX_Applications\shared\ticker_resources> Were placed in the directory images and animations in various formats. Through the **filename** (without extension), any image that resides in this directory can be used to create separator. The control characters used to describe images and animations are the **braces** (Alt +123, Alt +125).

Example: Setting the separator "{cat_cube}" any news will be separated from the next by a small cube in 3D animation, corresponding to the file "cat_cube.avi".

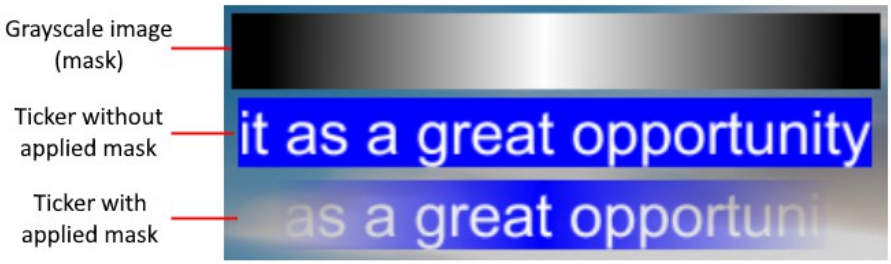
StyledTicker automatically recognize the file extension to upload, and you do not specify anything in the separator. The supported formats are GIF, PNG, AVI and MOV. This is also the sequence ordered by priority of file extensions that StyledTicker will open for uploading images.
If the image does not exist in any of the formats provided, the image is loaded by default which always has the name "Default.png" which is a gray bullet.

The sequence may contain multiple images and text mixed together:

Example: Setting the separator "{cat_cube} {my_logo} {cat_cube}" any news will be separated from the next by a small cube in 3D animation, my station logos and more from the cube.

Playing time shows, as you can guess, the time that each single file of the playlist will use before ending its execution and moving on to the next one. The button allows you to update every time a new file is added.

In the Styled Horizontal Ticker it is also possible to add a **mask**. In “Mask file”, by adding a black and white (or grayscale) image, the effect is calculated by the luminance of the image pixels. Black creates complete transparency. It is also possible to reverse the effect, thus bringing white to transparency. Just tick "**Inverted mask**".



6.13.1 The “HTML Settings”

StyledTicker allows you to control some important aspects of the rendering of these texts in HTML format in the title in the description of the news.

With "**Import fonts**" you decide whether to use the default font that indicated in the text or HTML. In this case, you can define the minimum and maximum font imported, avoiding to put text scrolls too small or too large.

With "**Import Pictures**" you enable support for loading and displaying images in any HTML text. When creating the RSS file to be displayed through the ticker, the link to the resource located on your hard drive or network, can be done with absolute and/or relative path. If you want indicate a localized resource in the shared folder from ClassX_Applications, you can write:
"<img Src="C:\ClassX_Applications\shared\picture\meteo\anim\lcd\x64\cloud_64.avi">, with an **absolute** path to the image or "<img src = "{shared.dir}\picture\Weather\anim\lcd\x64\cloud_64.avi">" using the **relative** path.

Choosing a **relative** path, the root of the absolute link may change dynamically in relation to the unit of choice for ClassX Applications boot (C drive or in an external hard drive or in a network) without the user having to worry about updating the path correctly every time.

With "**Import colors**" you enable support for loading and displaying HTML text in any color. If enabled, the default colors are ignored.

With "**Enable categories**" you enable support for loading and displaying of tags RSS Category, or we can differentiate each marking the information content based on a category (eg car, health, sports, gossip, politics, economics, etc.).
The logic implementation of the **separator** is the same text as above. In this case, who produces the news can use the **RSS Category** tags to introduce the sequence of characters that indicates the appropriate category. Unlike the separator text, no need for braces.

Check "**Align**" aligns the display of categories within the scrolling (top, middle, bottom or baseline of the text).

6.14 TextShow v.1.2

The plugin TextShow provides all the necessary tools to manage a slide show of information, consisting of texts and images obtained from data sources in text format html, xml, rss, csv and txt.



With full support of technology **RSS / XML** TextShow is able to interpret each of the components (TAG) "evidence" and "description" common in the standard RSS format file (for a thorough discussion of RSS is recommended to consult extensively documented: http://it.wikipedia.org/wiki/Really_simple_syndication).

TextShow is able to visually interpret the RSS tags, providing the opportunity to color the text differently depending on the type of membership.

TextShow is able to execute files in CSV (comma separated values) where, for each row, the first field identifies the title and identifies the following description.

Example of **CSV** file reports:

title, description
"Ferrari still in the lead," "The Ferrari still win at Monza ..."
"Elections 2008", "The 2008 elections are fast approaching ..."

TextShow supports text in HTML format and automatically takes care to render them in a timely manner. Support for HTML, however, limited to the most popular tags for formatting text and uploading images, both local and from the web. There are currently no external plugin or supported form.

An example of a CSV file with **HTML** formatted text (in red the title, the description in green):

```
"<P><FONT COLOR="#CC0000" FACE="Courier New, Courier"><B>Titolo A</B></FONT></P>" ; "<P ALIGN="CENTER"><FONT FACE="BernhardFashion BT">Descrizione A1<BR> Descrizione A2<BR> Descrizione A3</FONT></P>"
"<P><FONT COLOR="#CC0000" FACE="Courier New, Courier"><B>Titolo B</B></FONT></P>" ; "<P ALIGN="CENTER"><FONT FACE="BernhardFashion BT">Descrizione B1<BR> Descrizione B2<BR> Descrizione B3</FONT></P>"
```

The output of the plugin is organized into two parts: the upper part is devoted to the display of the title (if present), while the bottom displays the description. This information is appropriately formatted according to the parameters laid down by TextShow in the "Settings" panel.

Using appropriate controls, you can customize the look of the font, its size and style (bold, italics). Text formatting includes controls for horizontal and vertical alignment of text (depending on area assigned to the plugin). The justification of the text can be obtained using the appropriate HTML tags.
You have the option to force text to upper case, both by description or by title. TextShow automatically adjusts the size of the title if its length was larger than the preset.

In the same panel are also controls to set the color of the title, description and their backgrounds.

The button-shaped folder allows you to select the file that will interpreting TextShow. The field just to the right displays the path and file name referenced by the plugin. Even more visible on the right is an LED that lights up whenever the file is read or updated (automatically once per minute).

ⓘ In the case of text in HTML format, and formatting information are taken from the tracking tags in the text and take precedence over those set in the plugin.
The loading and/or update texts are automatically every minute during the execution of TextShow. For this reason, the reported changes to the properties of the plugin have no immediate effect, but you must wait about one minute of time to observe the effects. The preview area is updated yet again without waiting times.

The option of "**Update every (sec.)**" allows you to specify how often the file must be reloaded.

The "**Timeline**" section and "**Effects**" panel include the same controls used for the plugin **SlideShow**. See "Slide Show" to know about them.

6.15 Karaoke v.1.1

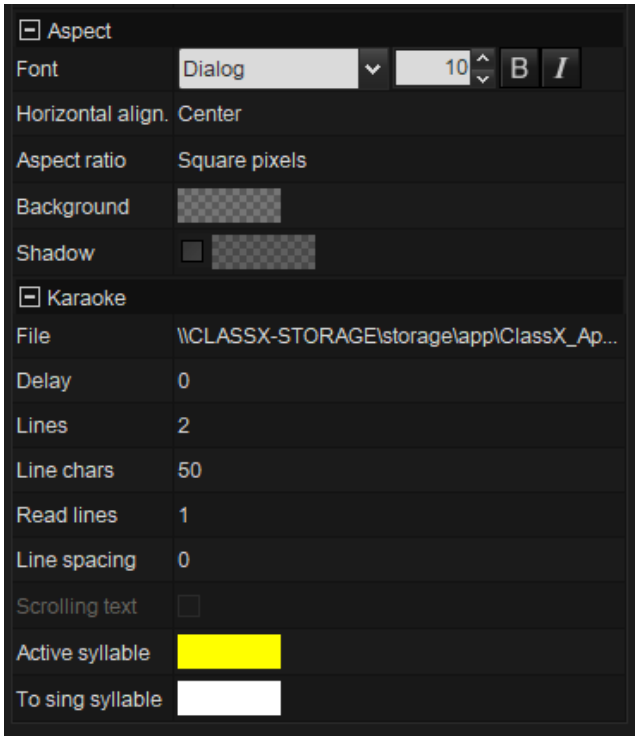
The karaoke plugin as the name suggests, manages and synchronizes the execution of an audio track associated with the text showing step by step and the syllables to sing. The plugin uses two file formats for its execution: **WAV, Midi/Kar, Mp3**. The karaoke plugin in fact, derives from these files synchronization time and the text to be displayed during the execution of the WAV file.
This mechanism has all the classic functions of a karaoke player but instead more midi track, running a digital audio track of the highest quality.



Warning: To function properly, both files must have the same name, lie in the same directory, and obviously have a different extension. For example:
- midi file: C:\audio\Popoff (Zecchino d'Oro).kar
- wav file: C:\audio\Popoff (Zecchino d'Oro).wav

The text of karaoke shows in the middle of the plugin. You can change the display features from the **Settings** panel using the Aspect and Karaoke sections . The first, in the "**Aspect**", you can define fonts, size, style, text alignment, background color and shadow. In "**Karaoke**" allows you to manage the following properties:

- **File:** here you can enter the necessary file that is audio or text.
- **Audio Delay**: Sets a delay in milliseconds to be applied to audio playback. The correct value to associate with this control varies depending on the display device chosen by the application that runs this plugin. In most cases the more correct value is about 400 milliseconds, but in any case it is advisable to carry out the necessary checks. This value can be changed during the execution of plugin to make it easier to set.
- **Lines**: defines the maximum number of lines of text to be displayed simultaneously.
- **Lines read**: defines the number of read lines of text to be displayed.
- **Characters**: defines the maximum number of characters per line (the format is the word).
- **Spacing**: Defines the vertical spacing in pixels between the lines.
- **Active syllables**: sets the color of the sung syllables.
- **To sing syllables**: defines the color of the syllables to sing.



The "**Preview**" panel displays the text that will display a preview of the text.

6.15.1 Manual Mode

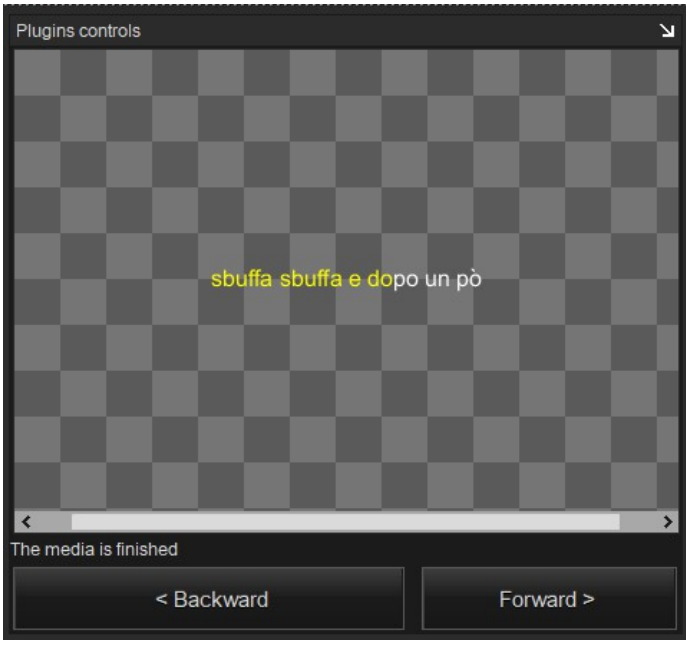
“plugins control” panel allows you to manually manage the progress of both text outlining the syllables singer. This mode is automatically activated when the selected file in text format. In this case the control panel will enable two buttons "forward" and "back" to help manage the progress of the sung syllables.

ATTENTION: In this mode the MediaKaraoke will perform no audio track.

The division into syllables of the text is done automatically. This automatic procedure is efficient but not always,

match them because the text by each singer is always different and subjective. Exist in this case, the special characters if properly placed in the .txt file can manage both the progress that the division into syllables.

The "_" allows us to consider two words as one.

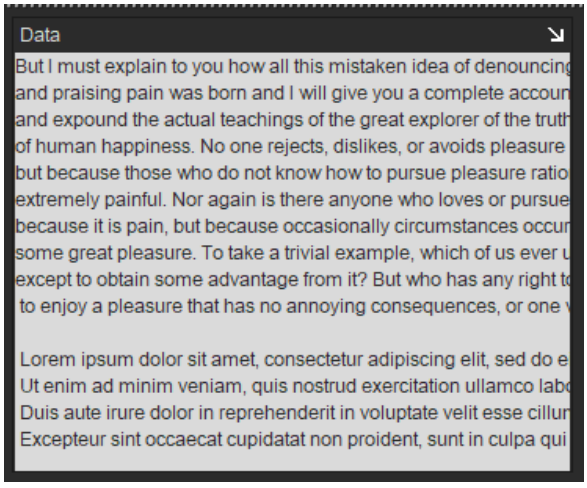


The character "~" manages the division into syllables in the word.
The character "#" will insert a pause between two subdivisions.

These special characters are of course invisible.

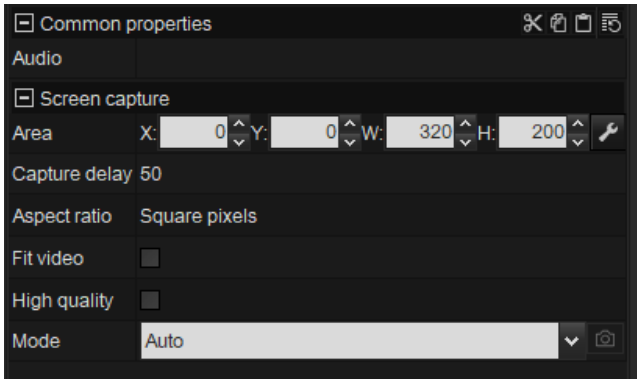
In manual mode, the option to "Text Flow" allows you to advance the text as you complete a row was the singer. Instead, if this option is disabled, to sing the next lines will be displayed only when those views were sung.

From the “**Data**” panel you can view all the text linked to the inserted file.



6.16 ScreenCapture v.1.1

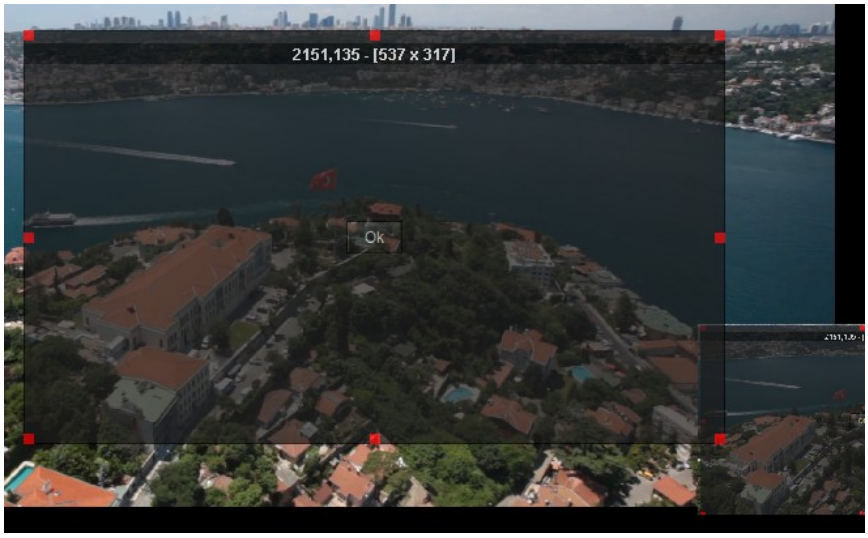
A call from Skype©, a web page, your desktop or work environment: with the plugin ScreenCapture, what you see on your PC is on air. The plugin ScreenCapture makes it possible to easily capture an area of your screen, which can accommodate for example a video captured from YouTube, a call from Facebook© or your favorite news site.
The plugin ScreenCapture has no limitations and can be used as a texture of an object (where possible) or as an image.



The ScreenCapture object is created with a size of 320x200 pixels with a capture area of coordinates X, Y (0,0).

The button shown in the figure, allows to choose the area to be captured and the size of the object. Alternatively, the fields at the side, allowing you to manually enter the data.

Through the settings area you can enable the HQ (High Quality) and choose the interval in milliseconds to capture. Another very important aspect to consider is the choice of an appropriate aspect ratio in according to the properties of the project or output video.

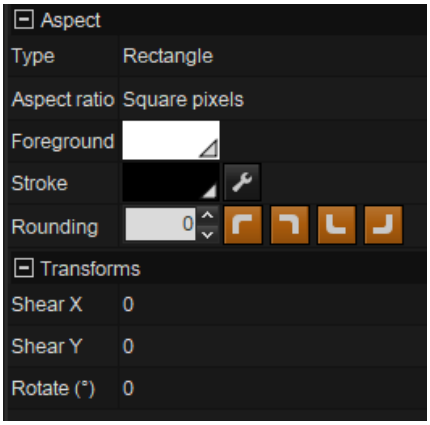


The capture of the area will be simultaneously clicking on the "OK" button.

From **Mode**, you can choose between "Auto" and "Screenshot". In "Screenshot" mode, by defining a capture area and clicking on you can create various screenshots that will be listed in the **Playlist** panel. From the latter, it will be sufficient to select the desired one or remove it by right-clicking.

6.17 Shape v1.0

The Shape plugin is one of the main components in the library developed by ClassX. Thanks to its flexibility, this component is used and called within applications in various forms.



The preset geometric shape is a rectangle white in color and borderless.
The first combo box in “Aspect”, that is “Type”, serves to select the type of graphics primitive (henceforth called the shape). Currently available are four types of shapes: **rectangle, ellipse**, **polygon** and **generic shape.**

Controls "**Aspect ratio**" are used to set the aspect ratio of pixels based on the output device, so as to correct any distortions introduced by the shape of the display.
The combo box lets you choose one of the most common presets, while the field on the right allows you to set the value of custom aspect ratio.

Notes: To better understand the aspect ratio, suppose you create a circle of size 300 x 300 pixels. The shape of the preview on PC monitor is perfect (square pixels), while on a video output to 16:9 (1.422) or 4:3 (1.067) circle takes the shape of an ellipse. The reason for this imperfection lies around in the aspect ratio that is set depending on the output device.

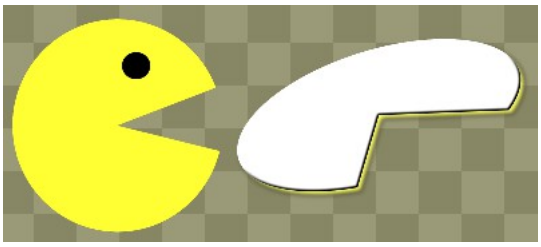
The “Transforms” section allows to edit the coordinates: the first two "**Shear X**", "**Shear Y**" allow you to change the tilt of the shape on the X and Y, respectively, from -60 ° to +60 °, the third "**Rotation**" you can change the shape of the Z-axis rotation from 0 ° to 359 °.

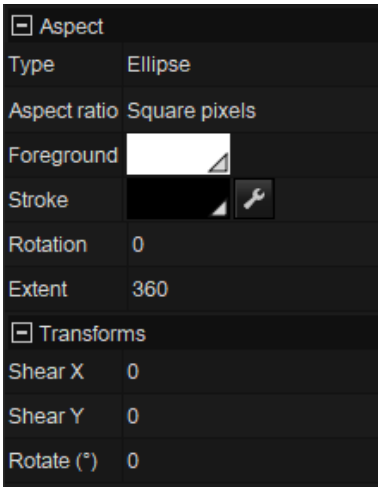
6.17.1 “Rectangle” Shape

To facilitate the creation of shape, the panel "Shape" was designed in order to configure itself according to the type of shape you selected. The "Rounding" field allows you to round the corners of the square. The higher the value, the larger the arc of rounding.

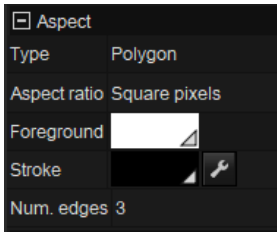
6.17.2 “Ellipse” Shape

Selecting the ellipse shape, two fields appear: "**Rotation**" and "**Extent**". The "Rotation" field allows you to rotate the shape on the Z axis from 0 ° to 360°, while the "Extent" field has the function of setting an opening angle on the shape so as to easily create "pieces" of a pie chart .



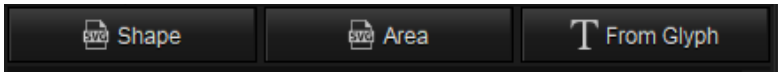


6.17.3 “Polygon” Shape

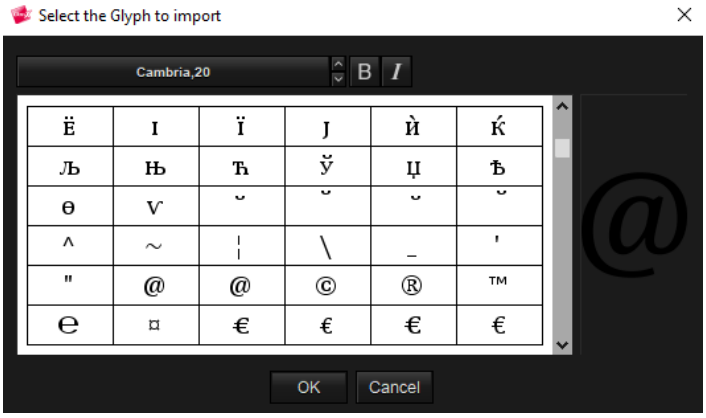


Selecting polygon shape, in “**Num.edges**” combobox, you can change the number of sides of your shape. The field is initially set to 3 (creating a triangle). By setting "5" in the field, it creates a pentagon, with "6" a hexagon, with "8" an octagon, etc.

6.17.4 “General path” Shape



“General path" shape allows the use of **.SVG** or **Glyph** resources. The latter, given the variety of fonts available, greatly amplifies the creativity of the user, providing symbols and images, vector completely. The import of the Glyph is not limited to just system fonts but, thanks to the use of drag and drop, you can drag the. TTF fonts directly in the preview of the characters and use it. The font isn't automatically installed on your system, but is redirected in real time and then lost at the end of the project (the object included in the project remains available for use).



As for the import of **.SVG** objects, you can choose whether to use the image area, or complete shape.
For all shapes, you can change the graphic appearance from the "Graphic Attributes" panel.



Note: Thanks to the elasticity of the characteristics of the object shape, you can create objects with particular shapes and sure effect. For example, you can create a circular text in this way:

- select the circle as the shape and set your preferred size;
- select "None" as the fill;
- select "Text" as the type of board;
- customize the border with a text as desired by selecting available cyclic;
- select "Gradient" as filler board;
- ended up here is a quick and effective "circle text"

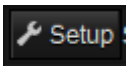


6.18 SharedFrame v.1.0

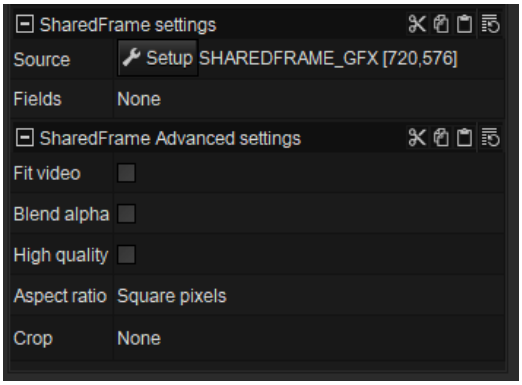
This plugin allows you to create video and graphics areas which external applications can share, access and write. SharedFrame is based on a particular technology, developed specifically by ClassX to enable cooperation and synchronization between multiple graphics applications. This technology, called SharedFrameHandler™ is available as a software developer's kit (SDK) for third party developers who want to integrate their applications with the platform ClassX.

The fields of application are numerous and range of SharedFrame interfacing to third-party software (eg programs can produce dynamic graphic that we want to integrate) to the connection between multiple applications ClassX cascade (e.g. in a plugin of LiveBoard) . In the official release we added a very simple example that allows you to play the video clips over a SharedFrame MediaPlayerClassic directly. The possibilities are really a lot.
One SharedFrame application establishes a connection between the "producer" of graphics and the "consumer" (the SharedFrame plugin is always a consumer). Although you can use the same SharedFrame between multiple producers and consumers, appropriate synchronization mechanisms prevent any overlap in which, for example, more producers write simultaneously.

6.18.1 Create a SharedFrame plugin



Each SharedFrame plugin is created or identified by a name and a reference given resolution. By default, the SharedFrame is created with the following settings:



SharedFrame Name: **SHAREDFRAME_GFX**
Size: **720x576** points

The name of the identifier SharedFrame is important because it allows applications to uniquely locate in memory. The name is case-sensitive, and therefore the names "foo" and "FOO" are considered different.

There are names reserved for SharedFrame that can be used by users for specific applications:

CSX_VIDEO reserved by SharedFrame Video Renderer by ClassX. Video is a virtual rendering device applications based on DirectShow (MediaPlayerClassic type) can use to view videos directly on a SharedFrame.

CSX_FRAMELINK reserved by SharedFrame FrameLink ClassX by drivers. With this driver you can create a sort of virtual slot between the video output of an application and a ClassX SharedFrame created a second application.

For visualization through **PiP** it is necessary to select the appropriate command in the "Preferences" → "Video" menu.

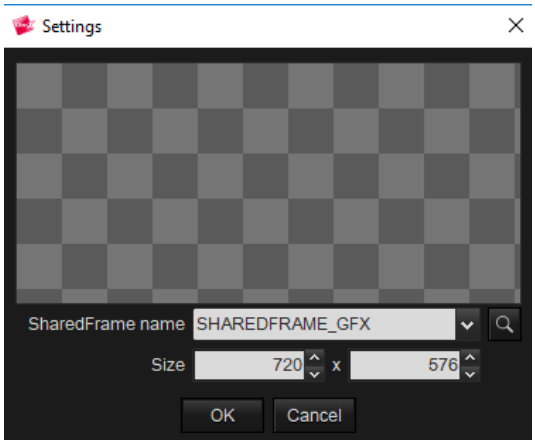
Warning: The dimensions are equally important in the definition of a SharedFrame: SharedFrame two can not exist with the same name but different resolutions.

The resolution of a SharedFrame can be set at will, but it is always advised not to exceed the full HD 1920x1080 resolutions in order to avoid possible problems related to excessive memory footprint system.

A specific function, connected to the LENS button allows you to automatically detect the resolution of an existing SharedFrame, such as one created by an application already running.

6.18.2 Use a SharedFrame plugin

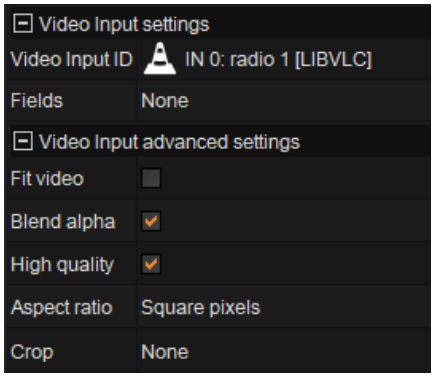
Frames received from an application by ClassX SharedFrame are treated like any other media, picture or graphic. The pixels that make up a SharedFrame are always in ARGB format, then with alpha channel. This lets you overlay it with any other plugin SharedFrame respecting the original transparencies.



The combination according to the alpha-channel transparency is controlled by the original "Compose alpha." With SGL disabled the Alpha channel management may require more computing time. For this reason, in the presence of sources that do not require transparency channel is best to disable this feature.

One SharedFrame can be resized at will enable the option "Fit Video." The quality of scaling is entrusted to the option "High Quality", which enables scaling using bilinear interpolation.

6.19 Video Input Source v.1.0



This plugin allows you to insert up to 10 different sources and to play them simultaneously without weighing down the machine or the output signal.

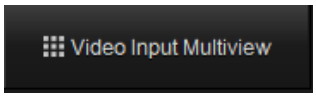
As we will explain later, it is possible to connect several sources from which to import the signal.

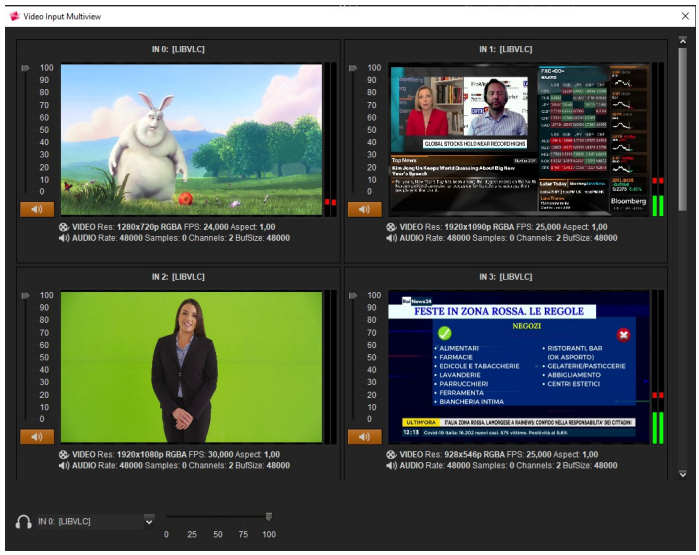
From the Settings panel, as for the other contents, you can choose the quality, the aspect ratio, the alpha compositing and adapt the size to the space chosen for playing.

In the "**Video Input Manager**" section it is possible to select the source ID.

6.19.1 Video Input Manager

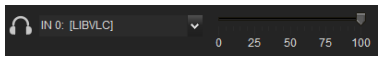
As already mentioned in the explanation for the preferences menu, the "Video Input Manager" tab allows you to manage and connect the various sources. By clicking on "**Video Input Multiview**" you have a global view of all the previews.

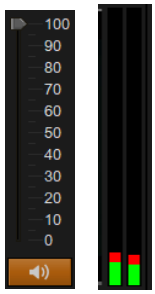




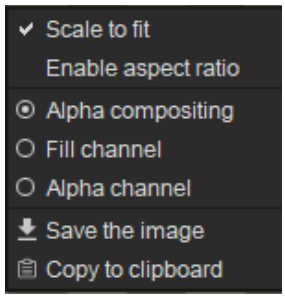
Each Preview, on the left, has its audio management for playout (**master volume**). This allows to uniform the volume of each source. For displaying the intensity of the audio frequencies, on the right there are two useful red and green vu-meters.

At the bottom of the Multiview, there is another bar for adjusting the volume. It allows the management of the **general volume** of the Video Input Multiview. From the drop-down menu, you can choose the preview panel on which to focus audio and adjust its volume. To deactivate this function, click on the symbol on the **toolbar** (refer to chapter 4.11).





By right-clicking over the area inside the preview, a small menu appears and you can select some actions that can be performed directly on each individual panel:



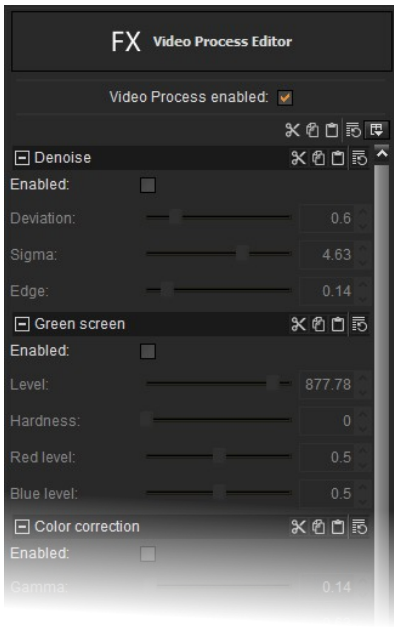
- **Scale to fit** and **Enable aspect ratio**: set the display proportion of the video source. By removing the setting “Scale to fit”, the content will be shown in its real size;
- **Alpha compositing**: displays the video signal in fill and key;
- **Fill channel**: displays only the “Fill” channel;
- **Alpha channel**: displays only the “Alpha channel”;
- **Save the image**: save the visible video frame at the time of saving in png format;
- **Copy to clipboard**: copy, to the "clipboard" of the operating system, the frame of the visible video at the time of saving.

6.19.1.1 The FX “Video Process Editor” panel



As already mentioned in chapter 4.5.7 on the Video Input Manager, by clicking on the “**FX**” button inside each single “**Video Input Monitor**” it is possible to open the “**Video Process Editor**”.
When the button “FX” is **orange** and at the top of the monitor, next to the source name, a small orange LED appears, it means that the Video Process Editor is enabled on that source.

The Video Input alone is already a complete and performing feature of **CBA** (ClassX Broadcast Application).
Anyway there are cases when the captured video may need to be further elaborated in order to create a perfect looking, broadcast production.
“Video Process Editor” was born **to correct** some imperfect luminance or chroma balancing or **to add** some special effects in real time to the incoming video, i.e. filters, masking, green screen, etc.
Thanks to a deep, clever use of the computer's GPU it is perfectly possible to process each captured Video Input frame at a very limited cost in terms of computational impact.



ClassX FX Video Process filters are organized as a sequence of video elaborations, applied (see list) on each frame.
Every filter is fully parameterized and can be operated in real time, for an immediate feedback.

The filters already have a suitable default setting for the most common applications. Of course it's possible to alter and fine-tune the settings according to your very-own production needs.

By checking the “Video Process Enabled” option it is possible to set multiple interesting effects in our video source. Each filter can be of course enabled or disabled individually. You can also enable multiple options at the same time.

Let's analyze each filter individually:

- **Denoise**: it's a very useful filter, indicated to remove unwanted noise from the incoming video (quite common when the video is coming from a low quality source and/or with a incorrect illumination).
  This filter can be easily used to "beautify" faces by "flattening" lines while keeping the original contrast and details of the image.

  It's possible to set three parameters:

  DEVIATION: the standard deviation from a correct color
  SIGMA: the sensitivity of the filter
  EDGE: the pixel-edge smoothing effect

- **Green screen**: it's a very popular filter aimed to remove the green screen background color of the incoming video. The Green screen is suited for special effects, virtual backgrounds and virtual sets.

  It's possible to set four parameters:

  KEY COLOR: the color to be filtered
  LEVEL: the sensitivity of the filter
  HARDNESS: the level of hardness of the alpha channel border computed by the filter
  DESPILL: level of elimination of any green border on the contrast areas of the filter
  SMOOTHNESS: level of pixel oversampling during filter calculation. It has the effect of proportionally eliminating any serrations present on the edge

The Green screen filter is able to filter any color, not just green.

- **Color correction**: as the name tells, this filter delivers real time color correction through the most-common, widely known params.

  It's possible to set four parameters:

  GAMMA: the gamma level
  BRIGHTNESS: the brightness level
  CONTRAST: the contrast level
  SATURATION: the saturation level
  HUESHIFT: the hue level

- **Mask**: video is generally shaped as a rectangular area. There are cases when you may want to cookie-cut this area or mask some parts. The mask filter just takes a GREYSCALE image and uses it as color-pass key. For each pixel in the mask: white is full source video color, black makes it completely transparent, gray makes it semi-transparent.

  It's possible to set three parameters:

  FILE NAME: the mask image file (any format supported, use greyscale images)
  GAIN: the gain of the mask, i.e. each pixel will be multiplied by this param in order to strengthen the mask effect
  OFFSET: the offset of the mask is used to fine-tune the mask and create a reverse of the mask itself

- **Pixelize**: due to popular demand, the pixelize effect comes from the eighties and basically renders the video pixels as "fat" pixels according to the efx params.

  Only one parameter:

  LEVEL:the pixelize level, i.e. the size of the pixel

- **Deinterlace**: in presence of interlaced video, scaling, rotation, transformation may introduce annoying visual flickers and artifacts. This filter will apply odd-even line deinterlacing in order to fix unwanted flickering. No other params.

- **Crop**: the crop filter allows you to crop the VideoInput via four sliders (pixel values).

  LEFT: crop from the left
  RIGHT: crop from the right
  TOP: crop from the top
  BOTTOM: crop from the bottom
  COLOR: sets the color (RGBA) outside the cropped area

- **Alpha premultiply**: video sources in Video Input are always processed as RGB+ALPHA. While the alpha channel information is normally set to full opaque, video coming from NDI, VLC, can have a valid alpha transparent information.
  In order to correctly overlay translucent video sources, we always suggest to use alpha-

premultiplication. No other params.

More info:
http://www.jhlabs.com/ip/blurring.html - https://ciechanow.ski/alpha-compositing/

- **Custom Process (GLSL)**: this function allows advanced users to develop custom filters.
The aim of the filter is to give users full access to video pixels and their coordinates by means of a specific GPU-specific programming language called GLSL.

GLSL is an industry standard shading language developed fron Khronos Group https://registry.khronos.org/OpenGL/index_gl.php
Its main aim is to give coders the ability to write gpu-level programs with very small effort and incredibly fast execution.
These gpu-level programs are generically referred as "Shaders".

The Custom Process filter is executed for each pixel of each videoinput (when enabled), automatically.
It supports 2 separate custom parameters (**value0** and **value1**, represented with sliders) where the user can modify the final effect.

The Custom Process filter comes with ready-to-use sample shaders like flip, mirror, old-film, sphere, stars, etc..
Advanced users may find very easy to port existing GLSL shaders like those from https://glslsandbox.com/ or https://www.shadertoy.com/ into VideoInput.

HOW TO USE

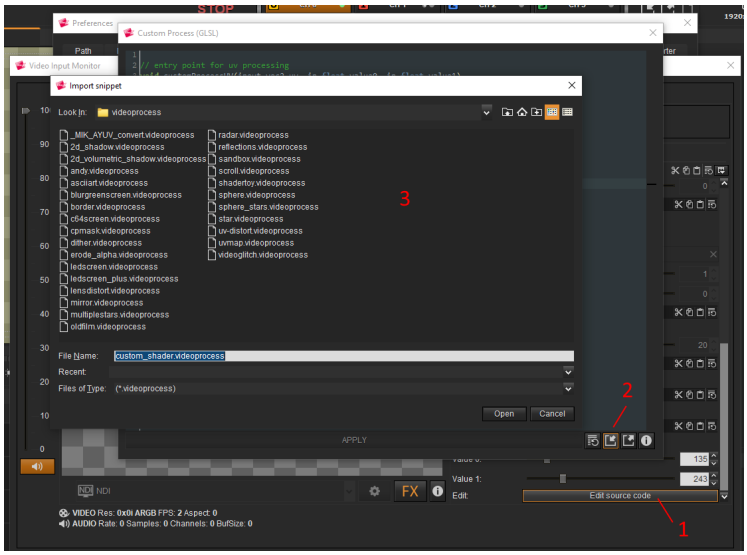
To make things easier to the users, the Custom Process filter has a builtin context-sensitive custom code editor with error-reporting, jump-to-error, syntax highlight and all coder-specific features.

Code modifications are completely interactive and they're applied in realtime to the video input, allowing quick and easy effect prototyping and development.

The shader code can be imported into the editor or exported both as snippet or entirely.

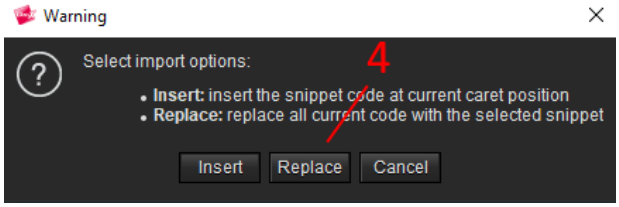
The shader code can be reset to the ClassX defaults, just in case we want to start from scratch.
The default code is just a do-nothing shader with comments and entry-points where coders will want to add their custom code.

The editor is completed with a info window where users will find a quick and easy guide in order to start and enjoy the power of Custom Process shaders.



It's possible to open, use or edit some default effects provided by ClassX just by follow these steps:

1. Click on **Edit source code**
2. Click on **Import**
3. Choose the effect
4. Click on **Replace**



6.19.2 Video input sources types

In the next paragraphs we will introduce all the types of video input sources that you can use on Liveboard.

Some video input types have a "**FIELD MODE**" option. This option is useful when it is not clear what field mode the video input resource you want to play has or when you need to set a specific field mode.

This setting allows you to select between:

- **FROM INPUT**: it maintains the original field mode of the video input;
- **ODD FIRST**: it orders the fields so that the first one is odd and therefore has the order "odd-even";
- **EVEN FIRST**: it orders the fields so that the first one is even and therefore has the order "even-odd";
- **DEINTERLACE**: it deinterlaces the video input source;
- **OFF**: it turns off the fields and the video input source becomes progressive.

Additional common options for some video inputs include:

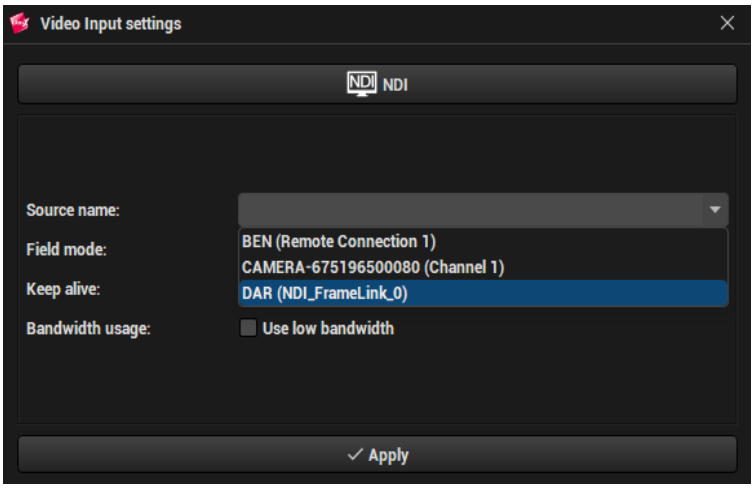
- **Keep Alive**: When this option is enabled, the video input remains active and does not stop even if it is not currently playing.
- **Synchronize to Output**: When enabled, the video input's source frame rate is aligned with the output frame rate set in LiveBoard.

6.19.2.1 NDI© source

NDI© (Network Device Interface) is an opensource protocol developed by **NewTek©** to allow you to share a video signal on a local network. NDI allows several video systems to identify and communicate with each other over IP and to transmit them in high quality with video and audio in real time. This protocol can take advantage of any video device connected to the network.
The NDI can operate bidirectionally on a local network, with many video streams on a shared connection. It supports all resolutions including 4K with multiple different audio channels. The protocol also includes tools that implement video access rights, grouping, bidirectional metadata and IP commands.



Any external NDI source can be connected in the "Video Input Source" plugin. After selecting the source from the drop-down menu, clicking on the gear will open a window that will allow you to select the desired source among those available. After selecting it, clicking on "Apply" will appear on the corresponding panel.



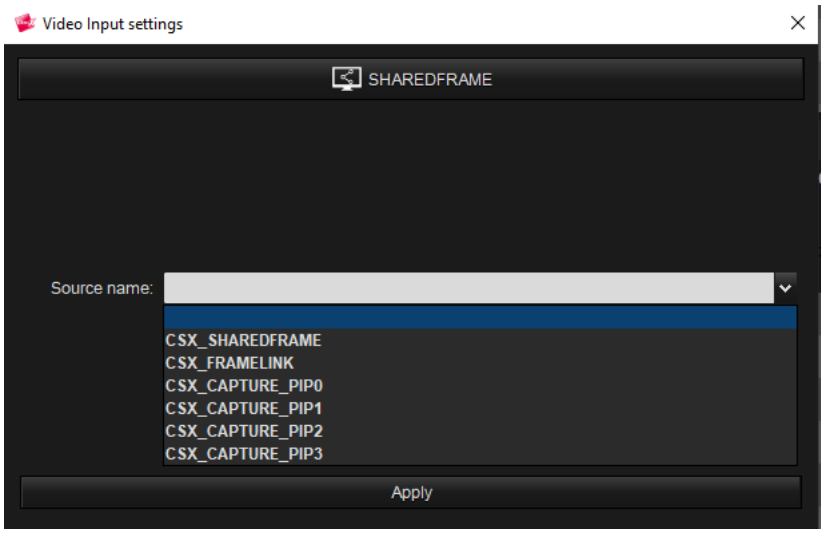
If the "**Bandwidth Usage**" option is enabled, the bandwidth used for the NDI stream will be reduced, resulting in lower signal quality.

6.19.2.2 The SharedFrame

SharedFrame is a virtual video output between multiple applications. Video Input allows you to create video and graphic zones that external applications can share, access and write. SharedFrame is based on a particular technology, developed specifically by ClassX to allow cooperation and synchronization between multiple graphic applications. This technology, called SharedFrameHandler ™, is also available in the form of software developer's kit (SDK) for third-party developers who want to integrate their applications with the ClassX platform.



After selecting SharedFrame from the drop-down menu, clicking on the gear will open a window that will allow you to select the desired source among those available. After selecting it, clicking on "Apply" will appear on the corresponding panel.



The **identification name** of the SharedFrame is very important as it allows applications to uniquely identify it in memory. The name is case-sensitive and therefore the names "pippo" and "PIPPO" are considered different.

There are reserved names for SharedFrame that can be used by users for specific applications:

CSX_FRAMELINK reserved by SharedFrame FrameLink driver by ClassX. Through this driver it is possible to create a sort of virtual connector between the video output of a ClassX application and a SharedFrame created in a second application.

To view via **PIP**, select the appropriate command in the "Preferences" → "Video" menu.



Warning: The dimensions are equally important in the definition of a SharedFrame: SharedFrame two can not exist with the same name but different resolutions.

The resolution of a SharedFrame can be set as desired, but it is always recommended not to exceed the full HD 1920x1080 resolutions in order not to run into possible problems related to the excessive use of system memory.

Frames received from a SharedFrame are treated by ClassX applications like any other media, image or graphic. The pixels that make up a SharedFrame are always in ARGB format, therefore with alpha channel. This allows the SharedFrame to be superimposed on any other plugin respecting the original transparencies.

The alpha channel combination according to the original transparency is controlled by the "Compose alpha" option. With SGL disabled, management of the Alpha channel may take longer to calculate. For this reason, in the presence of sources that do not require the transparency channel, it is advisable to disable this functionality.

A SharedFrame can be resized as you want by enabling the "Fit video" option. The scaling quality depends on the "High quality" option, which enables scaling via bilinear interpolation.

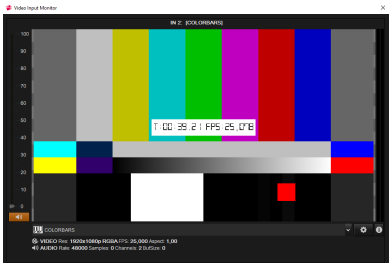
6.19.2.3 Connections with SDI video cards

On LiveBoard you can also connect audio/video sources via SDI connections.

Specifically, it supports: BlackMagic® DeckLink, Bluefish444 and AJA.

Video input resolution and color space are automatically detected. So, just select the input card but be careful: it could already be used as an output device.

6.19.2.4 Colorbars



This video input can be used to create simple test EBU color bars. It also shows the uptime and the current FPS of the device. It shows a small square that slowly bouncing from side to side, beeps every time it hits the edges of the screen.

6.19.2.5 Spout

Spout is an **ultra-fast**, **real-time video frame** sharing system for Microsoft Windows that allows applications to share **OpenGL textures**.
Spout technology allows you to share video signals with other applications within the same workstation. Audio is not currently supported.

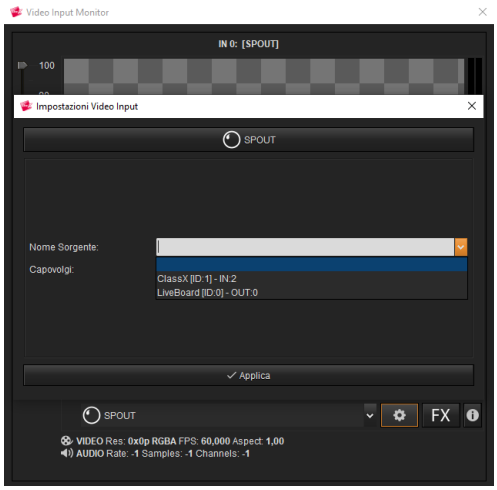
This technology is used both in **input** and **output**.

It is possible to play any video input on the Spout surface by enabling the appropriate flag (**Play to Spout**). The Spout sender name is automatically generated and it is possible to view it using the tooltip and copy it with a right-click "**Copy Spout sender name**". Otherwise it is simply possible to find it in the list among the available Spout input resources.

For the **Input** part, simply select the **Spout resource** in the list of available resources.



Once the Spout resource has been selected, by clicking on the button with the gear, it is necessary to select the name of the external video to be displayed in the input.



Through Spout it is possible to play for example a video resource coming from another Video Input. In this way it is possible to have two Video Inputs of the same resource **without burdening the CPU**. Even the SDI resources can be duplicated.
It can be useful, for example, if it is necessary to add effects from the "**FX**" panel on an input video in SDI but at the same time also have the original video available without any effect and therefore usable for another graphic contribution.

The "**Flip vertically**" option allows you to flip the display of the input video.

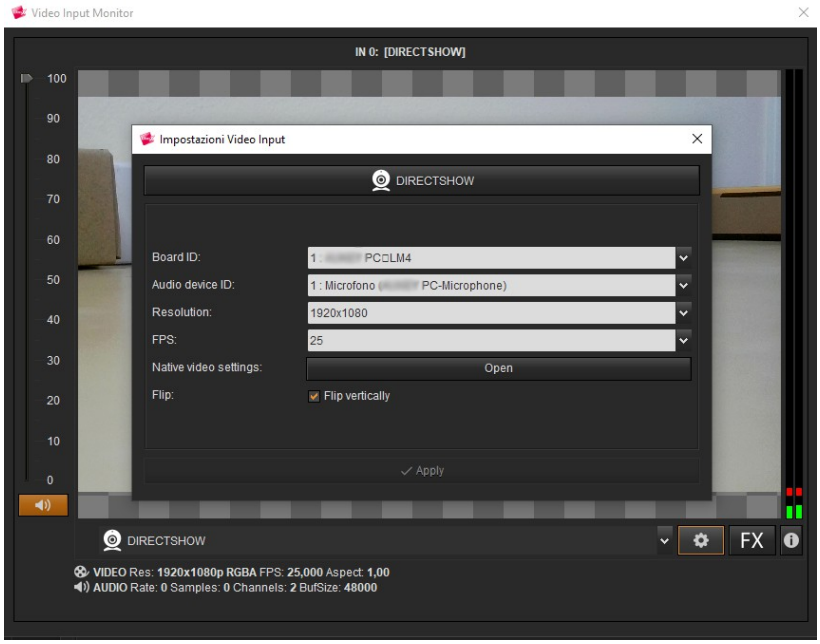
For further information on Spout technology, please consult the official website https://spout.zeal.co/

6.19.2.6 Directshow

Directshow is a Video Input Source that allows you to import and play video and audio on LiveBoard directly from Directshow devices connected to the workstation, such as a webcam and a microphone.

Its operation is very simple: once the devices are connected and correctly installed following the procedures required by Windows, simply set the parameters as required.

- **Board ID**: from the drop-down menu, select the video device to play on;
- **Audio device ID**: select the audio device to play on;
- **Resolution**: select the desired resolution while paying attention to the maximum resolution of the connected video device;
- **FPS**: source fps to play correctly on LiveBoard;
- **Native video settings**: if the selected Directshow device contains native configuration settings from the manufacturer, it is possible to inherit them directly on Video Input by clicking on "**Open**";
- **Flip**: Check the option if the video signal is not displayed in the correct direction.



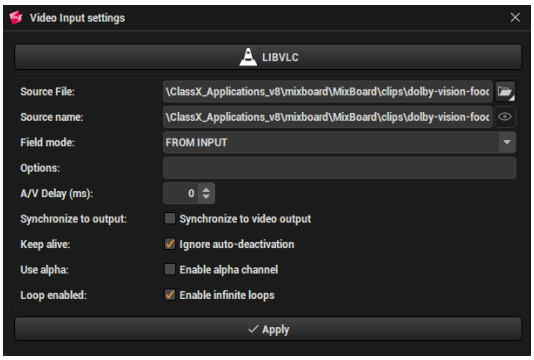
Warning: The VideoInput device is based on **Microsoft DirectShow API**, a media-streaming architecture for Microsoft Windows. This VideoInput has proven to be working very well under normal conditions. However due to the complex nature of media handling, codecs and streaming in general, you may encounter unexpected issues. Please remember that this VideoInput depends on third-part driver implementations. Therefore we cannot be held responsible for any problem that may arise from the use of Directshow video input.
If you experience malfunctions or crashes when using a specific capture device, please consider to install a better and updated DirectShow driver from the manufacturer's official website. **Use it at your own risk**.

6.19.2.7 LibVLC

This option is based on libVLC, the developer library behind the famous VLC player.

Please read the libVLC disclaimer on the official website for more information on features and limitations.

This Video Input has proven to work very well under normal conditions. However, due to the complex nature of media management, codecs and streaming in general, unexpected problems may occur. Therefore, we cannot be responsible for any type of problem that may arise with the use of libVLCVideo Input. ***Use it at your own risk.***



LibVLC can play with different video sources/resources. It is possible to both view local resources and streaming resources from the internet.



By clicking on the "**Source File**" icon, a file chooser will open where you can select a video file directly from your local archive or select an **m3u** playlist previously saved locally (https://en.wikipedia. org/wiki/M3U).
In the next chapter we will explain in detail how to use the m3u playlist.

Before playing a stream content, it is recommended to initialize it in the "Preview" panel. This is because the contents of VLC do not remain active in the background; they automatically deactivate after 60 sec of inactivity.

M3u playlist management

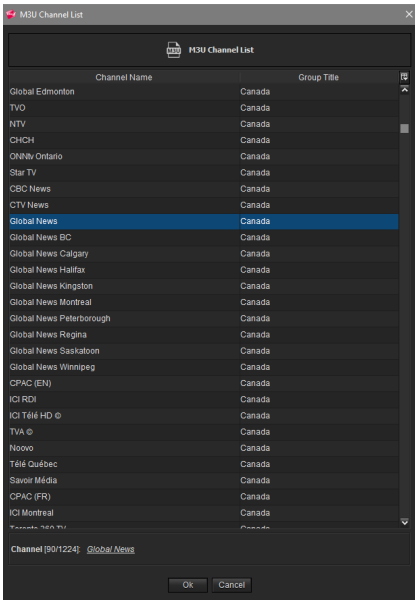
In the “Source file” area it is possible to insert a link or an m3u playlist file/URL. For this operation, drag & drop is also supported.

Once the m3u file has been inserted, the button with the eye will be enabled next to the "Source Name" text area. By clicking on it, the list of channels contained in the file will automatically be available. With a double-click on the chosen channel, it will be played inside the VideoInput.

This feature is extremely useful for quickly accessing pre-compiled lists of streaming channels and playlists in general.

Note: Since initializing a streaming channel is a background operation, playback may take a few moments.

M3U channel list example:



Advanced guide on using LIBVLC options

The options that can be set for the LIBVLC Video Input source allow you to use additional functions.

To have a complete view of all those available, you need to connect to the site:
https://wiki.videolan.org/VLC_command-line_help/

With the following guide, we give a brief explanation of the most useful functions for broadcast purpose.
i.e.: To facilitate understanding, a “Question / Answer” mode will be used.

Audio and video appear to be out of sync. Lip sync is missing. How to delay / unsynchronize the audio channel?

Among the Options use: `:audio-desync=<millis>`

<millis> is the desynchronization time in milliseconds (can be negative to reverse the A / V desynchronization)

Example: `:audio-desync=-200`

With a value of -200, the audio will be played 200ms earlier than the video.

Using a low quality internet connection, sometimes the stream stops working. Is there any way to force an automatic reconnect?

The Video Input will always try to automatically reconnect to the stream source.
You can force re-connection with: `:http-reconnect=65535`

This option will enable a best-effort reconnection. Also remember that unstable source streams may also cause reconnection problems.

Is there a way to loop a media stream (i.e. streaming media files)?

Yes it's very simple and just use the option: :input-repeat=<nloops>

<nloops> is the number of loops to execute. To have an infinite loop, you need to use 65535.

Example: `:input-repeat=65535`

It will loop the media forever.

Can I play a local media file instead of a network-based stream?

Yes, indeed. All you have to do is to place the path of the file into the "Source name" field.

Example:
C:\media\video.mp4

Can I play YouTube videos?

Yes, you can play YouTube as well as Twitch, Vimeo and other streaming sources, i.e. Anyway you cannot simply copy-paste the videoclip link.
Normally you need to use some simple browser addons in order to grab the REAL stream link.

There are lots of free addons for any browser. However, they must be installed.
On FireFox we selected:

- Video DownloadHelper - http://www.downloadhelper.net/
- The Stream Detector - https://github.com/rowrawer/stream-detector

Example 1 – to grab YouTube (also valid for Twitch, Vimeo, etc)

- open your YouTube video in your browser (i.e. https://www.youtube.com/watch?v=fn3KWM1kuAw)
- select your preferred resolution in the YouTube settings (i.e. 1080p HD)
- click on Video DownloadHelper icon and go to the video details
- locate the "videoUrl" field and copy the (long) link
- paste the link inside “Source Name” in LIBVLC of Video Input
- click on “Apply” and the video will start
- N.B.: the links are temporary and may become invalid after some time!

Example 2 – to grab Twitch streams (also valid for other streaming platforms)

- open your Twitch video in your browser
- The Stream Detector will start to notify you that there are streams playing in the page
- click on The Stream Detector icon and select the stream link you want to grab
- paste the link inside “Source Name” in LIBVLC of Video Input
- click on “Apply” and the video will start
- N.B.: the links are temporary and may become invalid after some time!

Example 3 – to use VLC to detect the YouTube stream (the desperate case of LIVE YouTube events)

- copy the YouTube live video link from the browser
- start VLC and open a network URL (hit CTRL-N)
- paste the link of the live video and click on "Play"
- goto Media Information (CTRL + I)
- copy the link of the "Location" at the bottom of the information window
- paste the link inside “Source Name” in LIBVLC of Video Input
- click on “Apply” and the video will start

Can I grab/play streaming RADIO/TV channels?

Yes, definitely. Just use The Stream Detector.

- open the streaming TV web page in your browser (i.e. https://www3.nhk.or.jp/nhkworld/en/tv/sumo/)
- The Stream Detector will start to notify you that there are streams playing in the page
- Click on The Stream Detector icon and select the stream link you want to grab (it should appear like ".m3u8" file)
- paste the link inside “Source Name” in LIBVLC of Video Input
- click on “Apply” and the video will start
- N.B.: the links are temporary and may become invalid after some time!

6.19.2.8 Chromium

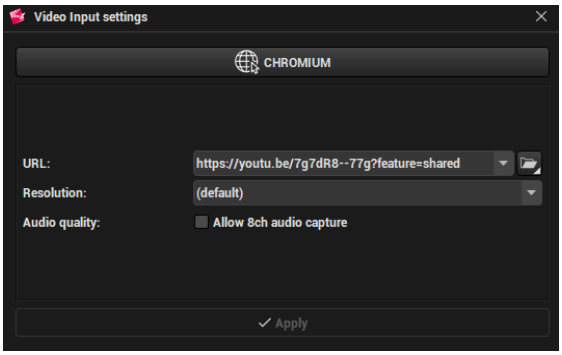
Chromium is a Video Input based on CEF (Chromium Embedded Framework).
See: https://github.com/chromiumembedded/cef

Chromium Video Input allows you to create an A/V source based on a specific URL or web page, using an HTML5 rendering engine with graphics acceleration support.

With Chromium you can easily create media players, include YouTube channels, ZOOM/Teams calls or render HTML5 pages with speed and flexibility.

This Video Input is a real window on the web. It supports real-time user interaction, audio and automatic synchronization with the output video device, for exceptionally smooth playout.

To open a URL, you need to insert it into the "URL" area and then click on "**Apply**" at the bottom.



Settings panel parameters:

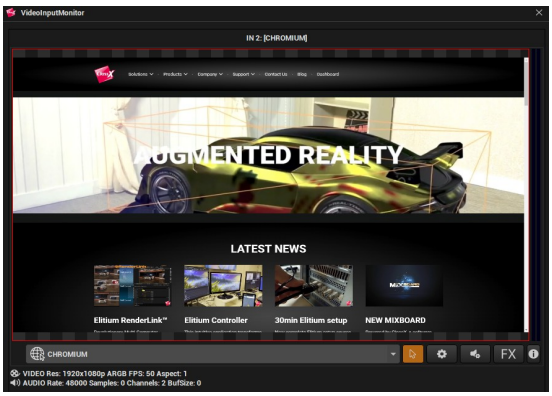
- **URL**: Load the web page from the specified address. Drag&drop is supported.
- **Resolution**: Set the size of the web page.
- **Audio quality:** you can enable the capture of 8 audio channels.

Direct interaction with Chromium



To interact within Chromium Video Input like a normal browser, first of all you need to click on the button representing the mouse arrow which will automatically turn orange.

Once enabled, the preview panel will be delimited by a red rectangle.
Like a normal web browser, interaction occurs through the use of a mouse and keyboard.
Using this function it is possible to navigate within web pages and select specific video content.



6.19.2.9 Dummy

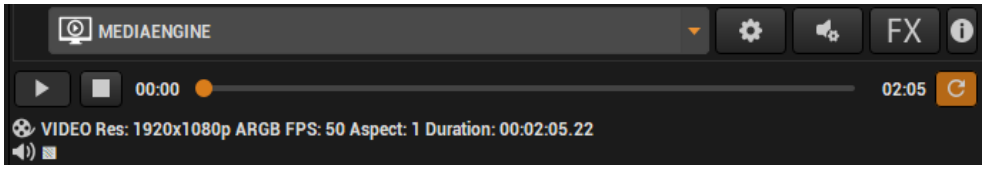
Dummy is an “empty” Video Input that generates a transparent video in HD resolution.

This Video Input takes advantage of the hardware acceleration of the graphics card and is generally used for the creation of customized **Shader** sources (see **GLSL** shader editor **FX panel** section).

Dummy Video Input automatically synchronizes with the framerate of the output device.

6.19.2.10 MediaEngine

MediaEngine is a tool designed for efficient video playback, capable of playing a wide array of hardware-accelerated video formats, accessing media files, URLs and live content.



MediaEngine returns the resolution, framerate and duration information of the video.

 This feature enables looping on the uploaded video.

When a MediaEngine video input is assigned to the **program** or **preview**, you can manually control its playback/Play, Pause or Stop directly from the input management panel or from the MediaEngine controller area.
A timeline slider is also available, allowing you to **scrub** through the video timeline and choose a specific point from which to trigger playback actions.

Additionally, you can define the video’s automatic behavior when it is played on the **program** or switched between **preview** and **program** with the “Video Input Switcher” (Par.6.1).
By clicking the gear icon to access the settings panel, you can choose from a range of actions to be automatically triggered in these scenarios.

To set up these automations, you need to configure two options: **Start Transition Action** and **End Transition Action**.



- **Start Transition Action:** You can choose between *PLAY* and *NOTHING*. Selecting *PLAY* will automatically start the video when it is assigned to the **program** output.

- **End Transition Action**: You can choose between *PAUSE*, *STOP* and *NOTHING*. This determines whether the video should be paused or stopped automatically when it is removed from the **program**.

In both cases, if *NOTHING* is selected, video playback control (PLAY/STOP/PAUSE) remains entirely manual.

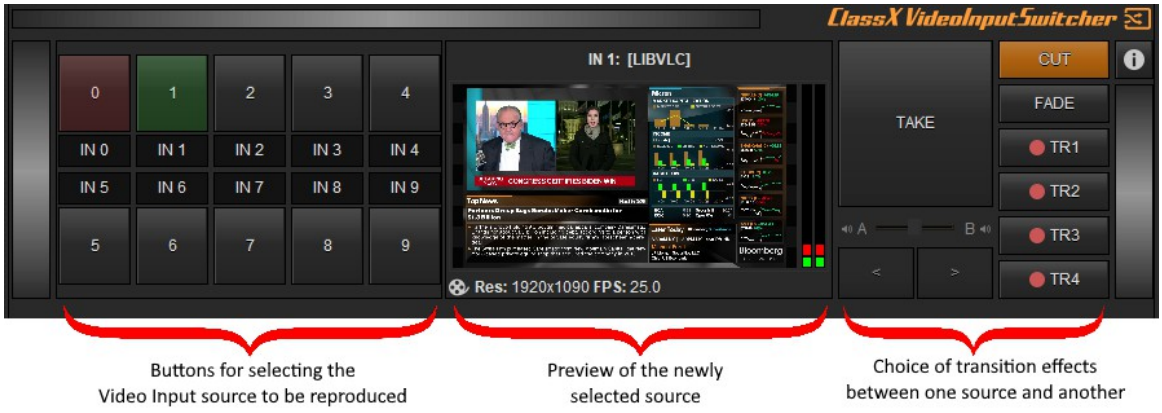
6.20 Video Input Switcher v.1.0

Video Input Switcher (VIS) is a plugin that allows you to have a real video mixer inside LiveBoard. It was created with the purpose of switching from one Video Input source to another with the full help of effects and transitions. The two video sources are called Video A and Video B or more simply A and B.

In the “Workspace” menu it is possible to select the “**Switcher**” workspace, recommended for an optimal experience for using the VIS.

6.20.1 The interface

As shown in the image below, it is possible to divide the interface of the Video Input Switcher into 3 macro-parts.



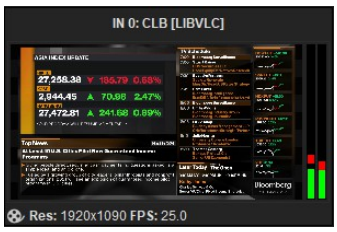
The 5 x 2 button matrix corresponds to the 10 sources set on **Video Input Manager** in the “Preferences” menu.



Each button can take on 2 colors in addition to the default one:

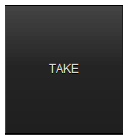
- **Red**: source that is or will be on air (**Video A**)
- **Green**: next source with which the transition will be performed (**Video B**)

The label under each button indicates the name of the source set on the Video Input Manager.



By selecting a source on the matrix, it is possible **to preview** it. The preview panel indicates at the top the **ID**, **name** and **type of source** (NDI, LIBVLC, DECKLINK, etc.) and, at the bottom, the **resolution** and **FPS** of the same. There is also a **vu-meter** that allows you to view the audio peaks of the selected source.

To make the transition from **Video A** to **Video B**, simply select the Video source from the buttons in the matrix and then press the "TAKE" button.

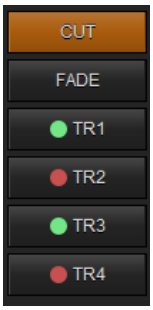


There are three default TAKE Modes:

- **CUT**: switch A / V from Video A to Video B direct with no effect
- **FADE**: switch A / V from Video A to Video B with a fade and is configurable from the Settings panel of the VIS



- **TR**: makes a transition from Video A to Video B with 2D / 3D effects based on graphic templates and created with ClassX editors. There are some default ones and released with the distribution, while others can be easily created by following the guide below.



Up to 4 distinct transitions can be imported. Just right-click on the transition button to load a graphic template.

The "Red Led" indicates a loading error or no template loaded, while the Green Led indicates that the transition has been loaded correctly.
By placing the mouse on the transition button just loaded, it is possible to display a "tooltip" with the path of the loaded template.

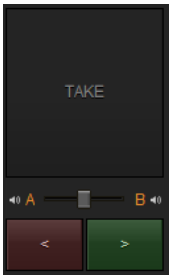
If the transition contains FrameMarker, the audio will be managed by the appropriate slider, otherwise an audio fade will be automatically applied to the source change.

6.20.2 "Dual video" display of A and B



In addition to the classic transitions between A and B in which Video B entirely replaces Video A at full screen during broadcast, it may happen that you need to keep for example A and B on video at the same time. To do this, just add a **FrameMarker** inside the transition template.

The transition enters in "paused state" whenever it encounters a FrameMarker in the transition template. Once the pause status is reached, the **audio** and **video control panel**, under the TAKE button, will automatically be enabled.



Audio control:

By sliding towards A, source B will be muted and vice versa. Clicking with the right mouse on the cursor will reset the set volumes and the cursor will return to the central position.

Video control:

- **Green button**: the forward transition to source B continues
- **Red button**: the transition back to source A continues

Shortcuts:
- **CTRL + NumPad:** to switch to Video B without pressing the TAKE button
- **NumPad:** to select the Video B source
- **Enter:** to press the TAKE button

6.20.3 Small guide on using Video Input Switcher

- Set the video sources in **Video Input Manager** and give them a name
- Insert the VIS plugin in the LiveBoard grid
- Upload transition templates to VIS (if needed)
- Select Video A and Video B on the VIS matrix
- Play VIS in the LiveBoard grid
- Select the effect to use for the transition (e.g. FADE or TR1)
- Press the TAKE button

Warning: be sure to enable **scaling** on Video Input from Video Input Manager if you are using video sources with different resolutions.

From the "**Video Input Multiview**" panel it is possible to view a small preview of all the Video Input sources selected through the Video Input Manager. For more information, see the chapter dedicated to the Video Input Manager.

6.20.4 How to create a transition for VIS with template editors

Requirements for a correct creation of the transition templates:

- The template can have any length and timeline with any level of complexity.
- The template should have the same project resolution as the LiveBoard output resolution.
- There is no limit to the number of objects that can be used in the template, except that of the common sense. We can therefore create templates enriched with animations of transition, texts, backgrounds and images.
- Two Video Input objects MUST be inserted and renamed exactly as Video_A and Video_B and these two objects MUST be set as DATASOURCE of MANUAL type.
- It is possible to use multiple copies of Video_A and Video_B in the same template. This allows you to create reflection and/or mosaic effects, etc. The use of copies of the input has no significant impact on VIS's performance during airing.
- It is possible to use without limitation any 2D/3D movement, transformation, scaling, rotation, mapping and texturing on 2D, 3D objects of any nature.

Warning: remember **NOT** to use Filters, Attributes, Masks on Video Inputs and any other animated object in the template, if you want to avoid overloading the graphics engine and thus taking the risk drop frames during the playout.

6.20.4.1 Software to be used for creating the templates

We recommend the use of **CastaliaCG** and or **CoralCG** due to their nature of vector graphic engines for VIS transitions. In fact, only CastaliaCG and CoralCG templates guarantee the perfect support for scaling, shifting, aspect ratio correction and LiveBoard Effects. (It is possible to use MoreCG but **only** if we plan to use VIS in full screen, without scaling and shifting).

6.20.4.2 The Timeline of a transition template

The structure of a generic transition timeline is as follows: the transition starts with Video_A full screen and ends with Video_B full screen. For a correct transition, it is important to add at least 10 initial frames paused on Video_A and 10 final frames paused on Video_B. Video Input Switcher supports FrameMarkers.

Back and forth and on a transition and “versus”

If a “trigger and pause” **FrameMarker** is inserted, the execution of the transition is paused so that the operator can decide whether to continue or return back to the previous marker with the dedicated arrows in the Video Input Switcher user interface. This feature allows you to easily create "A versus B" - type scenarios, where Video_A and Video_B are simultaneously present on the video.

In order to avoid the typical field-flickering of the stop of objects in movement with interlaced video, it is recommended to insert the “trigger and pause” FrameMarker into a minimum of 10-frame pause section on the timeline.

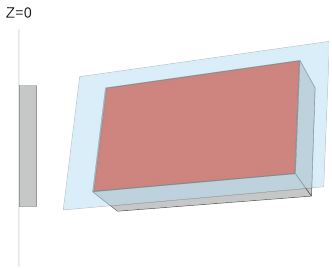
For a correct audio fade it is important to apply the right values to the keyframes; 100% volume on the first keyframe of Video_A and 100% volume on the last keyframe of Video_B. The intermediate keyframes will be modulated according to the logic of the movements and the effect we want to obtain.

6.20.4.3 Notes on 3D transitions made with CoralCG

CoralCG places its objects in a three-dimensional space. Objects can have thickness and be rendered with additional lighting effects, materials and textures, according to the settings that we choose.

In order to avoid visual differences in the transition between CUT / FADE to TRANS, it is important that the Video_A and Video_B Video Input objects respect the following settings in CoralCG:

- Light **NOT** enabled
- Default material
- No additional weaving
- Default setting of colour and transparency on the starting keyframe of Video_A and last keyframe of Video_B.
- Wapping on planar rectangular objects that respect the original size (e.g. 1920x1080) of the output resolution.
- If you use extruded objects, remember to position the front of the object on the coordinate **Z = 0**, corresponding "ideal" plane of the screen.

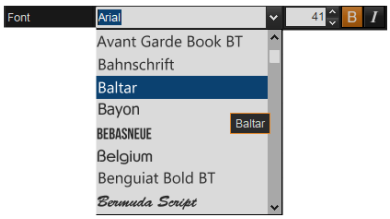


6.21 The functions on the text and the "Text-area"

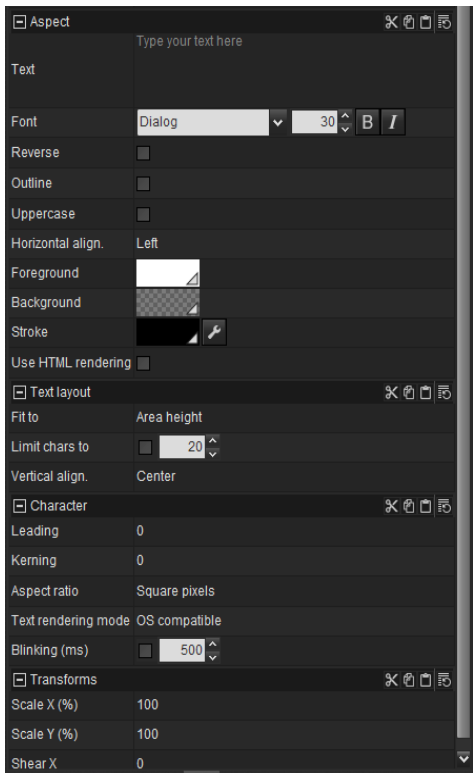
The possibility to edit the Text is one of the main components contained in the library developed by ClassX. The text customization panel offers all the tools necessary for editing each individual property. Thanks to its flexibility, from the **Settings panel**, this function is used and called up within plugins in various forms.

Clicking on **font** from the Setting panel, you open the window "**FontChooser**" from which you can select a new font to be attributed to the text.
In the FontChooser list the fonts are already rendered according to their style; this feature makes it easier and faster to select the required font.
To the right of FontChooser there are two buttons that allow you to set the font style (**Bold** - *Italic*) and the field that allows you to customize the font size with two arrow-shaped icons which permit to increase or decrease the actual size of the font.

Note: To import a new font into the ClassX software, it must be installed in the operating system "as administrator", ie by selecting the "**Run for all users**" option using the standard Windows procedure. When installing a new font, all ClassX software must be closed.



Apart from the now-classic controls for Bold and Italic text object allows you to create graphical text with the following styles:



- **Reverse** reverses the background with the body of the text, making the body transparent.
  Outline creates a border around the text, making the body transparent.
- **Uppercase** forces all characters to uppercase text.
- **Horizontal align.** respectively left, center, right, justified) of the various lines that make up the text.
- **Fit to** allows you to set how the text will fit in the area: area and area height. By defining the height and width of a specific area, any text regardless of its length will remain contained in the selected area.
- **Limit characters** by checking this option and defining the maximum number of characters to be displayed, regardless of the length of the text, the text will be truncated as soon as the number of characters defined is reached.
- **Vertical Align** allows you to set the alignment of the text vertically in top, middle or bottom.
- **Pixel Aspect ratio** which depends on the output device and allows you to correct any distortions. The combo box lets you choose one of the most common presets, while the field on the right allows you to set the value of custom aspect ratio.
- **Leading** line spacing of the text.
- **Kerning** horizontal spacing of the text.
- **Blinking (ms)** enables the flashing of the text and to determine the display time in milliseconds. This feature is not available in all applications.

Warning: For the correct reading of some fonts with special characters, it is preferable to select "**ClassX default**" as **rendering mode** and to save any external DataSource with **UTF-8** encoding.

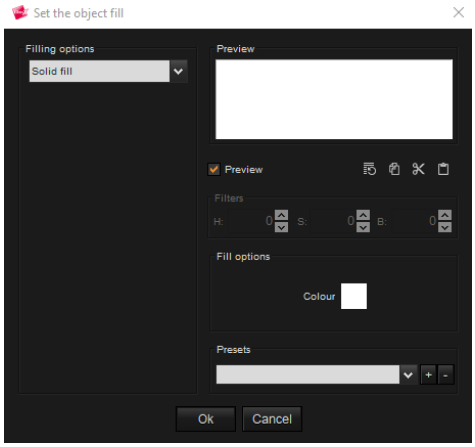
The "**Transformations**" section contains a series of sliders grouped by functionality:

- **Scale X** can "rescale" the text width.
- **Scale Y** can "rescale" the text height.
- **Shear X** allows you to tilt the text horizontally.

There are also three other functions which permit to customize the **foreground**, **background** and **stroke.** These functions are described in the next paragraph.

> **Notes:** To better understand the aspect ratio, suppose you create a circle of size 300 x 300 pixels. The shape of the preview on PC monitor is perfect (square pixels), while on a video output to 16:9 (1.422) or 4:3 (1.067) circle takes the shape of an ellipse. The reason for this imperfection lies around the aspect ratio that is set depending on the output device.

6.21.1 Filling of an object



In the text editing panel, colored boxes are visible for editing and displaying the fill of the text and its border. By clicking on it a window will appear in which it is possible to edit the type of filling desired.
This window is divided into several panels:

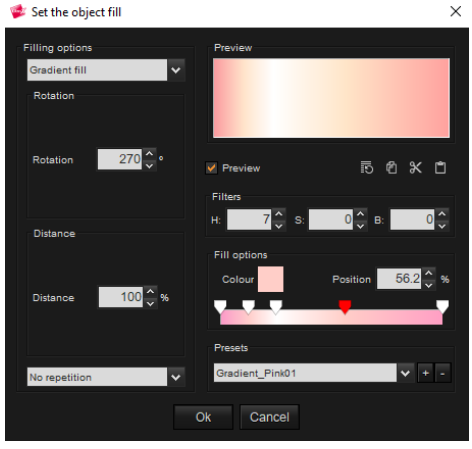
- **Filling options:** allows you to set the fill type (none, solid, gradient, radial) and its parameters
- **Preview**: Displays the selected fill.
- **Filters HSB**: Select and customize with filters
- **Fill options**: you can set depending on the type of coverage selected colors.
- **Presets**: allows the user to choose the filling from a list of presets.

6.21.1.1 Solid Filling

The default fill for an object type is solid, composed of a single uniform color. Through the customization panel you can edit that color: clicking on the button will show the **color-chooser** from which you can set the desired color and its opacity.
The editing of the solid fill color can be made directly, without the use of the panel for editing the fill. By clicking on the triangle located on the bottom of the button (see picture) or by clicking with the right mouse button on the color , **color-chooser** will appear directly for the desired color setting.



6.21.1.2 Filling and radial gradient



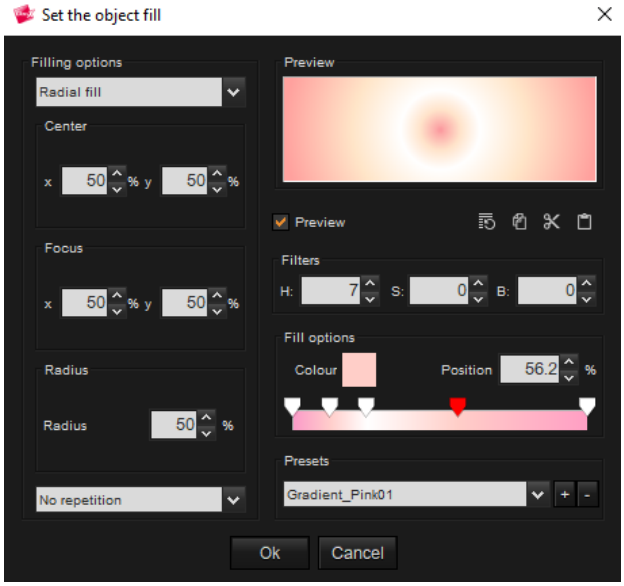
A gradient fill is a fill that has a progression between two or more colors and adds depth to the object. There are two types of gradient fills: Gradient, Radial. The passage of the gradient fill (or linear) occurs along a straight line through the object. The radial fill to proceed in concentric circles starting from the center of the object.

For both the customization panel, you can specify the attributes of the gradient fill. As can be seen in the figure for each color of the gradient is associated with a triangle, in particular, the selected color is indicated by a red triangle. For each component of the gradient you can follow these steps:

- editing of color through the **color-chooser**.
- setting of the position within the gradient through the appropriate component or by making a simple "**drag and drop**" the hot spot indicated by the mouse.
- insertion of a new color with a simple double-click the sensitive area.
- delete a color by double-clicking the mouse on the selected triangle.
  CAUTION: Because the gradient is a progression between two or more colors, the cancellation of a color can be up to the gradient will consist of a number of elements greater than two.

6.21.1.3 Fill options

Fill in the options panel displays the parameters of the two types of gradient fill and radial. For the gradient fill can set both the direction of the color cast, and the distance between the exterior colors. To fill the radial we can set the center point, the focus point and the length of the beam. For both types of coverage you can set the repeat mode color useful in all those cases where the fill itself unable to cover the entire surface of the text that has been applied.



6.21.1.4 Overview of the fill

The preview panel, as well as provide an instant preview of the filling, it also allows direct editing through the simple use of the mouse. For the gradient fill, you can set rotation and distance, while the radial fill, you can set the center point, fire point and radius.
Each changes made are instantly notified about filling out, causing the update of the current text we are editing the filling. Given that the immediate notification in certain cases can be quite costly in terms of performance, this feature can be disabled by unchecking the component called "Preview" located under the preview panel. In this case the filling of the current text will be updated at the close of the panel.

6.21.1.5 Presets

This panel handles a set of presets that you can use and customize.
The management is entrusted presets as you can see from the figure, respectively, three components that allow you to choose the preset you want from those available, save (+) or delete (-) a preset.

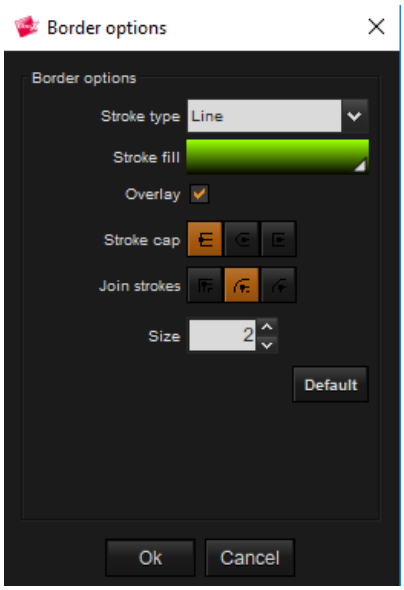
6.21.2 Edge of a text

To further enhance the look of your text, it's possible to create a border text. Pressing the button below the word "stroke" will open the panel for editing the text edge.

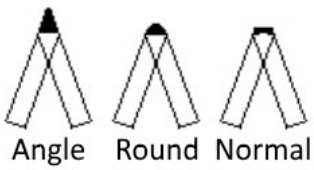


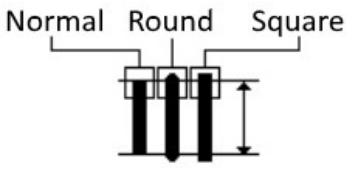
Currently there are five types of board, all customizable:

- **None**: no border;
- **Line**: border line type;
- **Dashed**: type edge dotted line;
- **Wave**: type edge wavy line;
- **Text**: text-edge.



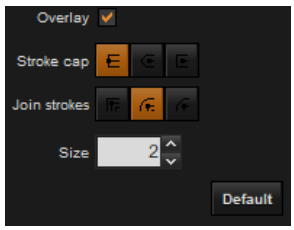
In each case, excluding the edge type "none", you can set the fill color, already discussed in detail in the previous subsection.
Through the option to decide whether we can superimpose the overlapping edge of the text or the opposite.



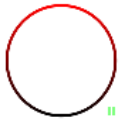


6.21.2.1 Edge type line

This is the classic board graphically represented by a line. As can be seen in the figure, we can set the thickness, and properties of the end line and the junction.
The icons "**Stroke cap**" act directly on the end of the line, giving it a rounded or square. The right figure shows the difference of styles in detail. Notice how the edge extends the segment compared to the original length of the polygon.



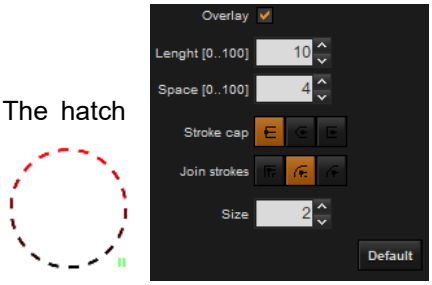
The three icons called "**Join strokes**", acting on the junction points of the edge. The three modes are: "Angle", "Round" and "Normal".
The right figure shows the difference between the styles: the first allows to combine the segments by creating a convex corner, the second creates an arc and finally the third creates a conjunction normal.



The button "**Default**" restores the default settings.

6.21.2.2 Edge dashed type

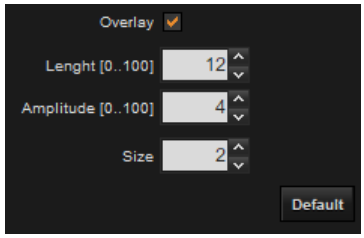
If the edge type is set to "**Dashed**", the window appears as in figure. The hatch options include customizing the "**Stroke Length**" which serves to set the length of individual traits and the "**Space**", which allows to determine the space between treatments.



As we mentioned earlier, the bottom panel is identical to the previous year and maintains the same operation.

6.21.2.3 Edge wade type

If the edge type is set to "**Wave**", the window appears as in figure. The wave-like edge options include customizing the "**length**" and the "**amplitude**" wave.

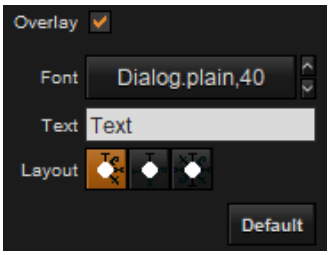




The button "**Default**" restores the default settings.

6.21.2.4 Edge text type

If the selected edge type is "**text**", the window appears as in figure.



The button "**Font**" allows you to select the font and its size through the **font-chooser**. To the right of this button are two arrow-shaped icons that serve to increase or decrease the current size of 10 point font.
In the "**Text**", you can edit the text to use in the border.
The three icons named "**Layout**" allow distribution of the text in three ways:

- **normal**: The text is distributed with their spacing as long as there are characters;
- **fits**: the text is distributed completely on the edge of extending the spacing between characters;
- **cyclical**: the text is distributed with its narrow spacing, and repeated until the entire board;



The button "**Default**" restores the default settings.

6.21.3 The “Text editor” of the MediaText

The **MediaText** plugin, compared to the others, has additional functions. Within the settings panel, in the **Aspect** section it is possible to write a free text by double-clicking on "**Text**" above "**Font**".





By clicking on the button with a “T” it is possible to have a more extensive view of the text-area.

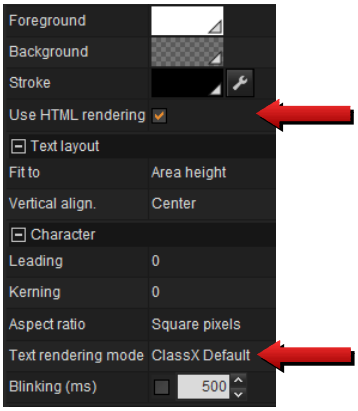
The entered text will be displayed by default without rendering the selected font and with a standard view from left to right. To change the writing direction, it is necessary to check "**Text right to left (RTL)**", while to display the text with the desired font, simply check "**Use font in the editor**".



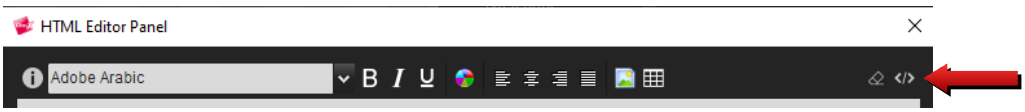
By clicking on this button it will be possible to open the **WYSIWYG HTML Editor** to customize and enrich any type of text with the support of HTML tags.

First of all, to use the functions of the HTML editor, you need to perform two steps:

- Set "**ClassX Default**" as the rendering mode
- Enable the "**Use HTML rendering**" flag



As can be seen from the image below, in addition to the classic “Bold, Italic and underlined” and the various text alignments, it is also possible to insert one or more images, tables, choose any type of color or manually customize the HTML tags.



For inserting images, you have to insert them as URL. Local images must also be entered via URL. The tab “Link” allows you to connect an external link to the image.



As for these two icons, the first deletes all content within the editor, while the second opens an HTML editor where you can manually insert or edit HTML tags.

Support HTML tags are:

- H1,H2,H3,H4,H5,H6
-
,
- ,<b>,<i>
- <font> with attributes font-family, font-size, font-fill
- with attributes font-face, font-size, font-fill
- <img>
- <table>, <tr>, <td>
- ,

TRICK: to insert a line break simply use the keyboard combination "shift + enter"

Warning: Being an editor not aimed at creating web pages but inserted exclusively to facilitate the composition of words / texts with different visual characteristics, some HTML tags may not match during rendering.

Another method to create HTML content within the text area is the traditional one already present in previous versions of LiveBoard. **HMTL mode** is automatically enabled when text starts with **<html>**.

Example:

```
<html><h2 style="text-align: center;">    Bello e <b><span style="face:Courier; color: #ff0000;">socievole</span> con tutti</b>. <br /> Interagire con i social network ora è più facile che mai.</h2>
<p style="text-align: left;"><span style="font-family: Verdana, Arial, Helvetica, sans-serif;"><strong> Interfaccia</strong> estremamente <strong>intuitiva</strong> per il <strong>controllo</strong> e la <strong>gestione</strong> di messaggi <strong>Twitter</strong>,</span> <br><br><FONT SIZE="20" FACE="Tahoma"><B><span style="font-family:Tahoma;">ClassX Logo</span></B>&#8482; <span style="font-family:Tahoma;">, per la generazione di logotipi e marchi di stazione statici ed animati con supporto di trasparenza.</span></FONT></p>
```



NOTE: all the unsupported HTML tags are ignored.

The HTML mode is automatically enabled when the text starts with **<html>**.

The tag ** ** replace **the white space**. A sequence of white spaces is ignored and collapsed into a single space. So we suggest to use this tag to preserve spaces between words and characters.

The tags H1,H2,H3,H4,H5,H6 set the only the font size: the value is automatically computed starting from the fot size of the Text Object, following this table:

H1 = font-size * 220 %
H2 = font-size * 200 %
H3 = font-size * 180 %
H4 = font-size * 160 %
H5 = font-size * 140 %
H6 = font-size * 120 %

7 International settings

Windows integrates an internal language management system that allows you to change the format used to display information to match the standards or language used in the user's country.

If you need to type and edit text in multiple languages, you can add input languages as shown in this video guide: http://www.youtube.com/watch?v=xtwnxkcNrE4.

When you add input languages or keyboard layout, Windows displays the language bar on the desktop.
Through the language bar or the Alt + Shift key combination, you can change the input language or keyboard layout.

By selecting the desired language, LiveBoard will display the user interface text, dialogs, menus and Help topics according to your preference.

If it is not possible to locate the text in the set language, English will be used.

7.1 Right to left text orientation

The LiveBoard text field displays the characters automatically from left to right. However, you can change the text orientation simply by pressing the Ctrl + Shift + O key.

8 Supreme Graphics Library

Among the various technologies developed by ClassX in the course of its business, **SGL** is perhaps the most meaningful summary of the experiences and skills that ClassX has gained to date in the graphics field.

SGL is the graphics system equipped to all the current ClassX solutions. **SGL** has been designed specifically to accelerate all graphics operations by exploiting the host machine's hardware and in particular the graphics card.

SGL is based on **OpenGL** and it draws all the power, efficiency and quality of tracking this important standard for the management of 2D and 3D graphics.

The performance of the accelerated programs with SGL greatly depends on the host machine: the faster the CPU-GPU system, the quicker the management and the creation of frames of the projects, to the benefit of the complexity and "lightness" of execution.

SGL supports all current graphics cards based on ATI and Nvidia, AGP bus, PCI-Express, PCI-X. The latest bus can dramatically increase the performance of SGL.

SGL at a glance:

Based on the OpenGL standard, it can guarantee superior quality and performance in all graphics operations.
It speeds up all 2D/3D graphics operations with AGP cards by several orders of magnitude.
It minimizes the workload on the CPU for graphics processing, using the maximum GPU.
Ready for PCI-Express, PCI-X, with the possibility of further acceleration thanks to the high-speed pixel read-back of these standards.
Supports broadcast resolutions, interlace and progressive scan.
Provides a high quality of graphics design.
SD-HD-ready HDV support high resolutions without any structural intervention.
It uses advanced OpenGL features. Compatibility and stability guaranteed with NVidia cards or ATI.

SGL is ready for the graphics of the future. We are currently proceeding with the implementation of:

Lighting, shading and depth of management via Z-buffer.
Vertex shaders and pixel shaders for real-time effects of light refraction, surface roughness, etc. polygonal distortion (depending on the current status of drivers for different cards).
Generation of hardware real-time volumetric light effects, distortions, displacement maps, feedback, motion blurs, etc..

8.1 Important notes on using SGL

8.1.1 Stability of use

SGL relies almost exclusively on the graphics card via the OpenGL graphics library. This type of implementation consequently means that more powerful graphics cards match, as a rule, more performance. Similarly, the quality of implementation of the card's drivers directly influence the quality and stability of SGL: a badly implemented driver may cause malfunction and even, in extreme cases, a system error. For this reason we invite you, if you encounter anything wrong, to check and possibly replace the VGA card drivers with a more stable and updated ones.

8.1.2 Quality and performance

The tracking quality through SGL depends on the graphics card settings and can be modified through its control panel. For details consult the operating manual of your graphics card. The voices 'Alias' and 'Texture Filter' are the most influential on the quality achievable with SGL. It may happen that the use of higher quality reduces the performance of SGL. Usually, the default settings of any graphics card are those that provide the best compromise between speed and quality.

8.1.3 Advanced Alpha management

One of the goals that we pursue more closely during the development of our applications is the quality of our graphics generation engine. The development and introduction of SGL in our application has brought enormous benefits in terms of performance, flexibility and productivity.

Today, **SGL** technology has been enriched by ClassX with an important feature called "**Advanced Alpha Management**", a mechanism for calculating the semi-transparent graphics that can produce unmatched quality results.

This mode uses the latest technologies and features introduced by the OpenGL 2.1 standard (GLSL, in particular). This means that the quality increase does not load the processor since all calculations are carried out solely in the VGA card GPU, at impressive speed.

Here's an example of the operating results of the **advanced alpha** feature:



As you can see, the edges of the "Diretta studio" animation show unsightly artifacts that make its borders appear lighter in colour. These are even more evident during the rotation of the object and particularly annoying because in in case of interlaced video (TV), this can cause plenty of flickering. The creation of artifacts is the result of an inaccurate calculation of the interpolated pixel transparency introduced during the rotation.

With the correct version (with enhanced alpha), the results are considerably better and the artifacts are completely gone.



The same concept also applies to texts, where the pixels on the edges of each letter are made transparent by both anti-aliasing algorithms (in supersampling rasterization of the curves that make up the characters), and motion tween:



Enabling "**advanced alpha management**" is optional and entrusted to a single check box in the Preferences panel of SGL. We recommend that you leave this feature enabled to achieve each time the highest quality of output.

8.1.4 Suggestion of use

SGL can be enabled on any MoreCG, CastaliaCG, CoralCG and LiveBoard project without any limitation. SGL accelerates all the graphics features by several orders of magnitude and the results obtained in terms of performance and quality are truly impressive. However, there is a limited set of cases in which SGL may also face difficulties in performing all the requested hardware acceleration:

- Large video clips and / or frequently updated animated textures
- Highly fractionated slide effects
- High quality wind effects
- Lyrics with very long path of movement
- Shape effects with large number of particles and movements on the path

9 LiveBoard Control

LiveBoard has a control interface called LBControl that allows you to interact with the graphic contents present in the current project. The LBControl interface offers all the tools to modify and execute the graphic contents but it does not allow to be able to add or remove them from the project. LBControl is used both from the Script content, both from the TemplateActions and from the remote protocol.
Centralizing LiveBoard in one single element provides security, performance and flexibility. A new feature introduced in the LBControl API, a modification or correction of a bug is immediately inherited by the technologies that integrate it. For this reason, every command available in the script environment corresponds orthogonally to a remote command available for the TemplateActions and for the remote protocol itself.

For example:

LBControl.stopContent(“B”); It corresponds to the command LBC_PLAYCONTENT(“A”);
LBControl.stopContent(“B”); It corresponds to the command LBC_STOPCONTENT(“B”);
etc...

The complete documentation can be consulted in the ClassX Developer Kit in the relative folder LBScripting\javadoc_lbcontrol.

For remote protocol and the Scriting environment, the LBControl interface offers a set of commands that can interact with the settings of the output device. The complete documentation of these commands can be consulted in the OutputDevice remote protocol section.

10 For the users

10.1 Programming Errors

LiveBoard was developed with the utmost care and has been tested on different configurations. The program should be free from problems and ready to be thoroughly exploited.
However, since it has been developed and tested by humans, could suffer from problems never encountered during the tests. It is an advantage for everyone if the user encountering such a problem sends a detailed written report to ClassX at support@classx.it bearing the following information:

- configuration
- steps to reproduce problem
- expected result
- obtained result
- workaround (if applicable)
- reproducibility (always / sometimes / rarely)
- level (show-stopper / urgent / low priority)

10.2 Collaboration

Do you have new ideas for improvement of LiveBoard?
Do you have advice and/or constructive criticism?
You want to develop a new effect or a new form of Export?
Any such communication will be seriously considered and evaluated for the development in a next version of LiveBoard.

10.3 How to obtain support via internet by ClassX

Please send an email to **support@classx.it**, remembering to include in the subject line the word "SUPPORT to **LiveBoard** ". Please allow us a week's time to answer your mail if you have no priority support-contract with ClassX (ClassX Maintenance Agreement).

All the latest news about LiveBoard and other products can be found at **www.classxcg.com**

Find tutorials, Tips&Tricks and more videos on our ClassX Dashboard.
If you don't know your credentials, please ask ClassX support, specifying your license number.

10.4 Important

Nothing contained in this document will give rise to representations or warranties, express or implied, with respect to the products described herein, the functionality, compatibility and availability.
In addition, the ClassX assumes no responsibility or liability for statements or assertions made by itself or by third parties or vendors in the prints reproduced here.
In any case, ClassX will never be liable for any direct, indirect, incidental or consequential damages resulting from any claims that may arise from allegations here, although there was no information on the possibility of such damages.

General Index

1 INTRODUCTION AND PROVISIONS 3
1.1 What is LiveBoard 3
1.2 Who is LiveBoard intended for 3
1.3 Technical features 3
1.4 Conditions of sale, installation and support 5
1.5 Software protection 5
1.6 Organization of files and directories 5
1.6.1 The ClassX conventions for naming files 5

2 LIVEBOARD INTERFACE 6
2.1 Docking/undocking panels 6

3 THE TOOLBAR 7
3.1 Loading a project 7
3.2 Saving a project 7
3.3 New project 7
3.4 Pack and Unpack a project 8
3.4.1 Pack Project (Export) 9
3.4.2 Unpack Project (Import) 12
3.5 Menu “File” 13
3.6 Preferences 13
3.6.1 Path 14
3.6.2 Misc 14
3.6.2.1 Positioning 14
3.6.2.2 Template 14
3.6.2.3 Graphics attributes 14
3.6.2.4 Remote connection 15
3.6.2.5 OSC protocol and integration with CuePilot 15
3.6.3 LiveBoard 16
3.6.3.1 Button size preferences (resume/loop) 16
3.6.4 SGL 16
3.6.5 Video 17
3.6.5.1 Audio output channels 17
3.6.5.2 Output video buffers 17
3.6.5.3 The Key mode 18
3.6.5.4 Spout output device 18
3.6.5.5 Preview output device 18
3.6.5.6 Video preferences import/export 19
3.6.6 Audio 19
3.6.6.1 Preview audio settings 19
3.6.6.2 Output audio track mapping 19
3.6.7 Video Input Manager 20
3.6.8 ClassX Issues Reporter 21
3.7 Menu Workspace 22
3.8 Menu Help 22
3.8.1 Info 22
3.8.1.1 Memory management 22
3.8.1.2 The “GC” button 22
3.9 Play to click 23
3.10 No-panic key 23
3.11 The Four channels 23
3.12 FPS meter 23
3.13 Import/Export/Clear channel and the resolution 24
3.14 Audio management of the preview 24
3.15 Frame Capture 24
3.16 NDI preview status 24
3.17 Tools 25
3.17.1 Import image sequence 25

3.17.2 Media Recorder *25*
3.17.3 Media Converter *26*
3.17.3.1 Utilization 26
3.17.3.2 Fill alpha channel 26
3.17.3.3 Auto Crop 26
3.17.3.4 Add Watermark 27
3.18 Export 28
3.18.1 Export to AVI with Alpha Matte *28*
3.19 Console 29
3.20 Automatic Start-up 29
3.21 Closing LiveBoard 29

4 CONTENTS GRID 30
4.1 Management of the Tabs 31
4.1.1 Move and rename a tab *31*
4.1.2 Operations on contents *31*
4.1.3 Content Preview in the Grid *32*
4.2 Contents 32
4.2.1 Creating a content *32*
4.2.2 Create a content from file system *33*
4.2.3 Deleting a content – Intelligent Dustbin *33*
4.2.4 Cutting, copying and replicating a content *34*
4.2.5 State of the content *34*
4.2.5.1 PLAY State 35
4.2.5.2 PAUSE/LOOP State 35
4.2.5.3 EDIT State 36
4.2.5.4 Rename a content 36
4.2.6 Common properties *37*
4.2.6.1 Icon and identification name 37
4.2.6.2 Keyboard Event 37
4.2.6.3 Mutual exclusion 37
4.2.6.4 Label 38
4.2.6.5 Execution in a group 38
4.2.6.6 Hide in positioning 38
4.2.6.7 Associated audio file 38
4.2.6.8 Verifying a content 39
4.2.6.9 Advanced controls 39
4.3 Common contents 39
4.3.1 File chooser and Tags browser *40*
4.3.1.1 Tag insertion and search 41
4.3.2 File chooser and Tags browser *42*
4.3.2.1 Tag insertion and search 43
4.3.3 Image content *44*
4.3.4 Media content *45*
4.3.4.1 MarkIn e MarkOut 46
4.3.5 Graphic Content *47*
4.3.6 Plugins Content *47*
4.3.7 Group Content *48*
4.3.8 Script Content *48*

5 THE PANELS 49
5.1 Main output device Panel 50
5.2 Positioning Panel 51
5.3 Preview Panel 54
5.3.1 FrameMarker *55*
5.3.2 Edit FrameMarker *55*
5.3.2.1 TemplateActions 56
5.4 Plugins controls Panel 57
5.4.1 Template Controller *57*
5.4.1.1 Control buttons 58

5.5 DataSource Panel....59
5.5.1 Status of a DataSource....59
5.5.2 Selected Record....59
5.5.3 Filter the available records....60
5.6 Data Panel....60
5.6.1 DataSource Options....61
5.6.1.1 Auto update....61
5.6.1.2 Update values on change....61
5.6.1.3 Keep user data....61
5.7 Script Editor Panel....62
5.8 Services Monitor Panel....63
5.9 Error Manager Panel....63
5.10 Shared Console Panel....63
5.11 Playlist Panel....64
5.12 Effects Panel....65
5.13 Video Input Multiview Panel....66
5.14 XR Settings Panel....67
5.15 Template Animations Panel....68
5.16 Graphic attributes Panel....69
5.16.1 Operations on attributes....69
5.16.2 Transparency....69
5.16.3 Solid....70
5.16.4 Background....70
5.16.5 Flip....70
5.16.6 Color correction....71
5.16.7 Buttonize....71
5.16.8 Outline....71
5.16.9 Emboss....71
5.16.10 Extrusion....72
5.16.11 Antialias....72
5.16.12 Shadow....72
5.16.13 Grab from Screen....72
5.16.14 Hardware accelerated Attributes....73

6 THE PLUGINS....74
6.1 The playlist....74
6.1.1 Playlist from folder....75
6.1.2 Playlist from file....75
6.1.3 Advanced options for Ticker and StyledTicker....75
6.2 Chart v.1.0....76
6.2.1 Pie Chart....76
6.2.2 Histogram chart....77
6.2.3 Line chart....77
6.2.4 Border and padding on the edge shape....77
6.3 ContentPlayout v.3.0....78
6.3.1 Assets and playlist....78
6.3.1.1 The asset status....79
6.3.1.2 Video asset....79
6.3.1.3 Gap asset....79
6.3.1.4 Image asset....79
6.3.1.5 Pause asset....79
6.3.2 Playlist execution....80
6.3.3 The Actions....80
6.3.3.1 Actions editor....80
6.3.3.2 FrameMarker....81
6.3.3.3 Playlist Import/Export....81
6.3.3.4 Timeline and FrameMarker....81
6.3.3.5 The FrameMarker Table....82
6.3.3.6 FrameMarker Actions table....82
6.3.3.7 Edit action values....82

6.3.3.8 Asset preview....82
6.3.4 Asset video settings....83
6.4 Image v.2.0....84
6.5 Simple MediaPlayer v.3.0....85
6.5.1.1 MarkIn e MarkOut....86
6.6 DigiClock v.1.6....87
6.7 Clock v. 1.4....88
6.8 LiveTimer v.1.1....89
6.9 MediaPlayer v.2.0....90
6.10 SlideShow v.1.3....92
6.11 MediaText v.1.3....93
6.12 Ticker Horizontal / Vertical v.3.0....94
6.13 Styled Ticker Horizontal v.2.0....95
6.13.1 The “HTML Settings”....96
6.14 TextShow v.1.2....97
6.15 Karaoke v.1.1....99
6.15.1 Manual Mode....99
6.16 ScreenCapture v.1.1....101
6.17 Shape v1.0....102
6.17.1 “Rectangle” Shape....102
6.17.2 “Ellipse” Shape....102
6.17.3 “Polygon” Shape....103
6.17.4 “General path” Shape....103
6.18 SharedFrame v.1.0....104
6.18.1 Create a SharedFrame plugin....104
6.18.2 Use a SharedFrame plugin....105
6.19 Video Input Source v.1.0....106
6.19.1 Video Input Manager....106
6.19.1.1 The FX “Video Process Editor” panel....107
6.19.2 Video input sources types....110
6.19.2.1 NDI© source....111
6.19.2.2 The SharedFrame....112
6.19.2.3 Connections with SDI video cards....113
6.19.2.4 Colorbars....113
6.19.2.5 Spout....113
6.19.2.6 Directshow....114
6.19.2.7 LibVLC....115
6.19.2.8 Chromium....118
6.19.2.9 Dummy....119
6.19.2.10 MediaEngine....119
6.20 Video Input Switcher v.1.0....120
6.20.1 The interface....120
6.20.2 “Dual video” display of A and B....121
6.20.3 Small guide on using Video Input Switcher....121
6.20.4 How to create a transition for VIS with template editors....122
6.20.4.1 Software to be used for creating the templates....122
6.20.4.2 The Timeline of a transition template....122
6.20.4.3 Notes on 3D transitions made with CoralCG....123
6.21 The functions on the text and the "Text-area"....124
6.21.1 Filling of an object....125
6.21.1.1 Solid Filling 125
6.21.1.2 Filling and radial gradient 125
6.21.1.3 Fill options....126
6.21.1.4 Overview of the fill....126
6.21.1.5 Presets....126
6.21.2 Edge of a text....126
6.21.2.1 Edge type line....127
6.21.2.2 Edge dashed type....127
6.21.2.3 Edge wade type....127
6.21.2.4 Edge text type....127

6.21.3 The “Text editor” of the MediaText 128

7 INTERNATIONAL SETTINGS 130
7.1 Right to left text orientation 130

8 SUPREME GRAPHICS LIBRARY 131
8.1 Important notes on using SGL 131
8.1.1 Stability of use 131
8.1.2 Quality and performance 131
8.1.3 Advanced Alpha management 132
8.1.4 Suggestion of use 133

9 LIVEBOARD CONTROL 134

10 FOR THE USERS 135
10.1 Programming Errors 135
10.2 Collaboration 135
10.3 How to obtain support via internet by ClassX 135
10.4 Important 135